FACULTY OF SCIENCE AND TECHNOLOGY
UNIVERSITAS JAMBI

4TH JAMBI INTERNATIONAL CONFERENCE ON ENGINEERING, SCIENCE AND TECHNOLOGY

BOOK OF ABSTRACTS

JICEST 2026

"Accelerating Green Innovation and Digital Transformation in Science, Technology, and Engineering for a Sustainable Future"

Wednesday, November 11, 2026 | University of Jambi | Online

Faculty of Science and Technology, University of Jambi
https://jicest.unja.ac.id fst_unja

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About The 4th JICEST 2026

The **Jambi International Conference on Engineering, Science, and Technology (JICEST)** is an annual international conference organized by Faculty of Science and Technology (FST), Universitas Jambi. Now in its fourth edition, JICEST brings together researchers, experts, practitioners, scholars, and students to exchange research findings, ideas, and practical solutions across engineering, science, and technology.

4th JICEST 2026 is held online from Universitas Jambi, Indonesia, on 11 November 2026. The conference is built around the theme "**Accelerating Green Innovation and Digital Transformation in Science, Technology, and Engineering for a Sustainable Future**," which places green innovation and digital transformation side by side. It asks how advances in science, technology, and engineering can be turned into outcomes that hold up in practice: cleaner processes, more resilient infrastructure, and tools that reach the people who need them.

The 2026 programme is organized into six sub-themes, each reviewed by its own scientific committee so that submissions are assessed by specialists in the relevant field:

1. Mathematical & Natural Sciences (chemistry, physics, biology, mathematics, environmental science, industrial chemistry, chemical analysis).
2. Earth Sciences & Mining Technology (geophysics, geology, mining engineering, sustainable resource management).
3. Civil, Chemical & Environmental Engineering (sustainable infrastructure, chemical process engineering, environmental technology, green engineering).
4. Electrical Engineering & Information Systems (smart technology, IoT applications, data analytics, digital innovation).
5. Educational Technology (digital transformation in education, STEM education).
6. Applied Science & Sustainable Innovation (technology transfer, innovation management, sustainable solutions for society).

Welcoming Remarks
From The Dean of Faculty of Science and Technology

Distinguished Speakers, Esteemed Guests, Presenters, and Participants,

It is my great pleasure and honor, on behalf of the Faculty of Science and Technology, Universitas Jambi, to warmly welcome you to the **4th Jambi International Conference on Engineering, Science and Technology (JICEST 2026)**

This year's theme, **"Accelerating Green Innovation and Digital Transformation in Science, Technology, and Engineering for a Sustainable Future,"** reflects the importance of scientific advancement and technological innovation in addressing today's global challenges. Issues such as climate change, environmental degradation, sustainable energy, and digital transformation require innovative approaches and collaboration across scientific disciplines

As an academic institution, the Faculty of Science and Technology recognizes its important role in advancing education, research, and innovation. We are committed to fostering scientific development and encouraging research that not only contributes to academic progress but also provides practical solutions to societal and environmental challenges.

Through JICEST 2026, we hope to create a valuable platform for researchers, academics, professionals, and students to exchange knowledge, share research findings, and establish meaningful collaborations across institutions and countries. We believe that the exchange of ideas and experiences through this conference will open new opportunities for interdisciplinary research and strengthen international academic networks.

I would like to express my sincere appreciation to the Conference Chair, keynote and invited speakers, presenters, reviewers, session chairs, the organizing committee, and all parties who have contributed their time and efforts to making this conference possible.

Finally, I hope JICEST 2026 will inspire meaningful discussions, encourage new research collaborations, and generate valuable contributions to the advancement of science, technology, and engineering. May this conference serve as a stepping stone toward a more innovative, resilient, and sustainable future.

Welcome to JICEST 2026. I wish you all a successful, productive, and rewarding conference.

Drs. Jefri Marzal, M.Sc., D.I.T.
Dean, Faculty of Science and Technology
Universitas Jambi

Welcoming Remarks from Conference Chair of 4th JICEST 2026

Distinguished Speakers, Esteemed Guests, Presenters, and Participants,

On behalf of the Organizing Committee, it is my great honor and privilege to welcome you all to the **4th Jambi International Conference on Engineering, Science and Technology (JICEST 2026)**.

This year's overarching theme, "**Accelerating Green Innovation and Digital Transformation in Science, Technology, and Engineering for a Sustainable Future**", addresses the most urgent technological and ecological imperatives of our era. Having recently endured one of the worst wildfire seasons in our region, we are acutely reminded of the devastating impacts of climate change, air quality degradation, and environmental fragility. These catastrophic events serve as a stark wake-up call that incremental progress is no longer enough.

At its core, the natural world operates through intricate balances of energy, matter, and entropy. Modern science and engineering must align with these fundamental principles to design clean technologies, optimize material efficiency, and build resilient digital infrastructures. Science, technology, and engineering must now stand at the very forefront of ecosystem restoration, disaster mitigation, and sustainable resource management to prevent such environmental crises and safeguard our shared environment for future generations.

JICEST 2026 serves as a vital global platform bringing together researchers, academics, industry leaders, and policymakers from diverse disciplines. Over the coming days, this conference will showcase groundbreaking research, foster interdisciplinary dialogues, and forge meaningful international collaborations to tackle these pressing ecological and technological challenges.

This Book of Abstracts compiles the collective intellect and innovative ideas presented at JICEST 2026. It reflects the dedication of researchers worldwide who strive to turn scientific insight into impactful, sustainable solutions. We extend our deepest gratitude to our keynote and invited speakers, authors, reviewers, session chairs, and the entire organizing committee whose hard work has made this event possible.

We hope JICEST 2026 inspires rich discussions, sparks new research partnerships, and fuels your drive toward green innovation and technological resilience.
Welcome to the conference, and I wish you all a highly engaging and rewarding experience.

Dr.rer.nat. Lucky Zaehir Maulana, S.Si., M.Sc.
Conference Chair, 4th JICEST 2026
Universitas Jambi

Scientific Program of The 4th JICEST

Date : Wednesday, 11th November 2026
Theme : Accelerating Green Innovation and Digital Transformation in Science, Technology, and Engineering for a Sustainable Future
Place : Graha Singedekane, 7th Floor of the Postgraduate Building, Universitas Jambi (UNJA) Telanaipura Campus, Jambi City

Keynote Speakers

1. Asst. Prof. Dr. Nattinee Thongdee (Khorat Fossil Museum, Thailand)
2. Asst. Prof. Dr. Worapat Paireekreng (King Mongkut's Institute of Technology Ladkrabang, Thailand)
3. Prof. Dr. rer. nat. Martin Dressel (University of Stuttgart, Germany)
4. Dr. Eng. Nur Hamid (King Fahd University of Petroleum & Minerals, Saudi Arabia)

Invited Speakers

1. Ilmi Fadhilah Rizki, M.Si. (Indonesian Oil Palm Research Institute, Indonesia)
2. Tika Restianingsih, M.Sc. (Institute of Science Tokyo, Japan)
3. Asmida Herawati, M.Sc., Ph.D. (Research Center for Photonics, National Research and Innovation Agency, Indonesia))
4. Dr. Rini Arianti (University of Debrecen, Hungary)
5. Prof. Drs. H. Sutrisno, M.Sc., Ph.D. (Universitas Jambi, Indonesia)

Biography of Keynote Speakers

1. Asst. Prof. Dr. Nattinee Thongdee

Current Position : Assistant Professor
Current Affiliation: 1) North Eastern Research Institute of Petrified Wood and Mineral Resources, Nakhon Ratchasima Rajabhat University
2)Tourism Industry and Hospitality Program, The Faculty of Management Science, Nakhon Ratchasima Rajabhat University
Email : nattineeoh@live.com

Education Background

- Bachelor's Degree (B.A.): Tourism Industry - Chiangmai Rajabhat Institute, 1993 - 1996
- Master's Degree (M.A.): Tourism Industry Management - Chiang Mai University, 1999 - 2000
- Doctor of Philosophy (Ph.D.): Architectural Heritage Management and Tourism - Silpakron University, 2002 - 2007

Research Focus

- Cultural heritage & tourism management
- Community-based tourism
- Geotourism

Professional Experience

- Lecturer and Researcher, Tourism Industry and Hospitality Program, Faculty of Management Science, Nanhon Ratchasima Rajabhat University, Thailand, 1993 - Present
- Lecturer and Researcher, Tourism Industry and Hospitality Program, Faculty of Management Science, Chandrakaserm Rajabhat University, Thailand, 1993 - 1994
- Deputy Director in Academic Service and Research, Purtified Wood Research Institute Nanhon Ratchasima Rajabhat University, Thailand, 2005 - 2006
- Planning and Development Dean Assistant, Faculty of Management Science, Nakhon Ratchasima Rajabhat University, Thailand, 2006 - 2009
- Academic Consultant in Tourism Development of The Tourism and Sport Development, Nakhon Ratchasima, 2008 - 2009

- The Expertise and Reader in Tourism Academic Article, Tourism Development Research Institute, Thailand Research Fund, 2009
- Director of Tourism and Hospitality Program, Faculty of Management Science, Nakhon Ratchasima Rajabhat University, Thailand, 2014 - 2020
- Director of Tourism Industry Curriculum, Faculty of Management Science, Nakhon Ratchasima Rajabhat University, Thailand, 2012 - 2020
- Director of Thai Tour Guide Professional Course Nakhon Ratchasima Rajabhat University, Thailand, 2012 - 2020
- Thai Commercial Organization, Nakhon Ratchasima Province, Thailand, 2014 - 2018
- Head Researcher of "Developing a Geopark and Geotourism Academy to Build Geoparker Capacity for Sustainable Human Capital and Community Economic Development in Khorat", 2025
- Head Researcher of "NRRU Creative Innovation University in Culture, Nature and Paleontology in Khorat Global Geopark", 2026

Publications

- Nattinee Thongdee (2012), Knowledge Management on Tourism Interpretation in Cultural Tourism Destination at River Basin Civilization, Korat Plateau, A Paper published in proceeding for International Conference The Inter-University Cooperation Program "ASEAN Knowledge Networks for the Economy, Society, Culture, and Environmental Stability", 8-12 July, South Korea.
- Nattinee Thongdee (2012), The Competitiveness Enhancement and Acknowledgement of Local Community Enterprises in Cultural and Local Wisdom Tourism Based on Suffciency at One Tambon One Product (OTOP) Villages, Nakhon Ratchasima Province, A Paper published in proceeding for International Conference The Inter-University Cooperation Program "ASEAN Knowledge Networks for the Economy, Society, Culture, and Environmental Stability", Srilangka.
- Nattinee Thongdee (2013), Cultural Tourism Management Based on River Basin Civilization at Korat Plateau lifestyle for Tourism Value Added and Local Cultural Identity Conservation, A Paper published in proceeding for International Conference 2012, Brunei.
- Nattinee Thongdee (2014), Tourism Interpretation Management of Cultural Heritage Sites at Silk Villages: A case of silk villages in "NAKHON CHAI BU-RIN" South-Northeastern Region, Thailand, A Paper published in proceeding for International Academic Conference on Social Sciences and Management, Kyoto, 2014.

Biography of Keynote Speakers

2. Asst. Prof. Worapat Paireekreng, Ph.D.

Current Position : Assistant Professor
Current Affiliation: Business School at King Mongkut's Institute of Technology Ladkrabang (KMITL)
Email : worapat_peng@hotmail.com
Web : https://www.kbs.kmitl.ac.th/en/person/dr-vorapat-pairigreng/

Education Background

- Bachelor's Degree (Bachelor of Art): Political Science - Ramkhamhaeng University, 1993 - 1995
- Bachelor's Degree (Bachelor of Business Administration): Money and Banking - Ramkhamhaeng University, 1996 - 2000
- Bachelor's Degree (Bachelor of Business Administration): Management Information System - Thammasat University, 1993 – 1997
- Master's Degree (Master of Science): Technology Management - Thammasat University, 2004 – 2007
- Doctor of Philosophy (Ph.D.): Information Technology - Murdoch University, 2008-2012

Research Focus

- Personalization & Recommendation Systems
- Data Mining & Machine Learning
- Mobile & Educational Technology

Professional Experience

- Lecturer, King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand, 2024
- Director of Ph.D. Program, Information Technology, Dhurakijpundit University, Bangkok, Thailand, 2015 – 2024
- Director of Master Program, Information Technology, Dhurakijpundit University, Bangkok, Thailand, 2014 – 2024

- Director of Datatech Academy, Dhurakijpundit University, Bangkok, Thailand, 2018 – 2020
- Deputy Dean, Academic Affair Faculty of Information Technology, Dhurakijpundit University, Bangkok, Thailand, 2013 – 2014
- Head of Software Development Center, Faculty of Information Technology, Dhurakijpundit University, Bangkok, Thailand, 2012 – 2013
- Head of department, Information Technology, Faculty of Information Technology, Dhurakijpundit University, Bangkok, Thailand, 2001 – 2007
- Consultant in IT Satisfaction Survey for Metropolitan Electricity Authority, 2024
- Consultant in OnePolice for Royal Thai Police , 2024.

Publications

- Apichaya Nimkoompai, Worapat Paireekreng, Suwannit Chareen Chit. “Enhancing User Experience for Mobile Learning Using Augmented Reality and Learning Style” Journal of University of Babylon, Pure and Applied Science, Iraq. vol. 27(1), pp 345-355, 2019.
- Atsawin Surawatchayotin, Worapat Paireekreng and Aurawan Imsombut, “The Predictive Model of Higher Education Guidance for Information Overload of Learner Groups Using Hybrid Ensemble Techniques”. TEM Journal, Vol 11(4) pp. 1792-1803. 2022.
- K. Umaji and W. Paireekreng, “A Study of the Remote Working Efficiency in IT Project Implementation during the COVID-19 Pandemic”, WSEAS Transactions on Business and Economics, 20, pp. 400-409, 2023.
- Wirun Wetsiri and Worapat Paireekreng. “Automating Community Pharmacy Workflows: The Impact of RPA on Operational Efficiency and Patient Care”, Journal of Mobile Multimedia, 21(1), River Publishers, pp. 113-147, 2025.
- Pempeeorn Wangchailert and Worapat Paireekreng, “Enhancing FOREX Market Predictions: A Comparative Study of Candlestick Patterns and the MIDDAM Patterns”, ECTI Transactions on Computer and Information Technology, Vol. 19 No. 1, pp 121-134, 2025.
- Rossukon Suwannakoot amd Worapat Paireekreng. “Component Analysis for Driving Factors of Social Media Usage in Elderly Tourism”, Journal of Mobile Multimedia, 21(2), River Publishers, pp. 197-220, 2025.
- Takorn Prexawanprasut amd Worapat Paireekreng. “Classifier-Agnostic Ambiguity Detection Framework for Targeted Oversampling in Imbalanced Learning”, IEEE Access, 14, pp. 62707 – 62725, 2026.

Biography of Keynote Speakers

3. Prof. Dr. rer. nat. Martin Dressel

Current Position : Professor of Physics and Head of the Institute
Current Affiliation: Physikalisches Institut, Universität Stuttgart Pfaffenwaldring 57, 70569 Stuttgart, Germany
Email : dressel@pi1.physik.uni-stuttgart.de
Web : https://www.uni-stuttgart.de/en/press/experts/Prof.-Martin-Dressel/

Education Background

- Vordiplom (Bachelor's Degree): Physics and Philosophy - Universität Erlangen, Nürnberg , 1979-1981
- Diplom-Physik (Master's Degree): Physics - Universität Göttingen & Drittes Physikalisches Institut, 1981-1986
- Doctoral Degree (Dr. rer. nat.): Physics - Universität Göttingen, 1986-1989

Research Focus

- Electronic, magnetic and optical properties of novel materials
- Solid state physics
- Molecular physics and organic materials

Professional Experience

- Chairman of 6. "Intern. Conf. on the Low-Energy Electrodynamics of Solids", Kloster Banz, 2004
- Chairman of 429. WE-Heraeus-Seminar: "Microwaves for Condensed Matter Physics", 2009
- Co-chair at 4th Bilateral German-Estonian Workshop on "Strong Nonlinear Vibronic and Electronic Interactions in Solids", Tartu, 2013
- Co-chair at "International School and Workshop on Electronic Crystals", Cargese, 2014
- Chair of DPG-Symposiums "Higgs Modes in Condensed Matter Physics and Quantum Gases", Berlin, 2015

- Co-chair at 11th “International Symposium on Crystalline Organic Metals, Super¬con¬ductors and Ferromagnets”, Bad Gögging, 2015
- Vice-chair of Gordon Research Conference “Conductivity and Magnetism in Molecular Materials”, Bryant University, 2018
- Co-chair at SPICE-Workshop: “New Spin on Molecular Quantum Materials”, Ingelheim, 2022

Publications

- M. Steigleder, M. Dressel. (2024).High-temperature setup for infrared reflection spectroscopy. Review of Scientific Instruments 95, 0214629, https://doi.org/10.1063/5.0214629
- Pal, P. Doležal, S.A. Strøm, S. Bertaina, A. Pustogow, R.K. Kremer, M. Dressel, P. Puphal. (2024). SrCu(OH) 3 Cl, an isolated equilateral triangle spin S = 1/2 model system. Physical Review Research 6, 033027, https://doi.org/10.1103/PhysRevResearch.6.033027
- R. Mathew Roy, X. Feng, M. Wenzel, V. Hasse, C. Shekhar, M.G. Vergniory, C. Felser, A.V. Pronin, M. Dressel. (2025). Interlayer Charge Transfer Induced by Electronic Instabilities in the Natural van der Waals Heterostructure 4Hb-TaS_2. Physical Review Letters 135, 116503, https://link.aps.org/doi/10.1103/sgz5-qj71
- L.Z. Maulana, A.A. Tsirlin, E. Uykur, Y. Saito, M. Dressel, M. Imai, A.V. Pronin. (2025) Optical conductivity and band gap in the double-Weyl candidate $SrSi_2$ at ambient pressure. Physical Review B 112, 155209, https://arxiv.org/abs/2510.04266
- O. Saritas, F. Bolle, Y. Lin, M. Dressel, R. Potjan, M. Wislicenus, A. Reck, M. Scheffler. (2025). THz electrodynamics and superconducting energy scales of ZrN thin films. Applied Physics Letters 127, 192601, https://arxiv.org/abs/2509.17272
- D. Chatterjee, P. Dolezal, F. Abbruciati, T. Biesner, K.M. Zoch, R. Khassanov, S.S. Islam, G. Kaur, S. Roh, F. Capitani, G. Garbarino, C. Krellner, P. Mendels, E. Kermarrec, M. Dressel, B. Wehinger, A. Pustogow, F. Bert, P. Puphal. (2026). Emergence of a fluctuating ground state in Y-kapellasite under pressure. Physical Review Letters 136, 136701, https://doi.org/10.1103/PhysRevLett.136.136701
- S. Priya, M. Dressel, J. Liebman, N. Drichko. (2026). Charge-sensitive vibrational modes in BEDT-TTF salts: Signatures of charge ordering and site charge. Journal of Chemical Physics 165, 144702, https://pubs.aip.org/aip/jcp/article/165/14/144702/3406285/Charge-sensitive-vibrational-modes-in-BEDT-TTF

Biography of Keynote Speakers

4. Dr. Eng. Nur Hamid

Current Position : Postdoctoral Fellow
Current Affiliation: Interdisciplinary Research Center for Smart Mobility and Logistics (IRC-SML), King Fahd University of Petroleum & Minerals (KFUPM), Dhahran 31261, Saudi Arabia
Email : nur.hamid@kfupm.edu.sa
Web : https://pure.kfupm.edu.sa/en/persons/dr-nur-hamid

Education Background

- Bachelor's Degree: Physics - Diponegoro University, 2011 – 2015
- Master's Degree: Computer Science - Universitas Indonesia, 2018 – 2020
- Doctoral Degree (Dr. Eng.): Electrical Engineering and Computer Science - Kanazawa University, 2021 – 2024

Research Focus

- Optimization
- Artificial Intelligence
- Robotics
- Machine Learning
- Reinforcement Learning

Professional Experience

- Teaching assistant for Prof. Wisnu Jatmiko Laboratory, Faculty of Computer Science, Universitas Indonesia, 2018 - 2020
- Research assistant at Intelligent Robots and Systems (IRoS) Laboratory, Faculty of Computer Science, Universitas Indonesia, 2020 - 2021
- Internship at Eftax Co. Ltd., Scribble Osaka Lab, Osaka, Japan (16 August – 5 September 2022) homepage: https://eftax.co.jp/en/
- Postdoctoral researcher at Interdisciplinary Research Center for Smart Mobility and Logistics (IRC-SML), King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia, 2024 - Now

Publications

- Hamid, N.; Dharmawan, W.; Nambo, H. Dynamic Path Planning for Unmanned Surface Vehicles with a Modified Neuronal Genetic Algorithm. Applied System Innovation Journal. 2023, 6, 109. https://doi.org/10.3390/asi6060109
- Dwiyanti, L., Nambo, H., & Hamid, N. (2024). Leveraging Explainable Artificial Intelligence (XAI) for Expert Interpretability in Predicting Rapid Kidney Enlargement Risks in Autosomal Dominant Polycystic Kidney Disease (ADPKD). AI (Switzerland), 5(4), 2037–2065. https://doi.org/10.3390/ai5040100
- Prayogo, R.D., Hamid, N. and Nambo, H., (2025). Hybrid CNN-Based Transfer Learning Enhances Brain Tumor Classification on MRI Images. IEEE Access. (13) 116654 – 116668 10.1109/ACCESS.2025.3584376
- Chahyadi, F., Uperiati, A., Pratiwi, R. A. I., & Hamid, N. (2025). Evaluating Lexicon Weighting and Machine Learning Models for Sentiment Classification of Indonesian Mangrove Ecotourism Reviews. Jurnal Teknik Informatika (Jutif), 6(6), 5679-5698. https://doi.org/10.52436/1.jutif.2025.6.6.5563
- Hamid, N., & Saleh, H. (2026). ETA-Hysteresis-Based Reinforcement Learning for Continuous Multi-Target Hunting of Swarm USVs. Applied System Innovation, 9(1), 7. https://doi.org/10.3390/asi9010007
- Hamid, N., Antariksa, G., Dharmawan, W., Jati, G., Saleh, H., El, S., 2026. AI-driven swarm USV operations: A comprehensive bibliometric and analytical review. Advanced Engineering Informatics 69, 103972. https://doi.org/10.1016/j.aei.2025.103972
- Dharmawan, W., Saleh, H., Hamid, N., Alindra, R., El-Ferik, S., & Jaqin, C. (2026). WaveSpectro-XAF: Dual-branch deep learning with STFT spectrograms for power seat DC motor noise classification. ICT Express. https://doi.org/10.1016/j.icte.2026.01.004
- Dharmawan, W., Hamid, N., Saleh, H., Nurwidya, A. I., & Widati, P. L. (2026). Explainable wavelet-scalogram learning for quasi-stationary faults in automotive DC motors using AERIS-Wave. Advanced Engineering Informatics, 71, 104394. https://doi.org/10.1016/j.aei.2026.104394
- Hamid, N., Dharmawan, W., Nambo, H., Saleh, H., & El Ferik, S. (2026). AdaNGA: Adaptive neuronal genetic algorithms for unmanned surface vehicle dynamic path planning in dynamic environment. Ocean Engineering, 352, 124692. https://doi.org/10.1016/j.oceaneng.2026.124692

Participant Technical Guidelines

A. Opening Session and Main Seminar Session

1. The application used for this international conference is **Zoom virtual meeting** (http://zoom.us/download/).
2. Zoom virtual meeting link:

Date	Sessions	Zoom ID, Passcode, and Link	Operator
11th November 2026	Opening Session, Main Seminar Session, Parallel Sessions, and Closing Ceremony	ID: 972 7293 4303 Passcode: 447455 Link: https://zoom.us/j/97272934303?pwd=42DxiaHZDi1WSTAw3Q3GzKau2i4q1Q.1	UPA TIK UNJA

3. The conference program, including the rundown and access links for online sessions, can be found in the **Book of Abstracts 4th JICEST 2026**.
4. Participants attending online are expected to join the main seminar room on time.
5. Please change your virtual background after joined the **Zoom virtual meeting**. Make sure to download the virtual background from the following link: https://drive.google.com/file/d/1EO483nl_56wG3diasnMi76YaqWkNXvUq/view?usp=sharing
6. Participants who are not presenting are prohibited from turning on their microphone, except during the Q&A session and only when permitted by the moderator.
7. During the sessions, participants may ask questions to keynote speakers using the Q&A feature in the Zoom chat box or using ‘rise hand’ feature in Zoom.
8. The moderator has full authority to manage and conduct the conference.

B. Guidelines for Parallel Sessions

1. The schedule, presentation order, and breakout room assignments for online parallel sessions can be found in the **Book of Abstracts 4th JICEST 2026**.
2. The presenter may choose to use video or live presentation during presentation.
3. If the presenter chooses a video presentation:
   - The presenter prepares a presentation video in English with a Power Point slides. The maximum duration is 10 minutes.

- Presentation video can be made using any platform such as zoom, Logitech capture, Power Point record slide show, or any other platforms.
- Please make sure of the quality and size of the video and audio before uploading them into Google Form.

4. Please upload the video and Power Point slides via Google Form: https://forms.gle/pb4WmuJSpwbckVin7 with a maximum file size of 10 MB for file presenter and 100 MB for video (optional).
5. Rename the file (PPT or video) using this format: "**Presenter Name_institution_title (3 words)**". E.g.: Susi Susanti_UNJA_Utilization Palm Leaves
6. Online participants are required to join the breakout rooms Zoom corresponding to the scope assigned by the committee.
7. Each scope in the online parallel sessions will also feature invited speakers. Q&A session will follow each presentation. Parallel sessions are divided into two parts: Session I: Invited presenters
   Session II: Presenters
8. Each presenter will be given a maximum of 10 minutes for the presentation and 15 minutes for discussion after 3 or 4 presenters have presented. The presenter must be present during the oral presentation; the video will not be aired if the presenter is absent, and they will not receive a certificate.
9. The session will be fully moderated, including the facilitation of discussion and selection of participants allowed to respond or ask questions.
10. Participants who are not presenting are prohibited from turning on their microphone, except during the Q&A session and only when permitted by the moderator.

Program Schedule

Wednesday, 11th November 2026

Agenda	The Allocation Time (WIB)	Duration	Chair Person
Registration and Zoom Room Open *(video: JICEST and Profile FST)*	07:00-07:55	**55'**	Committee
Opening Ceremony			
Opening Remark Ceremony by MC	08:00 – 08:05	**5'**	Rochmat Jaya and Lilian Putri Salsabila
Indonesia National Anthem: Indonesia Raya	08:05 – 08:10	**5'**	Committee Dirijen: Intan Sri Ayu Binti Darul Nizsham
Opening Prayer	08:10 – 08:15	**5'**	Riskal Fadli, S.T., M.Eng.
Opening Remark by Chairman of the Committee	08:15 – 08:20	**5'**	Dr.rer.nat. Lucky Zaehir Maulana, S.Si., M.Sc.
Opening Remark by The Dean of Faculty of Science and Technology, Universitas Jambi	08:20 – 08:25	**5'**	Drs. Jefri Mazal, M.Sc.,D.I.T
Opening Remark by Rector Universitas Jambi	08:25 – 08:30	**5'**	Prof. Dr. Helmi, SH.,MH
Handover of certificates and plaques to keynote and invited speakers and documentation	08:30 – 08.35	**5'**	Prof. Dr. Helmi, SH.,MH or Drs. Jefri Mazal, M.Sc.,D.I.T
Closing Remark Ceremony by MC	08:40– 08:45	**5'**	Rochmat Jaya and Lilian Putri Salsabila
Coffee Break	08:45 – 09:00	**15'**	Committee
Panel Session 1			

Agenda			The Allocation Time (WIB)	Duration	Chair Person
Panel Session 1					
No	**Keynote Speaker**	**The Title of Presentation**	**The Allocation Time (WIB)**	**Duration**	**Chair Person**
Opening of the Plenary Session by the *Session Chair* / Moderator			09:00 - 09:05	**5'**	Moderator: Akhiyar Waladi, S.Komp., M.Kom
1	Dr. Eng., Nur Hamid, S.Si., M.Kom. *Postdoctoral Fellow at King Fahd University of Petroleum and Minerals, Saudi Arabia*	Advancing Autonomous Robotic Systems via Multi-Agent Reinforcement Learning: Applications to Multi-USV Coordination	09:05 – 09:45	**30'**	
2	Asst. Prof. Worapat Paireekreng, Ph.D. *Assistant Professor KMITL Business School*	Influencing Factors for Green Transitions toward Net-Zero Pathways in Municipal Solid Waste Management (MSWM)	09:45 – 10:15	**30'**	
Discussion			10:15 – 10:30	**25'**	
Panel Session 2					
Opening of the Plenary Session by the *Session Chair* / Moderator			10:30 - 10:35	**5'**	Moderator: Dr.rer.nat. Lucky Zaehir Maulana, S.Si., M.Sc.
3	Asst. Prof. Dr. Nattinee Thongdee *Director Khorat Fossil Museum*	From Fossils to Future: Museum Innovation, UNESCO Global Geoparks and Sustainable Geotourism	10:35 – 11:05	**30'**	
4	Prof. Dr. rer. nat. Martin Dressel *Head of Institute Physikalisches Institut, Universität Stuttgart*	Condensed Matter Science: Building Blocks for the Future	11:05 – 11:35	**30'**	
Discussion			11:35 – 12:00	**25'**	
Break			12:00 – 13:30	**90'**	Committee
Parallel session			13:30 – finish		Committee

Asst. Prof. Dr. Nattinee Thongdee

Earth Sciences & Mining Technology

FROM FOSSILS TO FUTURE: MUSEUM INNOVATION, UNESCO GLOBAL GEOPARKS AND SUSTAINABLE GEOTOURISM

Nattinee Thongdee[1,2]

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Abstract

Museums and UNESCO Global Geoparks increasingly function as platforms for scientific learning, geological heritage conservation, public engagement, and sustainable regional development. Yet museum outreach, science communication, and geotourism are often presented as separate activities, obscuring how scientific collections, authentic geosites, and community participation can form a coherent learning and development system. This paper examines that integration through Khorat Fossil Museum and Khorat UNESCO Global Geopark, Thailand, and proposes an Integrated Museum–Geopark Engagement Framework.

The paper offers a practice-informed conceptual synthesis of documented institutional initiatives, organized around three interrelated dimensions: museum outreach, science communication, and sustainable geotourism. Designated in 2023, Khorat UNESCO Global Geopark covers approximately 3,167 km² and is identified as “The Land of Cuestas and World Paleontopolis.” Its cuesta landscapes, dinosaur fossils, ancient elephants, and petrified wood provide a distinctive scientific and educational foundation. Located within the geopark office complex, Khorat Fossil Museum integrates indoor exhibitions with a ten-rai (approximately 1.6-hectare) forested conservation area containing petrified wood preserved in situ. As one of the geopark’s geological sites, this outdoor area connects museum interpretation with direct observation of geological heritage.

Asst. Prof. Dr. Nattinee Thongdee

In museum outreach, documented programs include community conservation training for 190 participants in 2022, teacher capacity-building activities involving more than 100 participants during 2023–2026, and youth camps that extend learning beyond museum galleries. Science communication is supported by fossil storytelling, interactive interpretation, augmented-reality applications, digital learning, and citizen science initiatives involving mobile-assisted biodiversity observations. Sustainable geotourism links museum and geosite experiences to local guides, geoproducts, geofood, festivals, community enterprises, and opportunities for place-based livelihoods. Together, these practices illustrate how research, museums, schools, geosites, and communities can contribute to a connected heritage experience.

The proposed framework comprises four connected elements: (1) geological and paleontological heritage as a scientific knowledge base; (2) outreach, communication, and geotourism as complementary engagement mechanisms; (3) an integrated visitor and learning journey across museums, geosites, communities, and digital platforms; and (4) potential educational, conservation, social, and local economic outcomes. Smart, Green, and Inclusive Geoparks are proposed as future development directions encompassing digital accessibility, environmental responsibility, and equitable community participation.

These documented initiatives indicate the scope of institutional engagement but do not establish causal effects on learning or livelihoods. The framework therefore offers a basis for future evaluation using visitor learning assessments, community feedback, conservation indicators, and livelihood measures. The Khorat case contributes a transferable perspective on how museum innovation and UNESCO Global Geopark management can connect scientific heritage with meaningful public understanding and sustainable geotourism.

Keywords: Museum Outreach; Science Communication; Sustainable Geotourism; UNESCO Global Geoparks; Khorat Fossil Museum; Indoor–Outdoor Learning; Community Engagement

Prof. Dr. rer. nat. Martin Dressel

Mathematical & Natural Sciences

CONDENSED MATTER SCIENCE: BUILDING BLOCKS FOR THE FUTURE

Martin Dressel

1. Physikalisches Institut, Universität Stuttgart, Germany

Abstract

Using our current research activities on molecular quantum materials as one example, several important conditions are highlighted that are important for innovative and successful research in the field of condensed matter science. Organic charge-transfer salts are such a tailored matter: Through molecular engineering, the building blocks and molecular arrangements can be modified in order to create materials with a desired functionality. This way we can realize physical properties that do not exist in nature but have been predicted by theory, for instance quantum spin liquids. Advancing experimental methods, e.g. broadband magnetic resonance spectroscopy at ultralow temperatures, enables us to characterize the novel ground state. Another way is the use of ultrafast lasers to create and monitor physical states out of equilibrium: Transient superconductivity at elevated temperatures and light-induced condensation in quantum spin liquid Mott insulators are subject of current research.

Keywords: Molecular quantum materials, Organic charge-transfer salts, Quantum spin liquids, Broadband magnetic resonance spectroscopy, Ultrafast lasers

Dr. Eng. Nur Hamid

Electrical Engineering & Information Systems

ADVANCING AUTONOMOUS ROBOTIC SYSTEMS VIA MULTI-AGENT REINFORCEMENT LEARNING: APPLICATIONS TO MULTI-USV COORDINATION

Nur Hamid

Interdisciplinary Research Center for Smart Mobility and Logistics (IRC-SML), King Fahd University of Petroleum & Minerals (KFUPM), Dhahran 31261, Saudi Arabia

Abstract

The rapid advancement of artificial intelligence (AI) has opened new opportunities for developing intelligent and autonomous robotic systems capable of operating collaboratively in complex and dynamic environments. Among these technologies, Multi-Agent Reinforcement Learning (MARL) has emerged as a promising approach for enabling autonomous agents to learn cooperative decision-making strategies through interactions with their environments. This presentation explores the development and application of MARL in autonomous robotic systems, with a particular focus on swarm Unmanned Surface Vehicles (USVs) for maritime operations. Beginning with an overview of recent research trends and challenges in swarm USV systems, the presentation introduces fundamental MARL concepts, including Centralized Training with Decentralized Execution (CTDE) and Multi-Agent Proximal Policy Optimization (MAPPO). Several research case studies are presented to demonstrate the application of reinforcement learning in cooperative maritime tasks, including continuous multi-target hunting, target visitation, cooperative encirclement, and capacity-constrained evacuation. These studies highlight the potential of learning-based approaches to support adaptive coordination, autonomous navigation, and efficient multi-agent task execution. The presentation also discusses simulation-based development using Unity 3D, key challenges in multi-agent coordination, and future research directions toward robust, scalable, and real-world autonomous robotic operations.

Keywords: Multi-Agent Reinforcement Learning, Autonomous Robotics, Swarm Unmanned Surface Vehicles, MAPPO, Cooperative Multi-Agent Systems, Maritime Robotics.

Asst. Prof. Dr. Worapat Paireekreng

Applied Science & Sustainable Innovation

INFLUENCING FACTORS FOR GREEN TRANSITIONS TOWARD NET-ZERO PATHWAYS IN MUNICIPAL SOLID WASTE MANAGEMENT (MSWM)

Worapat Paireekreng

KMITL Business School

Abstract

Municipal solid waste management (MSWM) plays a critical role in achieving national net-zero emissions targets, yet municipalities frequently face structural constraints, low recycling rates, and high landfill dependencies that generate substantial methane emissions. This study investigates the key determinants influencing green transitions toward net-zero pathways in MSWM under a bio-circular-green economy framework. Employing a mixed-methods sequential exploratory design, the research combines a qualitative phase utilizing in-depth semi-structured interviews with key municipal stakeholders to build a theory-informed framework, followed by a quantitative phase utilizing Partial Least Squares Structural Equation Modeling (PLS-SEM) to test hypotheses and validate relationships among core constructs, including policy and regulatory support, technological innovation, institutional capacity, financial mechanisms, and stakeholder engagement. The expected outputs deliver an integrated conceptual model, validated measurement instruments, and empirical evidence quantifying greenhouse gas (GHG) reductions linked to specific municipal interventions. Ultimately, this research provides policymakers and practitioners with an evidence-to-action framework to accelerate low-carbon transitions, improve resource efficiency, and support sustainable, resilient cities.

Keywords: Municipal Solid Waste Management (MSWM), Green Transition, Net-Zero Pathways, Partial Least Squares Structural Equation Modeling (PLS-SEM), Circular Economy

Paralel Schedule

Room 1 (Online) 13:30-16:00 Moderator: Bunga Mardhotillah, S.Si., M.Stat.		
Time (WIB)	**Presenter**	**Title**
13.30-13.50	**Prof. Drs. H. Sutrisno, M.Sc., Ph.D. (Universitas Jambi)**	Challenging Coffee Fermentation: Integrating Sensor-Based Bioreactor, Ai-Assisted Modelling, And Spectroscopic Investigation For Sustainable Coffee Quality Assessment **(Invited Speaker)**
13.50-14.00	Diah Riski Gusti (Universitas Jambi)	Development of a Natural Cosmetic Product: "KENTARA GLOW" Peel-Off Mask from Potato Peels as a Business Opportunity
14.00.14.10	Muhammad Syafiq Alparizi (Universitas Jambi)	Utilization of Papaya Seeds for Hair Care Products in RT 13, Pematang Gajah Village
14.10-14.20	Q & A	
14.20-14.30	Ambrina Rosada (Universitas Sriwijaya)	Evaluation of Antioxidant Activity Using DPPH and FRAP Methods and Cytotoxicity Test by BSLT Method of Lausonia Inermus Leaves Exteact
14.30-14.40	Intan Lestari, Anisa Fitri (Universitas Jambi)	Modification of Palm Kernel Shell Biochar with Urea-Derived Nitrogen for the Adsorption of Naphthol Dye
14.40-14.50	Q & A	
14.50-15.00	Nita Astasya (Universitas Jambi)	Collaborative GIS Flood Hazard Mapping with Local Public Works for Spatial Planning and Community Disaster Preparedness in Pematang Jering Village, Muaro Jambi
15.00-15.10	Widya Twiny Rizki (Universitas Jambi)	Chitosan–Glucose-Assisted Modification Of Sno_2/Fe_3O_4 Nanocomposites For Methylene Blue Adsorption And Photodegradation
15.10-15.20	Q & A	
15.20-15.30	Diah Riski Gusti (Universitas Jambi)	Corrosion Rate and Inhibition Efficiency of Jernang Fruit Peel Extract (Daemonorops sp.) in Sulfuric Acid Solution
15.30-15.40	Zultriana (Universitas Sriwijaya)	Antibacterial Activity Of Endophytic Fungi Isolated From The Root Of The Mangrove Bruguiera Sexangula
15.40-15.50	Daniel Panggabean (Universitas Jambi)	The Effect of Sulfur Modification on POFA via Microwave-Assisted Impregnation in Mining Wastewater Treatment
15.50-16.00	Q & A	

Prof. Drs. H. Sutrisno, M.Sc., Ph.D. *Topic*

CHALLENGING COFFEE FERMENTATION: INTEGRATING SENSOR-BASED BIOREACTOR, AI-ASSISTED MODELLING, AND SPECTROSCOPIC INVESTIGATION FOR SUSTAINABLE COFFEE QUALITY ASSESSMENT

Heriyanti[1], Indra Lasmana Tarigan[2], **Sutrisno**[3*]

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[2,3]*Department of Chemistry, Faculty of Science and Technology, Universitas Jambi*

Abstract

Coffee fermentation is a complex process in which microbial activity and physicochemical conditions shape the flavor, aroma, and quality of the final product. Nevertheless, conventional fermentation often relies on empirical monitoring, limiting process reproducibility, adaptability, and quality consistency. This presentation outlines an integrated approach to precision coffee fermentation using a sensor-equipped mini-bioreactor, Internet of Things (IoT) data acquisition, mathematical modelling, artificial intelligence (AI), and spectroscopic quality assessment, with emphasis on Liberica coffee from Jambi. The bioreactor platform is designed to monitor fermentation-associated gas production, including carbon dioxide (CO_2), hydrogen (H_2), methane (CH_4), and ethanol. Time-course gas data are interpreted using Modified Gompertz and Modified Logistic models to characterize maximum production rates, lag phases, inflection times, and model fit. Both models described gas-production profiles with coefficients of determination (R^2) of 0.95–0.98. The predicted maximum production rates differed among gases, with CO_2 reaching its peak rate earlier (approximately 6–7 h), followed by CH_4 and ethanol (15.5–16 h), and H_2 (17–17.5 h). Simultaneously, AI-assisted modelling is considered for predicting coffee quality attributes and supporting data-driven fermentation decisions. Preliminary validation results showed strong performance for protein prediction (training R^2 = 0.999; leave-one-out R^2 = 0.989) and carbohydrate prediction (training R^2 = 0.976; leave-one-out R^2 = 0.889), whereas fat prediction requires further refinement (leave-one-out R^2 = 0.483). Spectroscopic investigation and sensory profiling provide complementary routes for assessing quality changes. Collectively, these components establish a framework for more controlled, predictive, and reproducible coffee fermentation, supporting innovation, responsible production, and sustainable development of regional coffee resources.

Keywords: Liberica coffee; precision fermentation; mini-bioreactor; IoT sensors; artificial intelligence; kinetic modelling; spectroscopy.

Room 1

DEVELOPMENT OF A NATURAL COSMETIC PRODUCT: "KENTARA GLOW" PEEL-OFF MASK FROM POTATO PEELS AS A BUSINESS OPPORTUNITY

Diah Riski Gusti*[1], Lenny Marlinda[2], Intan Lestari[3], Rahmi Rahmi[4], Dina Erliana[5], and Munifilia Ekasari[6]

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Abstract

The development of "KENTARA GLOW," a natural peel-off mask made from potato peels, represents an innovation that transforms household organic waste into an economically valuable product. This initiative targets Mendalo Indah Village, a location with significant market potential for beauty products due to its proximity to Jambi Province's two largest university campuses. Training sessions on producing this peel-off mask—held in RT 04/RW 01, Mendalo Indah Village—empower housewives and unemployed women, fostering a creative, innovative, health-conscious, and cost-effective economy. Potato peels are generally viewed as worthless waste and often discarded immediately, despite containing various bioactive compounds—such as antioxidants and vitamins—that benefit skin health. Meanwhile, the public continues to rely heavily on synthetic cosmetics, which carry the risk of adverse side effects with long-term use.

Keywords: potato peels, peel-off mask, unemployed, , waste

Room 1

UTILIZATION OF PAPAYA SEEDS FOR HAIR CARE PRODUCTS IN RT 13, PEMATANG GAJAH VILLAGE

Muhammad Syafiq Alparizi[1], Rahmi[1], Intan Lestari[1], Martina Asti Rahayu[1], Lenny Marlinda[1], Diah Riski Gusti[1], D.M. Magdalena Ritonga[1], and Anggi Deliana S[1]

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Abstract

The current extreme hot weather has sparked numerous complaints from the public, particularly among women who wear the hijab. The heat triggers excessive sweating, which negatively affects hair health leading to issues such as unpleasant odors, limpness, and dandruff. The women of the RT.13 group in Kali Aro Hamlet, Pematang Gajah Village, are active in community activities such as religious study groups and social gatherings (arisan); the majority are homemakers. Papaya is a popular fruit during hot weather due to its high water content (90%). However, the community typically consumes only the flesh, while the seeds are discarded as waste. Papaya seeds have potential as a hair care product ingredient because they contain the glucoside cacirin and carpaine. This community service initiative addresses the frequent complaints regarding hair care challenges during hot weather; if left unaddressed, these issues could lead to further hair damage. The training on producing hair masks from papaya seeds aims to provide a solution for hijab wearing women who feel uncomfortable during activities in the heat. Additionally, the initiative seeks to utilize papaya seed waste through a production process that is both affordable and easy for homemakers to implement. A participatory approach involving educational outreach was employed. The activity stages included presentations, information dissemination, and evaluation. Participants included the Community Service Team, the neighborhood head (Ibu RT), and the women of the RT.13 group. The activity demonstrated high levels of participant enthusiasm. Evaluations by the Community Service Team and the neighborhood head concluded that such activities are vital for the RT.13 women's group, as the community had previously lacked information regarding the utilization of papaya seed waste.

Keywords: Hair mask, Papaya seeds, cocoyl glucoside, Carpaine.

Room 1

EVALUATION OF ANTIOXIDANT ACTIVITY USING DPPH AND FRAP METHODS AND CYTOTOXICITY TEST BY BSLT METHOD OF LAUSONIA INERMUS LEAVES EXTEACT

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Abstract

Pacar kayu (*Lawsonia inermis* L.) is known to contain phenolic compounds, including quinones, coumarins, and xanthones, as well as flavonoids and terpenoids. Phenolic and flavonoid compounds generally exhibit various pharmacological activities. This study aimed to determine the antioxidant and cytotoxic activities of pacar kayu leaf extracts using various solvents, namely n-hexane, ethyl acetate, ethyl acetate:methanol (1:1), and methanol. Antioxidant activity was evaluated using the DPPH and FRAP methods, while cytotoxic activity was assessed using the BSLT (Brine Shrimp Lethality Test) method. The results showed that the highest antioxidant activity in the DPPH assay was exhibited by the ethyl acetate extract, with an IC_{50} value of 109.82 µg/mL. In the FRAP assay, the methanol extract showed the highest antioxidant capacity, with a value of 49.74 ± 0.27 mg AAE/g extract. Based on the percentage of Artemia salina larval mortality, the highest cytotoxic activity was observed for the ethyl acetate extract, with a mortality rate of 96.67% at a concentration of 1000 µg/mL. The results indicate that the type of solvent affects the antioxidant and cytotoxic activities of pacar kayu leaf extracts. The ethyl acetate extract exhibited the highest antioxidant activity in the DPPH assay and the highest BSLT larval mortality, whereas the methanol extract showed the highest antioxidant capacity in the FRAP assay.

Keywords: *Lawsonia inermis*, antioxidant, DPPH, FRAP, cytotoxicitu

Room 1

MODIFICATION OF PALM KERNEL SHELL BIOCHAR WITH UREA-DERIVED NITROGEN FOR THE ADSORPTION OF NAPHTHOL DYE

Intan Lestari[1], Diah Riski Gusti[1], Damris M[1], and Anisa Fitri[1]

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Abstract

Biochar is a promising adsorbent for removing dyes from aqueous solutions. This study aimed to modify palm kernel shell biochar using urea as a nitrogen precursor to improve its surface characteristics and adsorption performance toward naphthol dye. The biochar was prepared through carbonization, chemical activation using KOH, physical activation, and nitrogen modification using urea. The material was characterized using Fourier transform infrared (FTIR) spectroscopy and scanning electron microscopy (SEM). The FTIR results indicated changes in surface functional groups following nitrogen modification, while SEM analysis revealed changes in surface morphology and pore structure. Adsorption experiments were conducted to investigate the effects of pH, contact time, and initial naphthol concentration. The highest adsorption capacity in the pH variation experiments was 3.423 mg/g at pH 2. For contact time, the highest adsorption capacity of 3.456 mg/g was obtained at 30, 60, and 90 min. Furthermore, increasing the initial naphthol concentration from 25 to 200 mg/L increased the adsorption capacity from 2.456 to 19.89 mg/g. These findings indicate that urea-modified nitrogen-doped biochar has potential as an adsorbent for naphthol dye, with its adsorption capacity influenced by solution pH, contact time, and initial dye concentration.

Keywords: biochar, palm kernel shell, nitrogen doping, naphthol dye, adsorption.

Room 1

COLLABORATIVE GIS FLOOD HAZARD MAPPING WITH LOCAL PUBLIC WORKS FOR SPATIAL PLANNING AND COMMUNITY DISASTER PREPAREDNESS IN PEMATANG JERING VILLAGE, MUARO JAMBI

Ir. Dyah Kumalasari[1], S.T.[1], M.T.[1], Ir. Dila Oktarise Dwina[1], S.T.[1], M.T.[1], Ir. Ade Nurdin[1], S.T.[1], M.T.[1], Ir. M. Nuklirullah[1], S.T.[1], M.Eng.[1], Ir. Putri Nurul Kusuma Whardani[1], S.T.[1], M.T.[1], Nita Astasya[1], S.T. M.T.[1], Lisa Maharani[1], and M.Ars.[1]

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Abstract

Pematang Jering Village (Muaro Jambi Regency) experiences severe annual flooding due to its low-lying floodplain topography along the Batanghari River. Despite recurring damage to infrastructure and local livelihoods, the village lacks geospatial hazard maps, data-driven DDR systems, and adequate community disaster literacy. This program addresses these gaps by developing an integrated GIS-based flood hazard map and a village-level disaster information system. Furthermore, the initiative trains residents and village officials to leverage spatial data for land-use planning and flood mitigation. The methodology combines field data acquisition, spatial hazard modeling, information system development, and participatory capacity building. Key deliverables include the Pematang Jering flood hazard map, a digital disaster information platform, a map-based mitigation blueprint, and a trained village preparedness team. With core technical outputs deployed and training phases underway, this socio-technical framework establishes a replicable model for technology-driven rural flood resilience.

Keywords: GIS Mapping; Flood Hazard Modeling; Disaster Risk Reduction; Spatial Planning; Socio-Technical Systems.

Room 1

CHITOSAN–GLUCOSE-ASSISTED MODIFICATION OF SNO_2/FE_3O_4 NANOCOMPOSITES FOR METHYLENE BLUE ADSORPTION AND PHOTODEGRADATION

Widya Twiny Rizki[1], Defia Indah Permatasari[1], Melvi Muharmi[2], and Teguh Setyawan[3]

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Abstract

Methylene blue is a synthetic dye commonly used in various industries, and its presence in wastewater requires effective treatment strategies. Combining adsorption and photodegradation using multifunctional nanocomposites offers a potential approach for dye removal from aqueous solutions. This study aims to develop Fe_3O_4/chitosan–glucose/SnO_2 nanocomposites for methylene blue adsorption and photodegradation. Chitosan was synthesized from shrimp shell waste with a yield of 56%, while glucose was obtained from sugarcane bagasse through hydrolysis, with a hydrolysis yield of 87% and a glucose content of 1.12%. Fe_3O_4 was synthesized using the co-precipitation method, and SnO_2 was prepared via the hydrothermal method. The modification process produced a blackish-brown, gel-like nanocomposite. Scanning electron microscopy (SEM) revealed morphological changes, including an increase in granules or particles and the formation of more distinct pores following the incorporation of Fe_3O_4 and SnO_2. Further characterization using X-ray diffraction (XRD), Fourier-transform infrared spectroscopy (FTIR), vibrating-sample magnetometry (VSM), and UV–Vis diffuse reflectance spectroscopy (UV–Vis DRS), as well as methylene blue adsorption and photodegradation tests, is currently underway. These analyses will be used to evaluate the structural, functional, magnetic, and optical properties of the nanocomposite and assess its potential for methylene blue removal from aqueous solutions.

Keywords: Methylene blue; Fe_3O_4/chitosan–glucose/SnO_2; nanocomposite; adsorption; photodegradation.

Room 1

CORROSION RATE AND INHIBITION EFFICIENCY OF JERNANG FRUIT PEEL EXTRACT (DAEMONOROPS SP.) IN SULFURIC ACID SOLUTION

Diah Riski Gusti[1], Revis Asra[1], Lenny Marlinda[1], and Intan Lestari[1]

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Abstract

The development of environmentally friendly corrosion inhibitors from natural materials has attracted considerable attention as an alternative to conventional synthetic inhibitors. This study investigates the corrosion inhibition performance of methanolic extract from jernang fruit peel (Daemonorops sp.) on steel in sulfuric acid solution. The corrosion rate and inhibition efficiency were evaluated at extract concentrations of 0–2.5% and temperatures ranging from 30 to 60 °C. The results showed that increasing the extract concentration reduced the corrosion rate and improved the inhibition efficiency. The highest inhibition efficiency was 95.504% at an extract concentration of 2.5% and a temperature of 30 °C, with a corresponding corrosion rate of 0.659. In contrast, increasing the temperature generally increased the corrosion rate and decreased the inhibition efficiency. At an extract concentration of 2.5%, the inhibition efficiency decreased from 95.504% at 30 °C to 80.7316% at 60 °C. These findings indicate that jernang fruit peel extract has promising potential as a natural corrosion inhibitor in acidic environments. The inhibition performance may be associated with the adsorption of phytochemical constituents onto the metal surface; however, further electrochemical and surface characterization studies are required to establish the inhibition mechanism

Keywords: jernang fruit peel; Daemonorops sp.; green corrosion inhibitor; corrosion rate; inhibition efficiency; sulfuric acid

Room 1

ANTIBACTERIAL ACTIVITY OF ENDOPHYTIC FUNGI ISOLATED FROM THE ROOT OF THE MANGROVE BRUGUIERA SEXANGULA

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Abstract

Mangrove ecosystems are characterized by challenging environmental conditions that may select for microorganisms capable of producing diverse secondary metabolites. This study investigated the antibacterial activity of endophytic fungi isolated from the root of Bruguiera sexangula (Lour.) Poir. collected from Sungsang IV, South Sumatra, Indonesia. Six endophytic fungal isolates, designated BSA1–BSA6, were obtained and characterized based on their macroscopic and microscopic features. The antibacterial activity of the root extract (A) and the six fungal isolates was evaluated using the disc diffusion method against Bacillus subtilis, Escherichia coli, Staphylococcus aureus, and Salmonella typhi. Antibacterial activity was expressed as the percentage of the inhibition-zone diameter of each sample relative to that of the positive control. The root extract and all six fungal isolates showed antibacterial activity against the tested bacteria, with percentage values varying according to the isolate and bacterial species. BSA2 showed the highest activity against B. subtilis (77.384%), BSA6 against E. coli (93.375%), BSA5 against S. aureus (69.424%), and BSA4 against S. typhi (79.272%). These findings indicate that endophytic fungi associated with the root of B. sexangula are potential sources of antibacterial metabolites. Further studies are required to determine the minimum inhibitory concentration, minimum bactericidal concentration, and chemical constituents directly responsible for the observed antibacterial effects.

Keywords: Antibacterial activity; Bruguiera sexangula; Disc diffusion; Endophytic fungi; Mangrove.

Room 1

THE EFFECT OF SULFUR MODIFICATION ON POFA VIA MICROWAVE-ASSISTED IMPREGNATION IN MINING WASTEWATER TREATMENT

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Abstract

Mining wastewater containing heavy metals and sulfates requires effective and sustainable treatment alternatives. Palm Oil Fuel Ash (POFA), an agro-industrial waste, has potential as an adsorbent, but its efficiency needs to be enhanced through modification. This study aims to evaluate the effect of sulfur modification on POFA via microwave-assisted impregnation in mining wastewater treatment. POFA was activated using 0.1 M NH_4Cl and subsequently impregnated with thiourea as a sulfur precursor at various POFA-to-sulfur mass ratios (100:0; 97.5:2.5; 95:5; 92.5:7.5; 90:10; and 87.5:12.5 % w/w) using a microwave digester (150 °C, 500 W, 8 bar). The adsorbent was applied to real mining wastewater samples, and characterization was performed using Fourier Transform Infrared (FTIR) spectroscopy and measurements of physicochemical parameters (pH, TDS, EC, salinity, and turbidity). FTIR results confirmed the successful immobilization of sulfur, indicated by the appearance of C–S stretching bands at 700–600 cm^{-1} and the indication of C=S/C–N coupling in the 1563–700 cm^{-1} region, despite overlapping with the dominant Si–O–Si bands. The application to mining wastewater showed significant reductions in physicochemical parameters, with the 87.5%:12.5% ratio yielding the best results: a reduction of 80.4% in TDS (from 409 to 80 ppm), 80.0% in EC (from 778 to 156 µS/cm), and 98.74% in turbidity (from 748 to 9.40 NTU). Conversely, the solution pH increased from 7.8 to 8.15–8.80. This study is a preliminary evaluation; since quantitative heavy metal analysis (AAS/ICP) was not performed, the heavy metal removal efficiency and capacity have not been established. Nevertheless, sulfur modification on POFA proved capable of altering surface functional groups and significantly improving the physicochemical parameters of mining wastewater.

Keywords: POFA, sulfur modification, microwave-assisted impregnation, mining wastewater, FTIR, physicochemical parameters

Paralel Schedule

Room 2 (Online) 13:30-16:00 Moderator: Lydia Kartika Basaria Sitompul, M.T.		
Time (WIB)	**Presenter**	**Title**
13.30-13.50	**Dr. Rini Arianti (University of Debrecen)**	Human Cervical-Derived Adipocytes: From Patients to the Laboratory and Beyond **(Invited Speaker)**
13.50-14.00	Syaharani Adepia Novitri (Universitas Sriwijaya)	Evaluation of the Antioxidant Activity of Rambutan (*Nephelium lappaceum* L.) Peel Extract by DPPH (2,2-diphenyl-1-picrylhydrazine) and FRAP (Ferric Reducing Antioxidant Power) Methods
14.00.14.10	Nur Jaya (Universitas Jambi)	Antidiabetic Potential of *Syzygium myrtifolium* Leaf Ethanol Extract: Glucose Reduction and LC-HRMS Profiling
14.10-14.20	Q & A	
14.20-14.30	Ashif Irvan Yusuf (Universitas Jambi)	Modulation of Endogenous Antioxidant Activity and Oxidative Stress Markers by Kombucang (Secang Wood Kombucha) in a Metabolic Syndrome Model
14.30-14.40	Fadel ibnu muliya alqudrisyam (Universitas Riau)	Evaluation Of Damage And Structural Performance Of A Box Underpass Using Visual Inspection, Hammer Test, And Ultrasonic Pulse Velocity (UPV)
14.40-14.50	Q & A	
14.50-15.00	Edwin Permana (Universitas Jambi)	Utilization Of Dragon's Blood (*Daemonorops draco* (Willd.) Blume) Fruit Waste As A Natural Dye For Jambi Batik Based On Green Chemistry And The Creative Industry
15.00-15.10	Villa Fitri Yana Diah Pitaloka (Universitas Jambi)	Evaluation Of Double-Stage Treatment Integrating Electro-Fenton And Moringa Oleifera Biocoagulant For Palm Oil Mill Effluent Degradation
15.10-15.20	Q & A	
15.20-15.30	Riswan (Universitas Lambung Mangkurat)	Assessment of Coalbed Methane Content and Resource Potential Using Density Log Data in the Tanjung Formation, Indonesia
15.30-15.40	Ahmad Sazali (Universitas Jambi)	Optimization of Secang Wood Extract (*Caesalpinia sappan* L.) Formulation as an Herbal Fish Medicine: Stability, Toxicity, and Efficacy Studies in Aquaculture Systems
15.40-15.50	Rahma Tania Alya Susanti (Universitas Jambi)	Microwave-Digester POFA Sulfur Modification at 100°C for Hg^{2+} Removal from Multicomponent and Mining Wastewater
15.50-16.00	Q & A	

Dr. Rini Arianti *Topic*

HUMAN CERVICAL-DERIVED ADIPOCYTES: FROM PATIENTS TO THE LABORATORY AND BEYOND

Rini Arianti*[1]

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Abstract

The increasing global burden of obesity and its associated cardiometabolic complications represents a major challenge to public health, highlighting the urgent need for novel and effective therapeutic strategies. Although glucagon-like peptide-1 (GLP-1) receptor agonists have emerged as promising pharmacological interventions, their clinical application may be constrained by limited accessibility, cost, and potential adverse effects. In this context, human brown adipose tissue (BAT) has attracted growing interest as a potential therapeutic target for metabolic disorders, particularly following the identification of metabolically active BAT in adult humans. In adults, BAT is distributed across several anatomical depots, including the supraclavicular, deep cervical, paraspinal, and abdominal regions, where it contributes substantially to glucose utilization and whole-body energy metabolism. Unlike white adipose tissue, BAT possesses a specialized thermogenic capacity, converting stored chemical energy into heat through mitochondrial uncoupling protein 1 (UCP1), a process activated primarily by adrenergic stimulation. Our research has been focusing on human cervical-derived adipocytes and how they can take up nutrients from extracellular environment to fulfill high metabolic demands during thermogenesis. Our works have revealed that importance of ASC-1 amino acid transporter, thiamine transporters (SLC19A2 and SLC19A3), and iron transport proteins for efficient thermogenic response of brown adipocytes derived from human BAT biopsies. Pharmacological inhibition or deletion of these transporters results in compromised thermogenesis in human brown adipocytes. Investigating the molecular mechanisms by which essential metal nutrients contribute to the regulation of thermogenesis in human brown adipocytes may uncover previously unrecognized metabolic pathways that govern energy expenditure. Ultimately, these insights may contribute to the development of innovative approaches for combating obesity and its associated cardiometabolic complications by promoting energy dissipation rather than solely focusing on reducing energy intake or nutrient absorption

Keywords: Brown adipose tissue; Thermogenesis; Human adipocytes; Nutrient transporters; Mitochondrial uncoupling protein 1 (UCP1); Obesity

Room 2

EVALUATION OF THE ANTIOXIDANT ACTIVITY OF RAMBUTAN (NEPHELIUM LAPPACEUM L.) PEEL EXTRACT BY DPPH (2,2-DIPHENYL-1-PICRYLHYDRAZINE) AND FRAP (FERRIC REDUCING ANTIOXIDANT POWER) METHODS

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Abstract

Rambutan peel has been traditionally used in herbal medicine to treat various diseases, including diabetes, gout, and jaundice, as well as to help prevent cancer and maintain liver and kidney health. Rambutan peel (Nephelium lappaceum L.) has increasingly attracted attention as a potential source of antioxidant compounds. The antioxidant activity of aqueous rambutan peel extract has previously been reported, with a DPPH IC_{50} value of 0.797 µg/mL. This study investigated the antioxidant activity of rambutan peel extracts obtained using different solvents, namely methanol, ethanol, and ethyl acetate. Each extract was analyzed for total phenolic and total flavonoid contents, and its antioxidant activity was evaluated using the DPPH and FRAP methods. The results showed that the highest total phenolic content was found in the ethyl acetate extract, while the highest total flavonoid content was found in the ethanol extract, with values of 11.597 mg GAE/g and 6.553 mg QE/g, respectively. The methanol extract exhibited the highest antioxidant activity in both assays, with a DPPH IC_{50} value of 17.843 mg/mL and a ferric reducing antioxidant power expressed as an ascorbic acid equivalent antioxidant capacity (AAEAC) of 76.153 mg AAE/g. These findings indicate that methanol extract was the most active extract in terms of antioxidant activity among the extracts evaluated

Keywords: Rambutan peel, Antioxidant, DPPH, FRAP, Phenolics, Flavonoids.

Room 2

ANTIDIABETIC POTENTIAL OF SYZYGIUM MYRTIFOLIUM LEAF ETHANOL EXTRACT: GLUCOSE REDUCTION AND LC-HRMS PROFILING

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Abstract

Diabetes mellitus is a chronic metabolic disorder that highlights the need for alternative treatments derived from natural resources. This study aimed to evaluate the antidiabetic potential of ethanol extract of Syzygium myrtifolium Walp. leaves and characterize its metabolite profile using Liquid Chromatography–High-Resolution Mass Spectrometry (LC-HRMS). An in vivo experimental study was conducted using 24 male Wistar rats divided into six groups (n = 4): normal control, negative control, positive control, and three extract treatment groups receiving doses of 200, 300, and 400 mg/kg body weight (BW). Treatment was administered for 28 days, and blood glucose levels were measured before and after treatment. Data were analyzed using the Shapiro–Wilk, Levene, and Kruskal–Wallis tests, followed by Mann–Whitney testing where appropriate. All extract-treated groups exhibited reductions in blood glucose levels, with mean reductions of 355.25 mg/dL (77.18%), 376.50 mg/dL (79.49%), and 387.00 mg/dL (82.91%) at doses of 200, 300, and 400 mg/kg BW, respectively. Significant differences were observed among the six groups (H = 16.800; p = 0.0049). The highest mean reduction occurred at 400 mg/kg BW. LC-HRMS analysis annotated several compounds, including benzothiazole, (S)-1-methoxy-3-heptanethiol, and nonylthiol. The integration of blood glucose assessment and metabolite profiling provides preliminary evidence of the antidiabetic potential of S. myrtifolium leaf ethanol extract. Further studies are required to verify the identified compounds, elucidate their mechanisms of action, and establish the extract's safety and standardization.

Keywords: diabetes mellitus; Syzygium myrtifolium Walp.; red thorn leaves; ethanol extract; LC-HRMS

Room 2

MODULATION OF ENDOGENOUS ANTIOXIDANT ACTIVITY AND OXIDATIVE STRESS MARKERS BY KOMBUCANG (SECANG WOOD KOMBUCHA) IN A METABOLIC SYNDROME MODEL

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Abstract

Background: Metabolic syndrome is a complex metabolic disorder characterized by hyperglycemia, insulin resistance, dyslipidemia, and increased oxidative stress, all of which contribute to the risk of type 2 diabetes mellitus, cardiovascular disease, and other degenerative disorders. Oxidative stress contributes to the progression of metabolic syndrome through increased reactive oxygen species (ROS), decreased activity of endogenous antioxidant systems such as superoxide dismutase (SOD), and increased malondialdehyde (MDA) levels as a biomarker of oxidative damage. Kombucang (Secang Wood Kombucha), a functional herbal beverage produced through the fermentation of secang wood (Caesalpinia sappan L.), has the potential to be developed as a natural intervention for improving metabolic disturbances. Objective: This study aimed to analyze the potential of Kombucang to improve metabolic parameters and modulate endogenous antioxidant activity and oxidative stress in a mouse model of metabolic syndrome. Methods: This study employed an in vivo experimental design involving normal control, positive control, negative control, and Kombucang treatment groups receiving doses of 0.26, 0.52, and 0.78 mL/20 g body weight. Metabolic disturbances were induced using alloxan, chicken liver juice, propylthiouracil (PTU), and quail egg yolk. The measured parameters included blood glucose levels, uric acid levels, and total cholesterol levels before and after induction, as well as on days 7 and 14 following treatment. Results: The induction procedure increased blood glucose levels to 476–520, uric acid levels to 8.75–9.80, and total cholesterol levels to 211–258. After 14 days of Kombucang administration, the group receiving 0.26 mL/20 g body weight exhibited reductions of 62.0% in blood glucose, 54.5% in uric acid, and 41.9% in total cholesterol. At a dose of 0.52 mL/20 g body weight, the respective reductions reached 64.7%, 59.2%, and 52.6%, whereas the dose of 0.78 mL/20 g body weight resulted in reductions of 65.0%, 58.3%, and 51.7%, respectively. These percentages were calculated relative to the values recorded after induction. Conclusion: Kombucang demonstrated potential antihyperglycemic, antihyperuricemic, and total cholesterol-lowering activities in a mouse model of metabolic disturbances. The dose of 0.78 mL/20 g body weight produced the greatest reduction in blood glucose, whereas the dose of 0.52 mL/20 g body weight yielded the greatest numerical reductions in uric acid and total cholesterol. These findings support the further development of Kombucang as a candidate multitarget functional herbal beverage derived from local natural resources. Further investigations involving SOD and MDA measurements and statistical analyses are required to substantiate its antioxidant mechanisms of action and therapeutic efficacy.

Keywords: kombucang, secang wood, metabolic syndrome, antioxidant, oxidative stress

Room 2

EVALUATION OF DAMAGE AND STRUCTURAL PERFORMANCE OF A BOX UNDERPASS USING VISUAL INSPECTION, HAMMER TEST, AND ULTRASONIC PULSE VELOCITY (UPV)

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Abstract

A box underpass is a reinforced concrete structure that provides passage beneath a toll road embankment. Concrete structures may develop cracks due to inadequate curing, high temperatures, vehicle-induced vibrations, and construction-related factors. This study aims to evaluate the condition of the Pekanbaru Ring Road Toll box underpass, characterize existing cracks, and recommend appropriate rehabilitation methods. The research involved visual inspection and non-destructive testing (NDT) using the rebound hammer test and ultrasonic pulse velocity (UPV) test. Visual inspection was conducted to identify crack patterns, orientations, and widths, while the two NDT methods were used to assess concrete quality and homogeneity. The results revealed longitudinal cracks in the wall and bottom slabs, with no signs of spalling or exposed reinforcement. Rebound values ranged from 26.7 to 35.3, while the UPV test recorded a wave velocity of 4,075 m/s, indicating generally good concrete quality. However, cracks up to 0.55 mm wide and approximately 10 cm deep were identified, highlighting the need for appropriate repair measures. Crack injection and surface coating are recommended to seal the cracks and prevent water and aggressive substances from penetrating the concrete. Overall, the structure remains in relatively good condition; however, timely crack rehabilitation is necessary to maintain concrete durability and extend the service life of the box underpass.

Keywords: Box underpass; hairline crack; reinforced concrete; Non-Destructive Test (NDT); structural rehabilitation.

Room 2

UTILIZATION OF DRAGON'S BLOOD (DAEMONOROPS DRACO (WILLD.) BLUME) FRUIT WASTE AS A NATURAL DYE FOR JAMBI BATIK BASED ON GREEN CHEMISTRY AND THE CREATIVE INDUSTRY

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Abstract

This study characterized dragon's blood (jernang) fruit waste extracts obtained using three solvents, namely ethanol, methanol, and water, by UV-Vis and FTIR spectroscopy. The results show that the type of solvent affects the chemical components extracted. In the UV-Vis spectra, all three extracts began to absorb at around 280–300 nm and absorbed strongly up to approximately 400 nm, indicating the presence of compounds with conjugated electron systems that act as the chromophores of the dragon's blood pigments. The ethanol and methanol extracts displayed similar patterns, with maximum absorption at around 380 nm, indicating that both solvents extracted relatively similar chromophores; the higher absorbance of the methanol extract indicates a greater concentration of absorbing components. The water extract showed maximum absorption at around 320 nm, shifted toward shorter wavelengths, indicating differences in its components or their molecular environment. The FTIR results support these findings. All three extracts exhibited an O–H band at around 3300 cm^{-1}. The ethanol and methanol extracts both showed C–H bands (2900–2800 cm^{-1} and around 1400 cm^{-1}) as well as C–O bands (1113 and 1014–1018 cm^{-1}). The difference lies in the band at 1652 cm^{-1}, which was distinct in the ethanol extract but absent in the methanol extract, presumably because ethanol, being less polar, extracts more hydrophobic aromatic compounds and flavonoids. The water extract produced only two main bands, at 3299 and 1634 cm^{-1}, with the 1634 cm^{-1} band appearing sharp due to the overlap of the H–O–H bending vibration with the C=C vibration.

Keywords: dragon's blood (jernang), fruit waste, natural dye, Jambi Batik

Room 2

EVALUATION OF DOUBLE-STAGE TREATMENT INTEGRATING ELECTRO-FENTON AND MORINGA OLEIFERA BIOCOAGULANT FOR PALM OIL MILL EFFLUENT DEGRADATION

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Abstract

Palm Oil Mill Effluent (POME) is a highly polluting agro-industrial wastewater that poses severe environmental risks due to its complex organic load and intense turbidity. This study evaluates a novel double-stage treatment strategy integrating an Electro-Fenton Advanced Oxidation Process (AOP) with a subsequent Moringa oleifera seed-based biocoagulation stage. In the primary treatment stage, raw POME featuring a high pollutant baseline (COD 86,383 mg/L, turbidity 11,250 NTU, and pH 3.74) was treated in a 1 L batch Electro-Fenton reactor using Fe-Graphite electrode pairs operating at 0.4 A, H_2O_2/COD molar ratio of 0.35, and 300 rpm stirring speed for 60 minutes. The primary oxidation significantly degraded the organic matrix, producing an effluent with COD of 200,430.32 mg/L, BOD_5 of 66,500.63 mg/L, sulfate of 5,350.31 mg/L, pH of 3.11, and a residual dissolved iron (Fe) concentration of 200.42 mg/L alongside elevated turbidity (25,960.69 NTU). The novelty of this approach lies in strategically utilizing the generated in-situ iron hydroxide $Fe(OH)_3$ suspension and residual iron ions as nucleation seeds for post-biocoagulation. Active cationic proteins extracted from Moringa oleifera seeds facilitate charge neutralization and sweep flocculation, rapidly aggregating the Fe-organic colloids into large settleable flocs. This hybrid system offers a sustainable, cost-effective, and highly efficient eco-friendly solution for optimizing POME quality and advancing industrial wastewater management.

Keywords: Electro-Fenton; Moringa oleifera; POME; Double-stage treatment; Sweep flocculation

Room 2

ASSESSMENT OF COALBED METHANE CONTENT AND RESOURCE POTENTIAL USING DENSITY LOG DATA IN THE TANJUNG FORMATION, INDONESIA

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Abstract

Direct measurement of coalbed methane (CBM) content requires core sampling and specialized testing, motivating the evaluation of routinely acquired density logs as a complementary assessment tool. This study examines the relationship between coal density and methane content and estimates CBM resources in the Tanjung Formation, Asem-Asem Basin, South Kalimantan, Indonesia. Density logs, drilling records, direct desorption measurements, and gas-composition analyses were integrated for six coal seams intersected by a single exploration borehole within the PT XYZ concession. Coal resources were estimated volumetrically, and methane resources were calculated using measured methane contents. The seams occur at depths of 571.6-594.6 m, with thicknesses of 0.5-3.5 m and average densities of 1.70-1.90 t/m³. Methane contents range from 0.060 m³/t in Seam B to 0.195 m³/t in Seam F. An overall inverse density–methane association was observed, although contrasting methane contents at identical average densities indicate additional controls on gas occurrence. Estimated coal and methane resources total 1,371,500 tons and 213,714 m³, respectively. Seam F contributes the largest methane resource, 116,025 m³, owing to its greater coal tonnage and higher methane content. Nitrogen dominance and the presence of oxygen warrant verification of sampling integrity before interpreting gas origins. These findings support density logs as a preliminary screening tool for methane-bearing intervals, but the limited dataset and absence of independent validation preclude broader predictive application. Further borehole sampling and integrated reservoir characterization are required to assess regional applicability, methane recoverability, and development potential.

Keywords: Coalbed methane; Density logging; Methane content; Tanjung Formation; Asem-Asem Basin; Resource estimation.

Room 2

OPTIMIZATION OF SECANG WOOD EXTRACT (CAESALPINIA SAPPAN L.) FORMULATION AS AN HERBAL FISH MEDICINE: STABILITY, TOXICITY, AND EFFICACY STUDIES IN AQUACULTURE SYSTEMS

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Abstract

Freshwater aquaculture of catfish (Pangasianodon hypophthalmus) in Indonesia is threatened by Aeromonas hydrophila infection, the causative agent of Motile Aeromonas Septicemia (MAS). Overuse of synthetic antibiotiks raises concerns of antimicrobial resistance (AMR) and residues. Sappan wood extract (Caesalpinia sappan L.), rich in brazilin and flavonoids, has shown in vitro antibacterial activity, but its in vivo efficacy in catfish which is infected by A.hydrophila had not been tested. This study aimed to determine the effect of three formulated sappan wood extract formulas on the treatment of A. hydrophila-infected catfish, their effect on bacterial load, and the optimal formula in vivo. This experimental study used a Completely Randomized Design with six treatments (healthy kontrol, negative kontrol, positive kontrol/enrofloxacin, and three extract formulas differing in excipient: Tween 80, PEG 400, and glycerin), each with 3 replicates of 25 fish. Extract was obtained by 96% ethanol maceration (yield 4.02%) and applied by immersion at 550 ppm, after fish were injected with A. hydrophila at 10^3 CFU/mL. Survival rate, mortality, total plate count (TPC) in blood, clinical signs, and water quality were analyzed using ANOVA and Duncan's test. The results showed All three formulas significantly increased survival rate compared to the negative control (22.67%), reaching 46.67%, 41.33%, and 52.00% respectively, though still lower than the positive control (68.00%) ($F = 31,5$; $P<0.05$). Formula 3 (glycerin) performed best, with the highest survival rate and lowest bacterial load (2.67 log CFU/mL), compared to Formula 1 (3.64 log CFU/mL) and Formula 2 (4.78 log CFU/mL. In conclusion, formulated sappan wood extract significantly improved the survival of A. hydrophila infected catfish, with Formula 3 identified as the most promising candidate for further development as a natural, sustainable alternative to synthetic antibiotics in freshwater aquaculture disease management.

Keywords: Fish Medicine, Caesalpinia sappan, Stability, Toxicity, Aquaculture

Room 2

MICROWAVE-DIGESTER POFA SULFUR MODIFICATION AT 100°C FOR HG^{2+} REMOVAL FROM MULTICOMPONENT AND MINING WASTEWATER

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Abstract

This study investigates the characteristics of Palm Oil Fuel Ash (POFA) before and after sulfur modification using thiourea, and examines the physicochemical properties of filtrates from multicomponent and mining wastewater adsorption tests. Sulfur modification was performed in a microwave digester at 100°C, with a POFA to thiourea mass ratio of 90:10 for the modified sample. Functional groups were characterized using Fourier Transform Infrared Spectroscopy (FTIR), while filtrate quality was evaluated by pH, total dissolved solids (TDS), electrical conductivity (EC), salinity, temperature, and turbidity. The unmodified POFA spectrum exhibited absorption bands at approximately 3269, 1632, 1415, 994, 784, and 415 cm^{-1}. The sulfur-modified sample showed a dominant absorption band at 1002.50 cm^{-1}, compared with approximately 994 cm^{-1} for unmodified POFA, indicating a change in the vibrational environment of silicate-related bonds after modification. For the multicomponent system, filtrate values ranged from 511-716 mg/L for TDS, 1004-1427 µS/cm for EC, 8.30–9.25 for pH, 0.05–0.08 for salinity, and 0.96-3.12 NTU for turbidity. Mining wastewater filtrates showed TDS of 114-148 mg/L, EC of 224-293 µS/cm, pH of 8.51–8.77, salinity of 0.01, and turbidity of 4.88–21.10 NTU. These results demonstrate differences in filtrate characteristics between the two water systems.

Keywords: Palm Oil Fuel Ash, sulfur modification, thiourea, FTIR, mercury adsorption

Paralel Schedule

Room 3 (Online) 13:30-16:00 Moderator: Anggit Prima Nugraha, S.Si., M.Sc.		
Time (WIB)	**Presenter**	**Title**
13.30-13.50	**Asmida Herawati, M.Sc., Ph.D. (Research Center for Photonics, National Research and Innovation Agency/BRIN)**	Effect of Multi-Energy Photon Illumination on Charge Trap Deactivation in Perovskite Materials for Enhanced Photovoltaic Performance **(Invited Speaker)**
13.50-14.00	Fitra Wahyuni (Universitas Jambi)	Exploration And Inventory Of Tree Diversity At Kedaton Temple, Muaro Jambi, As A Basis For Developing A Natural Laboratory
14.00.14.10	Hanna Erika Marintan.S (Universitas Jambi)	Synthesis And Characterization Of Sulfonated Cellulose–Chitosan Hydrogel From Liberika Coffee Husk (*Coffea Liberica*) Using Sodium Tripolyphosphate (Stpp) As A Crosslinker For Metronidazole Delivery
14.10-14.20	Q & A	
14.20-14.30	Elvina Dhiaul Iftitah (Universitas Brawijaya)	Synthesis Of Carbon Dots From Patchouli Leaf-Waste For Metal Sensing, Antioxidant, And Antibacterial Applications
14.30-14.40	Eduwin Saputra (Universitas Bengkulu)	Reductive Desorption and Surface-Mediated Formation of Gold Nanoparticles from $AuCl_4^-$ Adsorbed on CTAB-Zeolite/Magnetite
14.40-14.50	Q & A	
14.50-15.00	Rasyid Syahridho Rahman (Universitas Jambi)	Effect Of Filtration Time On Cod Rejection And Antifouling Characteristics Of PVDF Membranes For Pome Treatment
15.00-15.10	Adriawan (Universitas Persada Indonesia)	Bridging the Intergenerational Communication Gap for Quality in Sustainable Construction Projects
15.10-15.20	Q & A	
15.20-15.30	Vidia Tanaya Putri (Universitas Jambi)	Enhancing Producer Gas Quality through Torrefaction Coupled Gasification of Oil Palm Fronds
15.30-15.40	Heriyanti (Universitas Jambi)	Analysis Of Relationship Between Soda Loss And Bleaching Chemicals Consumption In Pulp Bleaching
15.40-15.50	Miftahul Anwar (Universitas Brawijaya)	Pineapple Leaf Ash and NaOH Treatments of Resam Fiber (Dicranopteris linearis) for Renewable Biocomposites
15.50-16.00	Q & A	

Asmida Herawati, M.Sc., Ph.D. *Topic*

EFFECT OF MULTI-ENERGY PHOTON ILLUMINATION ON CHARGE TRAP DEACTIVATION IN PEROVSKITE MATERIALS FOR ENHANCED PHOTOVOLTAIC PERFORMANCE

Asmida Herawati

Research Center for Photonics, National Research and Innovation Agency (BRIN)

Abstract

Charge trapping and non-radiative recombination remain major challenges limiting the efficiency and stability of perovskite-based photovoltaic devices. Understanding how photon energy influences defect deactivation and charge-carrier dynamics is therefore essential for optimizing their optoelectronic properties. This study investigates photon-induced charge trap deactivation in methylammonium lead iodide ($MAPbI_3$) perovskite films, its implications for solar cell performance, and the influence of compositional modification on defect-related recombination in oxide perovskites. Steady-state photoluminescence (PL), time-resolved photoluminescence (TRPL), photovoltaic characterization, and impedance spectroscopy were employed to examine carrier relaxation, defect activity, and interfacial charge transport under controlled illumination. The results distinguish three trap populations with different energetic characteristics: low-energy photon-responsive traps (T_L), high-energy photon-responsive traps (T_H), and grain-boundary-associated traps (T_GB). Photon-induced deactivation of these traps enhances photoluminescence and modifies carrier relaxation dynamics, indicating reduced trap-assisted non-radiative recombination. In $MAPbI_3$ solar cells, controlled red-light illumination at 3 mW cm^{-2} improves photovoltaic characteristics, including short-circuit current density, open-circuit voltage, fill factor, and power conversion efficiency. The study highlights controlled photo-conditioning as a promising strategy for regulating defect activity and improving the optoelectronic performance of perovskite-based devices.

Keywords: Perovskite; $MAPbI_3$; photon-induced trap deactivation; charge-carrier dynamics; photoluminescence; defect engineering; solar cells

Room 3

EXPLORATION AND INVENTORY OF TREE DIVERSITY AT KEDATON TEMPLE, MUARO JAMBI, AS A BASIS FOR DEVELOPING A NATURAL LABORATORY

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Abstract

Indonesia is a country rich in cultural heritage and biodiversity a prime example is the Muaro Jambi Temple Complex, a cultural heritage site in Jambi Province. This complex is a significant cultural heritage site in Indonesia currently on the tentative list for UNESCO World Heritage status and is recognized as the largest Buddhist temple complex in Southeast Asia. Beyond its historical value, the area harbors diverse tree vegetation that plays a crucial role in maintaining environmental stability. To date, however, scientific information regarding the inventory and identification of tree species in the Candi Kedaton area remains limited. This study aimed to identify the tree species present in the Candi Kedaton area of the Muaro Jambi Temple Complex and to determine their scientific names. The research was conducted from April to August 2026 using a census method (complete enumeration), recording every individual tree within the study area. Identification was based on morphological characteristics, supported by field documentation, herbarium specimen collection, literature reviews, and appropriate taxonomic references. The results revealed the presence of 27 tree species in the Candi Kedaton area, representing 18 genera and 13 families, with a total of 84 individual trees. The Moraceae family exhibited the highest species richness, with eight species. Ficus racemosa was the most abundant species (15 individuals), followed by Durio zibethinus and Lansium domesticum (nine individuals each) and Archidendron jiringa (eight individuals). Several other species were represented by only a single individual. The calculated Shannon-Wiener diversity index yielded a value of H' = 2.87, indicating a moderate level of diversity. Scientific naming was determined based on morphological observations and the plant identification process, including verification against herbarium specimens in Bogor. The naming results are presented via QR codes containing information for each tree species. The information provided includes documentation, classification, general descriptions, uses, and conservation status based on the IUCN Red List. One species requiring particular attention is Artocarpus odoratissimus, which holds a global conservation status of Near Threatened (NT). This study is expected to provide baseline data on tree diversity in the Kedaton Temple area and support the management, conservation, and provision of information for visitors to the cultural heritage site.

Keywords: tree inventory, scientific naming, Kedaton Temple, morphology, species diversity, QR barcode

Room 3

SYNTHESIS AND CHARACTERIZATION OF SULFONATED CELLULOSE–CHITOSAN HYDROGEL FROM LIBERIKA COFFEE HUSK (COFFEA LIBERICA) USING SODIUM TRIPOLYPHOSPHATE (STPP) AS A CROSSLINKER FOR METRONIDAZOLE DELIVERY

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Abstract

Coffe husk is an abundant agro-indrustrial waste that contains cellulose and bioactive compounds with potential for the development of sustainable biomaterials. This study aimed to synthesize and characterize a chitosan-based hydrogel using sulfonated cellulose isolated from Liberika coffee husk (Coffea liberica) and evaluate its potential as a metronidazole delivery system. Cellulose was isolated through dewaxing, alkaline delignification, and hydrogen peroxide bleaching, followed by periodate oxidation and reductive sulfonation. Coffee husk polyphenols were extracted using hot-water extraction and utilized as a natural bio-crosslinker, while Sodium Tripolyphosphate (STPP) was employed as an ionic crosslinker for comparison. The hydrogels were characterized through swelling analysis, Fourier Transform Infrared Spectroscopy (FTIR), and Scanning Electron Microscopy (SEM), while metronidazole loading and in vitro release were evaluated using UV-Vis spectrophotometry. The cumulative drug release increased with increasing release time for all hydrogel formulations. The cumulative release values for A1 were 30.61%, 39.00%, 51.29%, 63.54%, and 72.35%, while A2 showed 25.08%, 42.04%, 58.07%, 79.57%, and 81.80%. Formulation A3 exhibited release values of 37.40%, 50.25%, 65.32%, 75.97%, and 84.57%, whereas A4 showed the highest cumulative release, reaching 96.36%. These findings demonstrate that the hydrogel formulations can facilitate metronidazole release and indicate that the crosslinker type and cellulose modification influence the release behavior. The study provides a potential approach for valorizing Liberika coffee husk into a functional hydrogel for drug delivery applications.

Keywords: Liberika coffee husk; sulfonated cellulose; chitosan hydrogel; sodium tripolyphosphate; metronidazole

Room 3

SYNTHESIS OF CARBON DOTS FROM PATCHOULI LEAF-WASTE FOR METAL SENSING, ANTIOXIDANT, AND ANTIBACTERIAL APPLICATIONS

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Abstract

Carbon dots (CDs) are emerging zero-dimensional carbon-based nanomaterials that have attracted considerable attention owing to their unique physicochemical properties, including tunable fluorescence, good biocompatibility, relatively low toxicity compared to inorganic quantum dots, and facile synthesis. These characteristics make CDs promising candidates for various applications, particularly in fluorescent metal-ion sensing and biological applications such as antioxidant and antibacterial activities. Furthermore, the utilization of biomass as a precursor offers a sustainable approach to CDs production while supporting the valorization of underutilized natural resources. Therefore, in this study, CDs derived from patchouli leaf waste were developed using a one-pot, microwave-assisted synthesis method using water as the sole solvent. The synthesized CDs exhibited fluorescence under UV illumination and demonstrated distinct fluorescence responses toward Fe3+ and Pb2+, with Fe3+ showing fluorescence quenching, whereas Pb2+ resulted in fluorescence enhancement. Furthermore, the CDs exhibited antioxidant activity, as evaluated by the DPPH radical scavenging assay, and antibacterial activity against Escherichia coli and Staphylococcus aureus.

Keywords: carbon dots, patchouli leaf waste, metal sensing, antioxidant, antibacterial

Room 3

REDUCTIVE DESORPTION AND SURFACE-MEDIATED FORMATION OF GOLD NANOPARTICLES FROM $AUCL_4^-$ ADSORBED ON CTAB-ZEOLITE/MAGNETITE

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Abstract

This study evaluates the adsorption of Au(III) ions using activated natural zeolite (ANZ) composites modified with magnetite (Mag) and functionalized with the surfactant cetyltrimethylammonium bromide (CTAB). The ANZ/Mag composite was functionalized with CTAB at a concentration equivalent to twice the cation exchange capacity. FTIR, SEM, and XRD confirmed the successful synthesis of the CTAB-ANZ/Mag composite. The optimum conditions for Au(III) ion adsorption were achieved using an adsorbent mass of 150 mg and a contact time of 60 minutes. At an initial Au(III) concentration of 50 mg/L, the adsorption percentage reached 94%. The adsorbed Au(III) ions were subsequently desorbed using ethylenediaminetetraacetic acid (EDTA) at various pH values and concentrations. Au(III) ions desorbed successfully at pH 4 using an EDTA concentration of 0.5%. The formation of gold nanoparticles from the CTAB-ANZ/Mag/Au system following desorption was evidenced by the appearance of a localized surface plasmon resonance (LSPR) peak at a wavelength of 428 nm.

Keywords: Adsorption-reduction, gold nanoparticles, CTAB, Magnetite, and zeolite.

Room 3

EFFECT OF FILTRATION TIME ON COD REJECTION AND ANTIFOULING CHARACTERISTICS OF PVDF MEMBRANES FOR POME TREATMENT

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Abstract

Palm oil mill effluent (POME) is a liquid by-product of palm oil processing characterized by its high organic content, requiring effective treatment to reduce pollutant concentrations. Membrane filtration offers a promising alternative for POME treatment by producing permeate with improved quality. Polyvinylidene fluoride (PVDF) membranes possess excellent chemical resistance and mechanical strength, although their hydrophobic properties make them prone to fouling. This study evaluated the influence of filtration time on chemical oxygen demand (COD) rejection and fouling reversibility using PVDF hollow fiber membranes. Coagulation–flocculation-pretreated POME was filtered using a dead-end system at 1 bar for 15, 30, 45, and 60 minutes. Subsequently, membrane cleaning was performed through backwashing with distilled water at 1.5 atm for 60 seconds. Pure water flux after cleaning was measured and combined with previous flux measurements to determine Flux Recovery (FR), Reversible Fouling (RF), and Irreversible Fouling (IF). COD rejection decreased slightly from 81.98% to 81.67% as filtration time increased from 15 to 60 minutes. Meanwhile, FR declined from 97.56% to 56.08%, while IF increased from 2.44% to 43.92%. These results indicate that prolonged filtration increases the accumulation of foulants that cannot be completely removed through hydraulic cleaning. Overall, filtration duration affects permeate quality and membrane fouling characteristics, providing a basis for optimizing PVDF membrane operation during POME treatment.

Keywords: POME, PVDF Membrane, Filtration Time, COD Rejection, Antifouling Characteristics.

Room 3

BRIDGING THE INTERGENERATIONAL COMMUNICATION GAP FOR QUALITY IN SUSTAINABLE CONSTRUCTION PROJECTS

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Abstract

Senior practitioners and younger, Generation Z professionals in construction projects often differ in their work patterns, technological proficiency, and modes of interaction. These disparities frequently lead to misalignment when overseeing quality and managing risks on-site, resulting in project delays and compromised construction quality. This study develops an integrated framework combining the risk management principles of ISO 31000 with the Japanese Project and Program Management (P2M) approach. This framework serves to reduce interaction barriers and ensure consistent project quality standards. The study employs a mixed-methods approach involving 150 practitioners—including engineers, senior managers, and field technicians—from various strategic infrastructure projects in Indonesia. Focus Group Discussions (FGDs) were conducted to map operational risks based on ISO 31000, while the P2M framework was utilized to translate the experience and technical wisdom of senior professionals into concrete operational guidelines for the younger workforce. Data indicate that unaddressed intergenerational miscommunication increases quality non-compliance on-site by 22.4%. ISO 31000 enables teams to detect communication bottlenecks early, while P2M facilitates a structured, cross-generational mentorship scheme. The integration of both approaches reduces rework caused by quality issues by 17.8% and strengthens on-site teamwork. This research concludes that the construction industry must integrate risk management with cross-generational knowledge management. Companies can utilize this framework as a practical guide to preserve technical expertise and maintain work quality within a multi-generational workforce.

Keywords: Keywords: Intergenerational Communication, Construction Quality Management, ISO 31000, P2M Framework, Knowledge Transfer.

Room 3

ENHANCING PRODUCER GAS QUALITY THROUGH TORREFACTION COUPLED GASIFICATION OF OIL PALM FRONDS

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Abstract

Oil palm fronds (OPF) are widely available biomass residues in Indonesia and offer considerable potential for renewable energy production. Gasification provides an effective route for converting OPF into producer gas, howefer its high moisture content can limit gasification performance and reduce producer gas yield and quality. Torrefaction offers a way to overcome this limitation by improving biomass fuel properties, particularly through carbon enrichment and the removal of oxygen containing compounds. This study examined how torrefaction affects OPF characteristics and the composition of gas produced during subsequent gasification. OPF was torrefied at 250 °C for 30 min and then gasified in an updraft gasifier at oxidation zone temperatures of 750–850 °C. Biomass properties were assessed through proximate and ultimate analyses, while producer gas composition (CO, CO_2, H_2, and CH_4) was measured using gas chromatography (GC). The results showed that torrefaction increased the carbon content from 44.50% to 52.60%. At 850 °C, torrefied OPF yielded higher CO and H_2 concentrations (21.15% and 13.24%, respectively) than fresh OPF (17.82% and 11.38%). In contrast, CO_2 decreased from 17.67% to 14.19%, while CH_4 declined from 6.48% to 5.86%. Similar trends were observed at 750 and 800 °C. Overall, torrefaction shifted the producer gas composition toward higher CO and H_2 concentrations, highlighting its potential to improve gas quality and support the conversion of oil palm biomass residues into renewable energy.

Keywords: Oil palm fronds, Torrefaction, Gasification, Producer gas, Renewable energy

Room 3

ANALYSIS OF RELATIONSHIP BETWEEN SODA LOSS AND BLEACHING CHEMICALS CONSUMPTION IN PULP BLEACHING

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Abstract

The consumption of chemicals during pulp bleaching depends on process conditions, including the amount of residual alkali carried over from the washing stage. This study examined the relationship between soda loss at the bleaching inlet and the consumption of chlorine dioxide (ClO_2) and hydrogen peroxide (H_2O_2). This analysis was based on 31 daily operational observations collected over a month at a pulp and paper mill in Indonesia. Descriptive statistics, Pearson correlation, and linear regression were used to evaluate the relationships. The results showed that the average soda loss was 9.27 kg Na_2SO_4/ADT, while the average ClO_2 and H_2O_2 consumptions were 18.16 and 2.29 kg/ADT, respectively. A strong positive correlations was found between soda loss and ClO_2 consumption ($r = 0.844$; $R^2 = 0.712$) and H_2O_2 consumption ($r = 0.842$; $R^2 = 0.709$). The regression equations obtained were $y = 0.581x + 12.776$ for ClO_2 and $y = 0.470x - 2.066$ for H_2O_2. These results indicate that higher soda loss was associated with greater consumption of both bleaching chemicals. The findings highlight the potential of soda loss as an operational indicator for monitoring bleaching chemical demand. Further studies incorporating additional process variables are needed to clarify these relationships and improve chemical consumption control.

Keywords: soda loss, pulp bleaching, chlorine dioxide, hydrogen peroxide, chemical consumption.

Room 3

PINEAPPLE LEAF ASH AND NAOH TREATMENTS OF RESAM FIBER (DICRANOPTERIS LINEARIS) FOR RENEWABLE BIOCOMPOSITES

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Abstract

Resam (Dicranopteris linearis), a fern widespread in tropical regions, has potential as a biocomposite reinforcement, but the wax, lignin, and hemicellulose on its fiber surface limit fiber strength and bonding with the matrix. Alkali treatment with NaOH is widely used to overcome this, but excessive concentrations damage the fibers and generate alkaline waste. Plant-waste ash, including pineapple leaf ash, offers a milder and more environmentally friendly alkali, but, to the authors' knowledge, it has not been tested on resam fiber. This study aims to compare the effects of 1% NaOH and a 10% pineapple leaf ash solution (both % w/w) on the tensile strength of single resam fibers relative to untreated fibers. Fibers are separated from mature resam stalks, soaked statically for 6 h at room temperature, rinsed to neutral pH, tested in tension according to ASTM C1557 with five valid specimens per group, and analyzed with one-way ANOVA. Previous studies reported that ash increased the tensile strength of bamboo and banana fibers by about 14–61% at optimum concentrations, whereas concentrated NaOH reduced it; these findings serve as the reference for the hypothesis and data interpretation. The novelty of this study is the use of pineapple leaf ash as a treatment medium for resam fiber, with single-fiber tensile strength as the main measure. For materials engineering, this study provides baseline data on resam fiber strength and a more environmentally friendly, waste-based alkali treatment option for designing biocomposites.

Keywords: resam fiber; Dicranopteris linearis; pineapple leaf ash; alkali treatment; tensile strength; natural fiber

Paralel Schedule

Room 4 (Online) 13:30-15:50 Moderator: Anna Emiliawati, S.T., M.Eng.		
Time (WIB)	**Presenter**	**Title**
13.30-13.50	**Tika Restianingsih, M.Sc. (Institute of Science Tokyo)**	Intrinsic Mechanism of Site-dependent Lattice Thermal Conductivity Reduction in Half-Heusler Alloys **(Invited Speaker)**
13.50-14.00	Nur Vivian Kontesa (Universitas Jambi)	A Comprehensive Workflow for Fe_2O_3 Structural Characterization via Origin Peak Fitting and VESTA Crystal Modeling
14.00.14.10	Ericson; Yurma Syaftiani (Universitas Jambi)	Laboratory-Scale Experiment of In-Situ Electro-Kinetic Leaching (ISEKL) for Rare Earth Elements and Critical Minerals Extraction from Granitic Rocks of the Merangin Geopark
14.10-14.20	Q & A	
14.20-14.30	Melvi Muharmi (Universitas Jambi)	Structural and Vibrational Properties of Hydrothermally Synthesized Fe-Doped $SrTiO_3$ for Potential Photocatalytic Applications
14.30-14.40	Andreyan Aditya Septian Nugraha (Bandung Institute of Technology)	Gamma-Ray Spectrometry And Portable XRF Screening Of Rare Earth Elements In Weathering Profiles At Kalan And Ella Hilir, West Kalimantan
14.40-14.50	Q & A	
14.50-15.00	Lidia Yohana Sianturi (Universitas Jambi)	Effects of TEMPO Oxidation and Crosslinker Type on Liberica Coffee Husk Cellulose–Chitosan Hydrogels for Metronidazole Delivery
15.00-15.10	Merry Martha Yohanna Sitindaon (Universitas Sriwijaya)	Determination of Antioxidant Activity of Rambutan (Nephelium lappaceum) Peel Fractions by DPPH, FRAP, and ABTS Methods
15.10-15.20	Q & A	
15.20-15.30	Rani Khoiriyah (Universitas Jambi)	Response Surface Optimization Of Phenolic Extraction And Antiokxidant Activity Form Curcuma Aeruginosa Rhizomes
15.30-15.40	Rara Fentari (Khon Kaen University)	Comparative In Vitro Anthelmintic Activity Of Emodepside And Ivermectin Against Strongyloides Ratti
15.40-16.50	Q & A	

Tika Restianingsih, M.Sc. *Topic*

INTRINSIC MECHANISM OF SITE-DEPENDENT LATTICE THERMAL CONDUCTIVITY REDUCTION IN HALFHEUSLER ALLOYS

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Abstract

Half-Heusler compounds are promising thermoelectric materials, but their high lattice thermal conductivity (κl) limits their performance. Alloying reduces κl effectively, but why this reduction depends on the alloying site remains unclear. We use first-principles phonon transport calculations to investigate why alloying at the X, Y, and Z sites reduces κl by different amounts in NiSn- and CoSb-based half-Heusler alloys. For nine ordered alloys selected by machine-learning screening, Z-site alloying gives the largest average reduction, while X- and Y- site alloying give comparable reduction. In the representative X-, Y-, and Z-site alloys, $Ti_{0.75}Zr_{0.25}CoSb$, $TiRh_{0.25}Ir_{0.75}Sb$, and $TiNiSn_{0.25}Pb_{0.75}$, κl is reduced by 34%, 38%, and 49%, respectively. This order follows the fraction of κl carried by the alloyed sublattice, which is 23%, 52%, and 70%, close to its mass fraction. Across the nine alloys, a larger mass variance weighted by this fraction generally gives a larger reduction. Alloying also softens and flattens the acoustic branches, which lowers the group velocities and shortens the phonon mean free paths. The acoustic–optical overlap and the three-phonon scattering phase space also increase. These results suggest that the site dependence κl reduction comes from the participation of the alloyed sublattice in the heatcarrying acoustic phonons, giving a simple guide for choosing the alloying site.

Keywords: Lattice thermal conductivity; half-Heusler; thermoelectrics

Room 4

A COMPREHENSIVE WORKFLOW FOR FE_2O_3 STRUCTURAL CHARACTERIZATION VIA ORIGIN PEAK FITTING AND VESTA CRYSTAL MODELING

Nur Vivian Kontesa[1], Zaskya Diva Ramadhani[1], Yogi Dwi Prasetyo[1], Tasya Amelia Putri[1], Rista Mutia Anggraini[1], Muhammad Ficky Afrianto[1], and Lucky Zaehir Maulana.[1]

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Abstract

Precise determination of crystallographic parameters from X-ray diffraction (XRD) data often faces challenges due to baseline noise, overlapping diffraction peaks, and complex indexing procedures. This study aims to establish a systematic and standardized computational workflow for analyzing the structural and crystallographic properties of iron (III) oxide (Fe2O3) by integrating OriginLab curve analysis and VESTA 3D spatial modeling. Experimental XRD diffractograms of Fe2O3 were processed in OriginLab, where baseline corrections, Gaussian peak fitting, and Bragg's law computations were executed to extract accurate 2θ positions, full width at half maximum (FWHM), and lattice interplanar spacings (d-spacings). The calculated Miller indices (hkl) were then utilized to construct and refine the 3D unit cell morphology, atomic positions, and bond geometries within VESTA. Peak fitting analysis in Origin successfully resolved subtle peak overlaps, enabling precise indexing of characteristic Fe2O3 diffraction planes, including (104), (110), and (113). VESTA crystal reconstruction verified the rhombohedral/corundum-type geometry, providing exact spatial distribution parameters and visual representations of Fe–O polyhedral linkages. The combined Origin-VESTA computational framework offers an efficient, reproducible, and robust protocol for crystallographic data processing and structural visualization, serving as a reliable reference for characterization of functional metal oxide nanomaterials.

Keywords: Fe2O3; X-ray diffraction; OriginLab; VESTA modeling; hkl indexing; peak fitting; crystal structure.

Room 4

LABORATORY-SCALE EXPERIMENT OF IN-SITU ELECTRO-KINETIC LEACHING (ISEKL) FOR RARE EARTH ELEMENTS AND CRITICAL MINERALS EXTRACTION FROM GRANITIC ROCKS OF THE MERANGIN GEOPARK

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Abstract

The global energy transition has accelerated the widespread adoption of low-carbon technologies, triggering an unprecedented demand for extreme-temperature-resistant components in electric motors and generators. Consequently, this phenomenon drives a critical need for Rare Earth Elements (REEs), particularly the Heavy REE (HREE) fraction. Jambi Province, specifically within the Tantan Granite Formation of the Merangin area, possesses a highly significant potential for primary HREE mineralization. However, the prospect's location within the UNESCO Global Geopark Merangin presents a critical paradox: conventional mining activities are strictly prohibited to preserve the geoheritage site. To address this constraint, this study proposes a sustainable, non-invasive extraction technology: In-Situ Electro-Kinetic Leaching (ISEKL). This method integrates the principles of In-Situ Recovery (ISR) with direct current (DC) electric field stimulation to mobilize metal ions from the low-permeability igneous rock matrix. The primary objective of this study is to empirically validate the geochemical concept of HREE desorption by determining the ligand affinity and optimal thermodynamic parameters required to liberate HREEs from the massive granitic matrix without compromising the rock's structural integrity. The experimental methodology comprises four sequential stages: (1) Geo-mineralogical characterization of the Tantan Granite samples via XRD, XRF, and petrographic analyses; (2) Hydrometallurgical batch leaching experiments to evaluate the efficacy of various leaching agents (sulfuric, chloride, and organic acids) under varied molarities and contact times; (3) Quantitative analysis of dissolved HREE concentrations utilizing ICP-OES instrumentation to calculate the recovery rate; and (4) Evaluation of mineral passivation phenomena through comprehensive solid residue observation. This research serves as a pivotal step in securing national strategic minerals while strictly adhering to global geoheritage conservation principles.

Keywords: Geopark Merangin; Insitu Electro-Kinetic Leaching; HREE ; Granit Tantan; Green Mining

Room 4

STRUCTURAL AND VIBRATIONAL PROPERTIES OF HYDROTHERMALLY SYNTHESIZED FE-DOPED SRTIO$_3$ FOR POTENTIAL PHOTOCATALYTIC APPLICATIONS

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Abstract

This study investigates the structural and vibration properties of Fe-doped $SrTiO_3$ ($SrTi_{1-x}Fe_xO_3$) synthesized via the hydrothermal method. Structural modifications and functional groups of the synthesized material were characterized using X-Ray Diffraction (XRD) and Fourier Transform Infrared (FTIR) Spectroscopy. XRD analysis revealed that the primary cubic perovskite structure of $SrTiO_3$ was successfully maintained following the doping process. the incorporation of Fe^{3+} ions substituting Ti^{4+} induced lattice expansion, evidenced by a distinct shift of the main (110) diffraction peak toward a lower 2θ angle near 32.4°. A minor secondary phase of $SrCO_3$ was also observed, attributed to the high surface reactivity of strontium ions with atmospheric CO_2 during synthesis and handling. Furthermore, FTIR spectra confirmed successful ionic substitution by showing modifications to the TiO_6 octahedral coordination, where Fe-O/Ti-O vibration bands overlapped and broadened around 518 cm^{-1}. Crucially, FTIR data confirmed the significant retention of surface hydroxyl groups (-OH) at ~3400 cm^{-1}. These hydroxyl sites are theoritically vital for trapping photogenerated holed to produce highly reactive hydroxyl radicals (•OH). The successful modification without phase degradation, combined with favorable surface properties, indicates that hydrothermally synthesized $SrTi_{1-x}Fe_xO_3$ is a highly promising photocatalyst candidate for advanced environmental remediation.

Keywords: Fe-doped $SrTiO_3$; Hydrothermal synthesis, Lattice expansion, Photocatalytic

Room 4

GAMMA-RAY SPECTROMETRY AND PORTABLE XRF SCREENING OF RARE EARTH ELEMENTS IN WEATHERING PROFILES AT KALAN AND ELLA HILIR, WEST KALIMANTAN

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Abstract

Kalan and Ella Hilir (Melawi, West Kalimantan) located in an uranium district with monazite-bearing ore in bedrock, but rare earth elements (REE) in the overlying weathering profiles are poorly documented.[ANH2.1] We tested whether field gamma-ray spectrometry and portable X-ray fluorescence (pXRF) can locate REE-enriched horizons for laboratory verification. Gamma-ray data (Gamma Surveyor Vario; 11 stations) were averaged per locality[ANH3.1], and 21 samples representing horizon O, A, and B from seven profiles on the Pinoh Metamorphics and Sepauk Tonalite were analysed in triplicate by SciAps X-555 pXRF. A certified-reference-material check of this instrument supports La and Ce but only screening use of Y and Th. Mean equivalent U (eU) and Th (eTh) were 134 and 64 ppm at Kalan and 61 and 34 ppm at Ella Hilir (eTh/eU 0.48 and 0.56), whereas pXRF U was below detection in 16 of 21 samples.[ANH4.1] Median La+Ce was 128 ppm at Kalan and 84 ppm at Ella Hilir; five samples exceeded 188 ppm (twice the upper crust), with profile maxima in limonite at Kln-2 (341 ppm), Kln-1 (288 ppm) and Ela-1 (195 ppm).[ANH5.1] Kln-1 and Kln-2 horizon A were Ce-dominated (Ce/La 3.9 and 7.4; upper crust 2.0), whereas at Ela-1 La rose from 18 ppm in topsoil to 92 ppm in horizon A (Ce/La 1.1). Both gamma data and La+Ce rank Kalan above Ella Hilir, but only horizon-level pXRF locates the enrichment: A horizons with two contrasting Ce/La signatures.[ANH6.1] These three horizons are prioritised for ICP-MS and exchangeable-REE analysis to test whether the enrichment is mineral-hosted or ion-adsorbed.

Keywords: rare earth elements, gamma-ray spectrometry, portable XRF, regolith geochemistry, Melawi, West Kalimantan

Room 4

EFFECTS OF TEMPO OXIDATION AND CROSSLINKER TYPE ON LIBERICA COFFEE HUSK CELLULOSE–CHITOSAN HYDROGELS FOR METRONIDAZOLE DELIVERY

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Abstract

This study investigated the effects of TEMPO oxidation and crosslinker type on the physicochemical properties and metronidazole release behavior of cellulose–chitosan hydrogels derived from Liberica coffee husk. Four hydrogel formulations were prepared using unmodified cellulose (H1 and H3) or TEMPO-oxidized cellulose (H2 and H4), combined with chitosan and crosslinked using either sodium tripolyphosphate (STPP) (H1 and H2) or Liberica coffee husk polyphenol extract (H3 and H4). The resulting hydrogels were characterized using Fourier transform infrared spectroscopy (FTIR), scanning electron microscopy (SEM), swelling analysis, metronidazole entrapment efficiency, and in vitro drug release in phosphate-buffered saline (pH 7.4). The results revealed that TEMPO oxidation and crosslinker type influenced hydrogel swelling, drug entrapment, and release behavior. H4 exhibited the highest swelling capacity of 737.39% after 240 min, while H2 showed the highest metronidazole entrapment efficiency of 63.05%. After 240 min, cumulative metronidazole release reached 96.36%, 55.39%, 84.57%, and 49.31% for H1, H2, H3, and H4, respectively. Hydrogels containing TEMPO-oxidized cellulose exhibited lower cumulative drug release than their unmodified counterparts, indicating improved drug retention. Furthermore, polyphenol-crosslinked hydrogels demonstrated higher swelling capacities than STPP-crosslinked formulations. These findings highlight the combined influence of cellulose oxidation and crosslinker selection on hydrogel performance, providing insights into the development of cellulose-based hydrogels for controlled metronidazole delivery.

Keywords: Biocrosslinker, Hydrogel, Liberica coffee husk, TEMPO-Oxidized Cellulose

Room 4

DETERMINATION OF ANTIOXIDANT ACTIVITY OF RAMBUTAN (NEPHELIUM LAPPACEUM) PEEL FRACTIONS BY DPPH, FRAP, AND ABTS METHODS

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Abstract

Rambutan (Nephelium lappaceum) peel contains phenolic compounds and flavonoids that have potential as natural antioxidants. This study aimed to assess the antioxidant activity of rambutan peel fractions using the DPPH ((2,2-diphenyl-1-picrylhydrazine)), FRAP (Ferric Reducing Antioxidant Power) and ABTS (2,2'-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid)) methods. Fractionation was carried out using ethyl acetate, ethyl acetate:methanol (1:1), and methanol as solvents. The total phenolic and flavonoid contents, as well as the antioxidant activity, were determined for each fraction. The results showed that the highest total phenolic content was found in the methanol fraction, with a value of 57.54 mg GAE/g, whilst the highest flavonoid content was found in the ethyl acetate:methanol fraction, at 30.92 mg QE/g. The methanol fraction exhibited the highest antioxidant activity in the DPPH assay with an IC_{50} of 56.51 µg/mL, in the FRAP assay with an AAEAC of 64.65 mg AAE/g, and in the ABTS assay, with an IC_{50} of 29.19 µg/mL. These results indicate that more polar solvents extract a greater number of phenolic compounds that contribute to antioxidant activity; however, a fraction that is active using one method is not necessarily active using another. The methanol fraction of rambutan peel shows the greatest potential as a source of natural antioxidants.

Keywords: Nephelium lappaceum, Antioxidants, DPPH, FRAP, ABTS.

Room 4

RESPONSE SURFACE OPTIMIZATION OF PHENOLIC EXTRACTION AND ANTIOKXIDANT ACTIVITY FORM CURCUMA AERUGINOSA RHIZOMES

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Abstract

Curcuma aeruginosa Roxb., commonly known as black turmeric, is a medicinal plant whose rhizomes contain various bioactive compounds, particularly phenolics, which are associated with antioxidant properties. The effectiveness of extracting these compounds depends on several operational parameters; therefore, optimizing extraction conditions is essential to maximize phenolic recovery and antioxidant activity. This study aimed to determine the optimal maceration conditions for extracting antioxidant compounds from C. aeruginosa rhizomes using Response Surface Methodology (RSM) with a Box– Behnken design implemented in Design-Expert version 12.0.3.0. Three independent variables were investigated: ethanol concentration, liquid-to-solid ratio, and extraction time. Total phenolic content and antioxidant activity were evaluated as response parameters to assess the effectiveness of the extraction process. The experimental findings indicated that variations in ethanol concentration, liquid-to-solid ratio, and extraction duration influenced the measured responses. A second-order polynomial model was developed to describe the relationship between the independent variables and the experimental responses, providing a suitable basis for identifying optimum conditions. The optimization results identified the optimal extraction parameters as an ethanol concentration of 73%, a liquid-to-solid ratio of 1:20 (300 mL of solvent per 15 g of rhizome), and a maceration period of one day. Under these conditions, the extraction process achieved the highest total phenolic content and antioxidant activity within the experimental conditions investigated. These findings demonstrate the potential of RSM and the Box–Behnken design as systematic approaches for optimizing the extraction of phenolic compounds from C. aeruginosa rhizomes and supporting further research on their potential applications as natural antioxidant sources.

Keywords: Antioxidant activity, Box-Behnken design, Curcuma aeruginosa, Optimization

Room 4

COMPARATIVE IN VITRO ANTHELMINTIC ACTIVITY OF EMODEPSIDE AND IVERMECTIN AGAINST STRONGYLOIDES RATTI

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Abstract

Strongyloides nematodes, which cause medical and veterinary disease, requires good anthelmintic treatment. Ivermectin is standardized treatment of strongyloidiasis, while emodepside is a promising parasitic nematode anthelmintic. Comparative evidence on their activity against Strongyloides larvae is limited. This study aims to evaluate the in vitro anthelmintic activity of emodepside and ivermectin against filariform larvae of Strongyloides ratti, an experimental model for Strongyloides infection. Larval motility was evaluated following exposure to varying concentrations of both pharmaceuticals for durations of 24, 48, and 72 hours. Half-maximal inhibitory doses (IC_{50}) were determined by nonlinear regression analysis. Survival analysis was conducted with the Kaplan–Meier technique, with group comparisons performed via the log-rank and Gehan–Breslow–Wilcoxon tests. Both substances diminished larval motility in dependent on concentration and duration. The IC_{50} values for Emodepside were 0.2008 ± 0.0096, 0.1923 ± 0.0138, and 0.1320 ± 0.0031 µM at 24, 48, and 72 hours, respectively. The associated ivermectin measurements were 0.2341 ± 0.0192, 0.2139 ± 0.0065, and 0.1466 ± 0.0128 µM. At a concentration of 2.5 µM, both substances resulted in total larval mortality within 72 hours. Survival curves exhibited considerable variation among treatment groups (log-rank, $p = 0.0060$; Gehan–Breslow–Wilcoxon, $p = 0.0046$). Both drugs shown in vitro efficacy against S. ratti. Emodepside consistently exhibited lower IC_{50} values compared to ivermectin. These results provide a comparative in vitro standard for assessing anthelmintic efficacy against Strongyloides larvae and encourage a further study of pharmacological reactions in this parasitic model.

Keywords: Strongyloides ratti; emodepside; ivermectin; anthelmintic activity; infective larvae

Paralel Schedule

Room 5 (Online) 13:30-15:50 Moderator: Alrizal, S.Pd., M.Si.		
Time (WIB)	**Presenter**	**Title**
13.30-13.50	**Ilmi Fadhilah Rizki, M.Sc. (Indonesian Oil Palm Research Institute (IOPRI))**	Green Innovation in Palm Oil Downstream Processing Through Eco-Friendly Extraction and Functional Lipid Production **(Invited Speaker)**
13.50-14.00	Martina Asti Rahayu (Universitas Jambi)	Box–Behnken Design Assisted Optimization of Colorimetric Histamine Detection Using Modified Silver Nanoparticles
14.00.14.10	Athaya Razan Zaidan (UPN "Veteran")	Digital Transformation Of Physical Chess Analysis Using Computer Vision Services And Stockfish Engine
14.10-14.20	Q & A	
14.20-14.30	Asla Khairunnisa (Universitas Jambi)	Stirred Adsorption And Regeneration Of Rice Husk Ash-Derived Silica For Free Fatty Acid Reduction
14.30-14.40	Regna Tri Jayanti (ATI Padang Polytechnic)	Engineering Application of Palm Oil Fly Ash and Natural Zeolite in a Fixed-Bed Adsorption Column for Textile Wastewater Treatment
14.40-14.50	Q & A	
14.50-15.00	Syamsuddin Mas'ud (Universitas Hasanuddin)	Contraction Principles In Controlled Metric Type Spaces: A Focused Survey
15.00-15.10	Defia Indah Permatasari (Universitas Jambi)	Synthesis and Characterization of Natural Magnetic Particle/Chitosan Composites Impregnated with Copper Nanoparticles
15.10-15.20	Q & A	
15.20-15.30	Micko Marius (Universitas Persada Indonesia Y.A.I.)	Cost–Function Traceability for HVAC Value Engineering in MEP Design-and-Build Projects
15.30-15.40	Evan Vria Andesmora (Universitas Bengkulu)	Multivariate Phenotypic Characterization And Preliminary Selection Of Potential Local Coconut Trees On Enggano Island
15.40-16.50	Q & A	

Ilmi Fadhilah Rizki, M.Si. *Topic*

GREEN INNOVATION IN PALM OIL DOWNSTREAM PROCESSING THROUGH ECO-FRIENDLY EXTRACTION AND FUNCTIONAL LIPID PRODUCTION

Ilmi Fadhilah Rizki*[1], Frisda Rimbun Panjaitan[1], Manda Edy Mulyono[1], Brahmani Dewa Bajra[1]

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Abstract

Crude Palm Oil (CPO) is a rich source of essential phytonutrients, including tocotrienols, tocopherols, and carotenes. However, conventional extraction methods often rely on toxic organic solvents. Aligning with the sustainable future paradigm, this presentation highlights recent green innovations in palm oil downstream processing. First, the application of Natural Deep Eutectic Solvents (NADES), specifically choline chloride-based mixtures, is presented as an eco-friendly alternative for extracting tocotrienol-rich fractions (TRF) from CPO. Second, the development of High-Oleic Low-Palmitic Red Palm Super Olein (HOLP-RPSO) through solvent-free dry fractional crystallization is discussed. This method preserves heat sensitive phytonutrients while producing a functional lipid profile with enhanced oxidative stability and favorable sensory attributes for food applications. By integrating these green technologies, the industry can minimize its environmental impact while maximizing the recovery of high-value bioactive compounds.

Keywords: Crude palm oil; Natural deep eutectic solvents (NADES); Tocotrienol-rich fraction (TRF); Dry fractionation; Red palm super olein; Green extraction.

Room 5

BOX–BEHNKEN DESIGN ASSISTED OPTIMIZATION OF COLORIMETRIC HISTAMINE DETECTION USING MODIFIED SILVER NANOPARTICLES

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Abstract

Histamine accumulation in fish and fishery products represents a significant food safety concern, highlighting the need for rapid and reliable detection methods. This study aimed to optimize the colorimetric detection of histamine using ethylenediaminetetraacetic acid (EDTA)-modified silver nanoparticles (AgNPs) through response surface methodology employing a Box–Behnken design (BBD). The colorimetric response was monitored using UV–visible spectrophotometry and expressed as the absorbance ratio at 648 and 397 nm (A648/A397). Three experimental factors, including pH, AgNP concentration, and detection time, were systematically investigated through 17 experimental runs. Analysis of variance (ANOVA) demonstrated that the quadratic model was highly significant ($F = 147.24$, $p < 0.0001$), with a coefficient of determination (R^2) of approximately 0.995. Detection time exhibited the strongest influence on the analytical response ($F = 735.79$), followed by AgNP concentration ($F = 524.16$), whereas pH showed no significant linear effect within the investigated range ($p = 0.7294$). A significant interaction between AgNP concentration and detection time was observed ($p = 0.0492$), while the nonsignificant lack of fit ($p = 0.3439$) supported the adequacy of the fitted model. Numerical optimization identified pH 7.28, an AgNP concentration of 154 µM, and a detection time of 15 min as the predicted optimal conditions, yielding an absorbance ratio of 1.3391. These findings demonstrate the applicability of Box–Behnken design for identifying influential experimental factors and optimizing the colorimetric response of EDTA-modified AgNPs toward histamine.

Keywords: Histamine, silver nanoparticles, Colorimetric detection, Box–Behnken design, Response surface methodology

Room 5

DIGITAL TRANSFORMATION OF PHYSICAL CHESS ANALYSIS USING COMPUTER VISION SERVICES AND STOCKFISH ENGINE

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Abstract

Evaluating physical over-the-board chess positions using analytical engines typically requires manual piece transcription, which is inefficient and error-prone during training sessions. This study aims to design and implement a two-tier web-based information system that integrates automated Forsyth–Edwards Notation (FEN) extraction services with the Stockfish chess engine. The system employs a decoupled client-server architecture communicating via a REST API. The FastAPI backend encapsulates a modular computer vision service—combining U-Net board segmentation, homography perspective transformation, and YOLO26 piece detection trained on the ChessReD2K dataset—utilizing in-memory model loading to minimize inference latency. Since visual inference alone cannot capture non-visual game states and remains sensitive to severe perspective occlusions, the React.js frontend provides auxiliary FEN parameter controls and multi-stage pipeline visualization alongside an interactive 2D board. The complete FEN string is then transmitted to Stockfish via the Universal Chess Interface (UCI) protocol to compute centipawn evaluations, best moves, and principal variations. The primary contribution of this work lies in the seamless system integration bridging deep learning inference services, transparent pipeline inspection, and UCI engine analysis within a unified web workflow. Functional testing on ChessReD2K test samples confirms that the system effectively streamlines physical chess position analysis through standard web browsers, contributing to digital transformation in sports training tools.

Keywords: Digital Transformation, Web Information System, Computer Vision Service, Stockfish Engine

Room 5

STIRRED ADSORPTION AND REGENERATION OF RICE HUSK ASH-DERIVED SILICA FOR FREE FATTY ACID REDUCTION

Asla Khairunnisa[1], Yety Anggraini[1], Rosmawati Sipayung[1], Rahma Amalia[1], Chindy Mauliza Duana[1], Rara Ayu Lestary[1], and And Nita Widyastuti[1]

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Abstract

Waste cooking oil is a potential biodiesel feedstock, but its free fatty acid (FFA) content can interfere with base-catalyzed transesterification. This study evaluated rice husk ash-derived silica for FFA reduction under stirred batch adsorption conditions and assessed its reusability after regeneration. The adsorbent was prepared by calcining rice husks and extracting silica with NaOH. Fourier transform infrared (FTIR) spectroscopy was used to characterize its functional groups. Adsorption experiments were conducted at 60°C, with a 5 wt% adsorbent dosage, 300 rpm stirring speed, and contact times of 10–90 min. The highest FFA removal during contact-time evaluation was 42.33% at 45 min, corresponding to an adsorption capacity of 30.080 mg/g. In a separate adsorption evaluation, FFA removal reached 39.15%, with a capacity of 27.824 mg/g. After regeneration, adsorption capacity decreased to 20.680 mg/g in the first cycle and 3.760 mg/g in the second, reaching -10.528 mg/g in the third cycle, indicating that the adsorbent no longer achieved net FFA removal under the tested conditions. FTIR spectra showed characteristic bands associated with Si–O–Si bonds and hydroxyl-containing groups. These findings demonstrate the potential of rice husk ash-derived silica for FFA reduction, although its declining adsorption capacity limits reusability.

Keywords: Rice husk ash-derived silica; Stirred adsorption; Adsorbent regeneration; Free fatty acids; Waste cooking oil.

Room 5

ENGINEERING APPLICATION OF PALM OIL FLY ASH AND NATURAL ZEOLITE IN A FIXED-BED ADSORPTION COLUMN FOR TEXTILE WASTEWATER TREATMENT

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Abstract

Textile industries are potential contributors to the global economy particularly in Asia. However, it is also recognized as a major source of environmental pollution and affect human health. Wastewater containing synthetic dyes represents one of the most persistent categories of water pollutants, posing significant ecotoxicological risks. Consequently, adequate treatment is required before discharging such waste into water bodies. This study aimed to design a fixed-bed adsorption column to evaluate the efficacy of palm oil fly ash (POFA) and natural zeolite in reducing COD and BOD levels via adsorption. Experiments involved varying the arrangement of adsorbents within a small-scale column to determine the optimal layering sequence, with clarity and pH serving as key monitoring parameters. Subsequent testing using a fixed-bed adsorption column (40 cm height, 25 cm diameter, 20 L volume) demonstrated that the highest efficiency was achieved with the following layer configuration: (1) coarse zeolite, (2) fine zeolite, (3) POFA, and (4) cotton fiber. The resulting effluent characteristics—pH 7, COD 55.55 mg/L, BOD 47.62 mg/L, TSS 0.2, and chromium 0.065 mg/L—met the quality standards stipulated in the Regulation of the Minister of Environment and Forestry No. P.16/MENLHK/SETJEN/KUM.1/4/2019.

Keywords: Palm Oil Fly Ash (POFA); Natural Zeolite; Adsorbent; Textile Wastewater; Fixed-Bed Adsorption Column

Room 5

CONTRACTION PRINCIPLES IN CONTROLLED METRIC TYPE SPACES: A FOCUSED SURVEY

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Abstract

Contraction Principles in Controlled Metric Type Spaces: A Focused Survey. Background: Controlled metric type spaces (CMTS), introduced in 2018, scale each term of the triangle inequality by a control function, and fixed point theory in this setting has expanded rapidly. Objective: This survey organizes recent contraction results in CMTS and clarifies the role of the control function in them. Methods: A purposively selected set of nineteen studies published between 2020 and 2025 was examined; their contractive inequalities, conditions on the control function, and conclusions were tabulated and grouped into two families. Results: In most linear and classical-type contractions, the control function appears only through a limit condition on the Picard sequence; in a few cases it enters the inequality itself. In nonlinear and function-based contractions, most papers keep this condition with a modified bound, some build the control function into a class of comparison functions, and two do not use it. Continuity assumptions are common, and several numerical examples are defined on three-point sets. Conclusions: In many of the reviewed papers, a new contractive form is transferred to the CMTS setting by verifying one common condition, but the variants of this condition are rarely compared with one another. Removing continuity-type assumptions without substituting other regularity conditions, and treating Hardy–Rogers-, Ćirić-, and Suzuki-type contractions, appear to remain largely open in the reviewed papers.

Keywords: Controlled metric type space, Fixed point, Contraction mapping, Control function

Room 5

SYNTHESIS AND CHARACTERIZATION OF NATURAL MAGNETIC PARTICLE/CHITOSAN COMPOSITES IMPREGNATED WITH COPPER NANOPARTICLES

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Abstract

Natural magnetic particle/chitosan/copper nanoparticle (NMP/Chi/CuNP) composites were successfully prepared through the impregnation of chemically synthesized copper nanoparticles (CuNPs) into a natural magnetic particle/chitosan (NMP/Chi) matrix. The CuNPs were produced via chemical reduction using ascorbic acid as the reducing agent and chitosan as the coating agent under microwave irradiation. UV–Vis spectrophotometric analysis revealed an absorption peak at 580–590 nm, confirming the formation of CuNPs, with the highest absorbance obtained at a precursor concentration of 20 mM. Fourier-transform infrared (FTIR) spectroscopy was employed to investigate the interactions between the NMP/Chi matrix and CuNPs in the resulting NMP/Chi/CuNP 20 composite. X-ray diffraction (XRD) analysis identified the most prominent characteristic peak of Fe_3O_4 at a 2θ value of 35.07°. The intensity of this peak was lower in the NMP/Chi/CuNP 20 composite than in the NMP and NMP/Chi samples. The average crystallite sizes of NMP, NMP/Chi, and NMP/Chi/CuNP 20 were 7.98, 67.03, and 53.95 nm, respectively. Scanning electron microscopy coupled with energy-dispersive X-ray spectroscopy (SEM–EDX) provided evidence of CuNP incorporation into the NMP/Chi matrix through the detection of carbon (C), nitrogen (N), oxygen (O), iron (Fe), and copper (Cu). Transmission electron microscopy (TEM) further revealed that the particles in the NMP/Chi/CuNP 20 composite exhibited a predominantly spherical morphology similar to that observed for CuNPs.

Keywords: Copper nanoparticles, composite, natural magnetic particles, chitosan

Room 5

COST–FUNCTION TRACEABILITY FOR HVAC VALUE ENGINEERING IN MEP DESIGN-AND-BUILD PROJECTS

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Abstract

Value Engineering (VE) requires auditable cost–function evidence before HVAC design alternatives are evaluated. Previous studies have addressed HVAC selection using multi-criteria assessment and integrated VE, FAST, and life-cycle cost methods; however, the literature reviewed provides limited item-level evidence of how contractor tender bill-of-quantities (BOQ) costs are traced to HVAC functions, particularly for multifunctional equipment and shared-scope works. This study establishes a cost–function baseline for an anonymized office-building MEP Design-and-Build project in Jakarta. Using the tender BOQ, room cooling requirements, and equipment schedules for a non-inverter direct-expansion single-split system, the VE Information Phase applies Cost Model and Pareto analyses, followed by function classification and Technical FAST in the Function Analysis Phase. A line-by-line audit reconciles 416 BOQ entries. Air-conditioning and ventilation works amount to IDR 6,438,217,000 (32.189% of the pre-tax MEP tender value). The baseline includes 73 AC units serving aggregate room cooling requirements of 2,319,244 BTU/h, not necessarily a simultaneous peak load. Identifiable cooling works, including dedicated AC power cables, total IDR 4,851,009,000; IDR 763,661,000 (15.74%) is partially traceable to functions, while IDR 4,087,348,000 (84.26%) remains associated with multifunctional or shared-scope items. Technical FAST identifies temperature control as the basic function and seven supporting functions. This auditable baseline supports transparent subsequent HVAC design comparisons without claiming complete function-cost allocation, verified energy savings, or selection of an optimal alternative.

Keywords: Value Engineering, Construction Management, HVAC, Cost-Function Traceability, Technical FAST

Room 5

MULTIVARIATE PHENOTYPIC CHARACTERIZATION AND PRELIMINARY SELECTION OF POTENTIAL LOCAL COCONUT TREES ON ENGGANO ISLAND

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Abstract

The local coconut of Enggano Island is a genetic resource with potential for development and conservation; however, phenotypic characterization and the identification of superior individuals remain limited. This study aimed to evaluate phenotypic diversity and identify potential coconut tree candidates based on a multi-trait approach. A total of 30 local coconut trees distributed across four villages on Enggano Island—Kaana, Kahyapu, Meok, and Banjar Sari—were evaluated based on six traits, including copra thickness, husk thickness, shell thickness, mature fruit diameter, young fruit diameter, and coconut water sweetness (°Brix). Data were analyzed using descriptive statistics, Pearson correlation, Principal Component Analysis (PCA), hierarchical clustering, and a multi-trait selection index. The results revealed phenotypic variation among individuals, with the coefficient of variation ranging from 11.09% for mature fruit diameter to 25.89% for husk thickness. PCA demonstrated that PC1 and PC2 explained 36.29% and 21.75% of the total variation, respectively, with PC1 mainly associated with fruit size and structure, whereas PC2 was predominantly influenced by coconut water sweetness. Hierarchical clustering analysis grouped the 30 trees into two phenotypic clusters, with the second cluster generally displaying higher copra thickness and fruit size. Based on the multi-trait selection index, six individuals—K5, M9, M8, BJ4, BJ7, and M10—were identified as potential candidate trees. These results indicate that integrating phenotypic characterization and multivariate analysis serves as an effective initial approach for screening potential individuals within the local Enggano coconut population.

Keywords: Enggano Island, Hierarchical clustering, Local coconut, Phenotypic diversity, Selection index.

Paralel Schedule

Room 6 (Online) 13:30-15:30 Moderator: Rizka Shafira, S.T., M.T.		
Time (WIB)	**Presenter**	**Title**
13.30-13.40	Despita (Universitas Jambi)	Anurans Species Composition Across Different Habitat Types In Rantau Kermas Village, Jambi, Indonesia
13.40-13.50	Cut Multahadah (Universitas Jambi)	Statistics for Startups: A Student Entrepreneurship Downstreaming Program at Koperasi G2, Jambi City
13.50-14.00	Q & A	
14.00-14.10	Pradita Eko Prasetyo Utomo (Universitas Jambi)	Automation for Handling error runtime using AI Tools
14.10-14.20	Ulfa Khaira (Universitas Jambi)	Deep Learning-Based Classification of Tomato Leaf Diseases Using MobileNetV3-Large
14.20-14.30	Q & A	
14.30-14.40	Reni Aryani (Universitas Jambi)	Smart Village System Analysis in Teluk Village Using an Object-Oriented Analysis (OOA) Approach
14.40-14.50	Cut Multahadah (Universitas Jambi)	Muaro Jambi Temple Ecotourism: Optimizing Tourist Visits Using K-Medoids and SEM-PLS
14.50-15.00	Q & A	
15.00-15.10	Nurza Purwa Abiyoga (Universitas Jambi)	Empowering Rural Communities through Simple Smart Farming Technologies for Productive Home Gardens and Household Food Security: A Community-Based Approach in Pematang Jering Village, Indonesia
15.10-15.20	Nurza Purwa Abiyoga (Universitas Jambi)	Accelerating Green and Digital Competencies in STEM Education: A Demand-Driven Talent Supply Chain Framework for Sustainable Industry Transformation
15.20-15.30	Q & A	

Room 6

ANURANS SPECIES COMPOSITION ACROSS DIFFERENT HABITAT TYPES IN RANTAU KERMAS VILLAGE, JAMBI, INDONESIA

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Abstract

Rantau Kermas Village, a buffer zone of Kerinci Seblat National Park in Merangin Regency, Jambi, comprises a mosaic of customary forest, agroforestry and riverine habitats, yet its anuran fauna has so far been documented only within the customary forest. This study aimed to inventory anuran species in three habitat types: natural forest, coffee–cinnamon agroforestry, and the Batang Langkup River. Data collection was conducted in July-August 2026. Sampling used the Visual Encounter Survey combined with line transects (200 × 5 m; nine transects, two repetitions each) at night (18:00–22:00 WIB). Totally 173 individuals of 18 species from five families were recorded. The natural forest harboured ten species representing all five families, seven of which were found exclusively there, including Leptobrachium waysepuntiense and Rhacophorus achantharrhena. The river also supported ten species, mainly ranids such as Hylarana chalconota and H. parvaccola, with Limnonectes blythii and Pulchrana sundabarat recorded only along the river. The agroforestry supported nine species and was dominated by Fejervarya limnocharis and F. cancrivora (45 of 68 individuals), with Polypedates pseudotilophus found only in this habitat. The river and agroforestry shared eight species, whereas the forest shared only three species with each of them (F. cancrivora, Odorrana hosii, and Polypedates macrotis), which occurred in all habitats. Five recorded species are endemic to Sumatra. These findings exceed the 11 species previously reported from the customary forest alone and show that each habitat supports a distinct anuran assemblage, underlining the importance of conserving both natural forest and riparian habitats.

Keywords: Anura; Habitat Type; Species Composition; Rantau Kermas; Sumatran endemic species

Room 6

STATISTICS FOR STARTUPS: A STUDENT ENTREPRENEURSHIP DOWNSTREAMING PROGRAM AT KOPERASI G2, JAMBI CITY

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Abstract

The development of the digital economy has increased the need for data analysis competencies in student entrepreneurship development. The community service program entitled "Statistics for Startups: A Student Entrepreneurship Downstreaming Program at Koperasi G2, Jambi City" aims to integrate research findings on students' entrepreneurial interest through statistical training, workshops, and mentoring. The activities were conducted through instructional sessions, demonstrations, hands-on business data processing using Microsoft Excel, and evaluation through questionnaires. The evaluation results showed that 81.82% of participants agreed that statistical materials were useful and applicable to business activities, while 18.18% selected other response categories. Furthermore, 80.91% stated that the program provided valuable additional knowledge, 77.27% reported an improved understanding of how to utilize business data, and 76.36% considered the materials relevant to their entrepreneurial needs. Meanwhile, the relevance of the program theme received 72.73% agreement, and the suitability of the implementation methods reached 73.64%. These findings indicate that the program received positive responses, particularly regarding the usefulness of the learning materials. However, further evaluation is needed to identify participants' needs and improve the mentoring methods. This program is expected to encourage data-driven decision-making and foster the development of innovative and sustainable student-led startups.

Keywords: Downstreaming, Koperasi G2, Statistics, Startups

Room 6

AUTOMATION FOR HANDLING ERROR RUNTIME USING AI TOOLS

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Abstract

In the software ecosystem, the emergence of bugs—defects, failures, or errors in source code that produce unintended output—is unavoidable. The impact of these service disruptions not only causes significant financial losses but also drastically reduces user trust. Currently, the process of identifying and monitoring system errors in service applications at the University of Jambi is still reactive and manual. This process creates a long communication chain, where technical information needed by developers is often lost or incomplete when forwarded to third parties. The impact of this lack of a centralized monitoring system is slow response times for repairs, which directly degrades the quality of the university's digital services. The inability to detect errors early also risks data integrity if database transaction failures are not promptly addressed. If left unchecked, this will reduce user productivity and negatively impact the reliability of the institution's information technology infrastructure. This research aims to develop an Integrated Error Monitoring Dashboard Application System. This system will automate error capture through application exception handlers, send data asynchronously via APIs, present real-time statistical visualizations, issue alerts via Telegram bots, and utilize artificial intelligence to accelerate problem diagnosis and resolution. In addition, this system is designed as an architecture that is ready to be developed towards a hybrid approach based on embedded AI in the next stage.

Keywords: Handling, Error, Runtime, AI, Dashboard

Room 6

DEEP LEARNING-BASED CLASSIFICATION OF TOMATO LEAF DISEASES USING MOBILENETV3-LARGE

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Abstract

Tomato leaf diseases can significantly affect plant growth and productivity, making accurate and timely disease identification essential for effective crop management. This study aims to develop a deep learning-based model for classifying tomato leaf diseases using MobileNetV3-Large. The study utilized the tomato subset of the PlantVillage dataset, consisting of 18,160 images across 10 classes. The dataset was divided into 13,069 training images, 1,452 validation images, and 3,639 testing images. MobileNetV3-Large was implemented using ImageNet pre-trained weights and trained with the AdamW optimizer and OneCycleLR learning rate scheduler. The experimental results show that the model achieved an accuracy of 99.37% and a macro F1-score of 99.12% on the test dataset. The resulting model contains 4.21 million parameters and was exported to ONNX format with a file size of 16.8 MB. Dynamic INT8 quantization was also evaluated to reduce the model size and computational requirements. However, the quantized model experienced an accuracy decrease of 27.21 percentage points compared with the original float32 model. Therefore, the float32 model was retained as the deployment model. These results indicate that MobileNetV3-Large provides high classification performance for tomato leaf diseases while maintaining a relatively lightweight architecture suitable for further deployment on edge computing devices. The developed model provides a strong baseline for subsequent evaluation using field-acquired tomato leaf images.

Keywords: Tomato leaf disease, Disease classification, Deep learning, MobileNetV3-Large, Image classification, Edge computing

Room 6

SMART VILLAGE SYSTEM ANALYSIS IN TELUK VILLAGE USING AN OBJECT-ORIENTED ANALYSIS (OOA) APPROACH

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Abstract

Digital transformation is one of the efforts to improve the quality of governance and public services in rural areas. Teluk Pemayung Village has achieved the status of an Independent Village; however, the implementation of technology for information provision, administrative services, MSME management, and community interaction remains unintegrated. This study aims to analyze the requirements and system structure of the Smart Village system in Teluk Pemayung Village using the Object-Oriented Analysis (OOA) method, focusing on four pillars: Smart Governance, Smart Economy, Smart Community, and Smart Living. The analysis was conducted through problem identification, business process analysis, stakeholder identification, feature requirements analysis, and object-oriented modeling using Use Case Diagram, Use Case Specification, Sequence Diagram, and Class Diagram. The results indicate that the system requires key functions for providing village information, administrative services, community complaints, MSME management, educational and facility information, and integrated government data management. Object modeling also provides an overview of the attributes, behaviors, and relationships among system objects. These results can serve as a structured basis for the further development of the Smart Village System in Teluk Pemayung Village.

Keywords: Teluk Pemayung Village, Smart Village, System Analysis, Object-Oriented Analysis, Village Information System

Room 6

MUARO JAMBI TEMPLE ECOTOURISM: OPTIMIZING TOURIST VISITS USING K-MEDOIDS AND SEM-PLS

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Abstract

Tourism is a strategic sector that contributes to economic growth, improves community welfare, and preserves cultural heritage and the environment. In recent years, the tourism paradigm has shifted toward ecotourism, which emphasizes environmental sustainability, local community empowerment, and educational value for visitors. One destination with significant potential for cultural-based ecotourism development is the Muaro Jambi Temple Complex, the largest temple complex in Southeast Asia, with considerable historical and ecological value. Despite its substantial potential, tourist visits to the area remain suboptimal and tend to fluctuate. This situation is influenced by various factors, including destination attractiveness, service quality, accessibility, and the limited optimization of digital promotion. Therefore, a comprehensive analytical approach is needed to gain a deeper understanding of tourist behavior. This study adopts a data-driven approach by integrating the K-Medoids algorithm and Partial Least Squares Structural Equation Modeling (PLS-SEM). K-Medoids is employed to segment tourists based on their characteristics and preferences, particularly age and distance from their place of residence, while PLS-SEM is used to analyze the relationships among variables influencing tourist visitation decisions. The study examines five exogenous variables: ecotourism attractiveness (X1), accessibility of tourist attractions (X2), cleanliness of facilities (X3), infrastructure conditions (X4), and security of the tourist area (X5), with tourist visit optimization as the endogenous variable (Y). Primary data are collected through questionnaires distributed to visitors of the Muaro Jambi Temple Complex. This study is expected to identify tourist segments visiting the complex through clustering algorithms and determine the dominant factors influencing tourist visits. The expected outputs include an article published in a national journal indexed in SINTA 1–3 and the establishment of collaborative partnerships between the Mathematics Study Program, Faculty of Science and Technology, Universitas Jambi, the Jambi Tourism Office, and the Jambi Cultural Heritage Preservation Agency to promote the ecotourism potential of the Muaro Jambi Temple Complex.

Keywords: K-Medoids, Muaro Jambi Temple Ecotourism, Optimizing Tourist Visits, SEM-PLS

Room 6

EMPOWERING RURAL COMMUNITIES THROUGH SIMPLE SMART FARMING TECHNOLOGIES FOR PRODUCTIVE HOME GARDENS AND HOUSEHOLD FOOD SECURITY: A COMMUNITY-BASED APPROACH IN PEMATANG JERING VILLAGE, INDONESIA

Fetty Febriasti Bahar[1], Oki Alfernando[1], Nurza Purwa Abiyoga[1], Sitti Sausan Maulida[1], Nadya Bestnissa[1], Alissia Westy Putri[1], and Rahmad Ashar[1]

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Abstract

Home gardens have considerable potential to serve as a source of household food and contribute to food security in rural communities. This community service program aimed to empower the community of Pematang Jering Village through the implementation of simple technology-based smart farming practices to optimize home garden utilization. The program was conducted through educational activities, the provision of a smart farming guidebook, practical demonstrations, and hands-on assistance, with the involvement of the Muaro Jambi Regency Department of Food Crops and Horticulture. A community-based approach was adopted to enhance residents' knowledge and skills in home garden management, food crop cultivation, and the application of simple agricultural technologies. The program is expected to encourage the development of productive home gardens and contribute to sustainable household food security.

Keywords: Smart farming, community empowerment, food security, Accessible technology.

Room 6

ACCELERATING GREEN AND DIGITAL COMPETENCIES IN STEM EDUCATION: A DEMAND-DRIVEN TALENT SUPPLY CHAIN FRAMEWORK FOR SUSTAINABLE INDUSTRY TRANSFORMATION

Lenny Marlinda[1], Oki Alfernando[1], Nurza Purwa Abiyoga[1], Niken Rarasati[1], and Togi Siholmarito Simarmata[1]

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Abstract

The convergence of digital transformation and green innovation has fundamentally reshaped competency requirements in Science, Technology, Engineering, and Mathematics (STEM) industries. However, higher education institutions struggle to align curricula with the rapid demands of Industry 4.0 and Environmental, Social, and Governance (ESG) standards, creating a critical skills gap. This study proposes a Demand-Driven Talent Supply Chain Management (TSCM) framework to accelerate green and digital competencies in STEM education for sustainable industry transformation. Employing a mixed-methods sequential exploratory design, the initial phase involved an Industry Demand Analysis via a survey of 15 practitioners across energy, mining, and infrastructure sectors. Preliminary findings reveal that the industry equally prioritizes three transversal competencies (mean = 4.53/5): Data Analytics (Digital), Communication, and Sustainability/ESG compliance (Green), alongside significant skills gaps in digital tool proficiency and green regulatory compliance. Future projections further emphasize data engineering, ESG integration, and adaptive leadership. These demand signals are systematically translated into a Competency Planning Layer comprising eight domains, notably Digital & Data and Sustainability & Governance. Subsequent phases will employ Delphi, Fuzzy AHP, and ISM to prioritize and map core competencies. By treating industry needs as continuous demand signals, this framework provides an evidence-based mechanism for curriculum optimization. Ultimately, this research accelerates digital transformation and green innovation by ensuring STEM higher education produces a future-ready workforce equipped to drive sustainable industry transformation.

Keywords: Talent Supply Chain Management, Digital Competencies, Green Skills, ESG, STEM Education, Curriculum Optimization, Sustainable Industry

Paralel Schedule

Room 7 (Online) 13:30-15:30 Moderator: Munifilia Ekasari, S.Si., M.Si.		
Time (WIB)	**Presenter**	**Title**
13.30-13.40	Novita Sari (Universitas Jambi)	Microwave-Assisted Urea Delignification Of Opefb: Effect Of Holding Time On Solid Yield
13.40-13.50	Icha Imelda Purba (Universitas Jambi)	Green Synthesis of Silver Nanoparticles Using Garcinia Species: Phytochemical Roles and Synthesis Parameters
13.50-14.00	Q & A	
14.00-14.10	Muhamad Alvin Reagen (Universitas Jambi)	Combination of Silica from Oil Palm Leaves with Barium in the Adsorption Process of Congo Red Dye
14.10-14.20	Alifya Lucky Nugraheni (Universitas Jambi)	Synthesis of a Smart Film from Cassava Starch and Chitosan Functionalized with Anthocyanins as a Food Freshness Biosensor
14.20-14.30	Q & A	
14.30-14.40	Zuli Rodhiyah (Universitas Jambi)	Education And Mentorship On Medical Hazardous Waste Governance For Healthcare Providers In Jambi City Toward Green Healthcare
14.40-14.50	Yoravika Dwiwibangga (Universitas Jambi)	Physicochemical Characterization and Evaluation of The Multi-Bioactivity of Virgin Coconut Oil (VCO) from Local Bengkulu Coconut Varieties
14.50-15.00	Q & A	
15.00-15.10	Arief Surya Darma (Universitas Jambi)	A Natural Pediastrum duplex-Dominant Consortium Persists Through Prolonged Nutrient-Unamended Storage
15.10-15.20	Annisa Amalia Ulfah Ulfah (Universitas Jambi)	Surface Functionalization Of Oil Palm-Derived Cellulose Nanocrystals: Implications For Cu(Ii) Adsorption Performance
15.20-15.30	Q & A	

Room 7

MICROWAVE-ASSISTED UREA DELIGNIFICATION OF OPEFB: EFFECT OF HOLDING TIME ON SOLID YIELD

Novita Sari[1], Ra'ida Raudhatussya'rifah[1], Etika Kristiana Gulo[1], Rara Ayu Lestary[1], Sri Mundarti[1], Ira Galih Prabasari[1], Nita Widyastuti[1], and Lenny Marlinda[1]

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Abstract

Oil palm empty fruit bunches (OPEFB) are solid waste from palm oil processing that decomposes slowly, despite containing relatively high levels of cellulose. Cellulose is bound to hemicellulose and lignin within the lignocellulosic structure, making delignification necessary. A microwave-assisted urea hydrotropic system is a more environmentally friendly option, in which the process time consists of heating time and hold time; however, studies on the effect of hold time on OPEFB remain limited. This study aims to determine the effect of hold time on the solid yield of OPEFB during microwave-assisted urea delignification. OPEFB powder (60 mesh) was treated with a 36.8% w/v urea solution at a solid-to-liquid ratio of 1:35, a microwave power of 450 W, and a temperature of 90 °C. The heating time was set at 5 minutes, while the hold time was varied at 10, 20, 30, 40, and 50 minutes; the solid yield of the residue was then calculated. The solid yield decreased gradually from 80.0% at 10 minutes to 75.0% at 50 minutes, consistent with the gradual dissolution of lignocellulosic components during longer hold times. The novelty of this study lies in the use of hold time as the sole variable, while other conditions were kept constant. These preliminary findings help in selecting an appropriate hold time and support environmentally friendly and energy-efficient biomass processing using microwave heating, although compositional analysis is still needed to determine the extent of delignification.

Keywords: Oil palm empty fruit bunches, Delignification, Urea hydrotropic; microwave, Hold time

Room 7

GREEN SYNTHESIS OF SILVER NANOPARTICLES USING GARCINIA SPECIES: PHYTOCHEMICAL ROLES AND SYNTHESIS PARAMETERS

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Abstract

The green synthesis of silver nanoparticles (AgNPs) has gained considerable attention as a sustainable alternative to conventional nanoparticle production due to its use of renewable biological resources and reduced reliance on hazardous chemicals. The genus Garcinia represents a promising plant source for AgNP synthesis because its species contain diverse bioactive phytochemicals, particularly phenolic compounds, flavonoids, xanthones, and tannins, which can participate in the reduction of Ag^+ ions and stabilization of the resulting nanoparticles. This review summarizes recent studies on the green synthesis of AgNPs using Garcinia species, with emphasis on the contribution of phytochemical constituents and the influence of synthesis parameters. The roles of plant part, extract preparation, pH, temperature, reaction time, silver precursor concentration, and extract-to-precursor ratio in controlling nanoparticle formation and characteristics are discussed. Particular attention is given to the relationship between phytochemical composition and synthesis conditions with changes in surface plasmon resonance, particle size, morphology, and colloidal stability. The review also identifies similarities and variations among reported Garcinia species and highlights research gaps that may support the development of more reproducible and sustainable AgNP synthesis strategies. Overall, this review provides an integrated perspective on the potential of Garcinia species as biological resources for the controlled synthesis of AgNPs.

Keywords: Green synthesis, Silver nanoparticles, Garcinia species, Phytochemicals; Synthesis parameters

Room 7

COMBINATION OF SILICA FROM OIL PALM LEAVES WITH BARIUM IN THE ADSORPTION PROCESS OF CONGO RED DYE

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Abstract

The synthesis of barium chloride with silica from oil palm leaves has resulted in the production of barium silicate. This was confirmed using XRD, which identified BaO, $BaSiO_3$ and Ba_2SiO_4, and was further supported by FTIR spectra and SEM. Barium silicate has been successfully utilised in the adsorption of Congo red. The adsorption equilibrium time was 100 minutes. The adsorption kinetics follow pseudo-first-order kinetics with a constant of 0.033 min^{-1}, indicating that the reaction rate is proportional to the amount of adsorbate. The adsorption isotherm follows the Freundlich isotherm with a constant of 0.187. The thermodynamics of adsorption showed ΔHo values of 36.37 kJ/mol and ΔS of 0.11 kJ/mol K, indicating that the adsorption occurring is physical (physisorption). This experiment demonstrates the potential of oil palm leaves to produce material that can be used as an adsorbent for dyes. Further development is still ongoing to produce renewable materials.

Keywords: Oil palm leaves, Barium silicate, Congo red, Adsorption, Advanced materials

Room 7

SYNTHESIS OF A SMART FILM FROM CASSAVA STARCH AND CHITOSAN FUNCTIONALIZED WITH ANTHOCYANINS AS A FOOD FRESHNESS BIOSENSOR

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Abstract

Fresh food products of animal origin, such as chicken meat, are highly susceptible to spoilage during storage. Microbial activity can cause spoilage, altering the chemical properties of the product. Changes in the pH of food products during storage serve as important indicators of quality deterioration. A colorimetric biosensor can detect these pH shifts using color changes in response to pH variations. Anthocyanins are pH-sensitive compounds that make them suitable natural sensors. Integrating anthocyanins into a biofilm matrix can result in a safe biosensor for detecting food freshness. This study aimed to synthesize a cassava starch–shrimp chitosan-based biofilm enriched with purple cabbage anthocyanins and evaluate its color response as an indicator of chicken meat freshness. Anthocyanins were extracted from purple cabbage by maceration using 96% ethanol. Biofilms were prepared by the solvent-casting method using cassava starch, shrimp chitosan, and glycerol, with anthocyanin concentrations of 2%, 4%, 6%, 8%, and 10%. The films were dried at 50°C for 24 hours. The films were dried at 50 °C for 24 h. Characterization included UV-Vis spectrophotometry, FTIR spectroscopy, and color measurements using the Color Grab application based on the CIELAB parameters (L*, a*, and b*) at pH 4, 7, and 10. The selected formulation was applied to chicken meat stored at 4°C for seven days. UV-Vis analysis of the anthocyanin extract showed a maximum absorption peak at 534 nm (absorbance 1.40). The film with anthocyanins showed absorption peaks at 507 and 567 nm, while the non-anthocyanin film showed peaks at 508 and 571 nm. FTIR peaks shifted from 3408.79 to 3434.88 cm^{-1} and from 1635.51 to 1643.84 cm^{-1} between days 0 and 3. During storage, chicken meat pH increased from 6.1 to 7.3, while the film a* value changed from +6.2 to −4.0. The film showed potential as a biodegradable visual indicator of chicken meat freshness through color changes associated with pH variation.

Keywords: Smart film, Cassava starch, Shrimp chitosan, Purple cabbage anthocyanin, pH-sensitive indicator, Chicken meat freshness

Room 7

EDUCATION AND MENTORSHIP ON MEDICAL HAZARDOUS WASTE GOVERNANCE FOR HEALTHCARE PROVIDERS IN JAMBI CITY TOWARD GREEN HEALTHCARE

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Abstract

The growth of small- and medium-scale healthcare facilities—such as primary clinics, independent laboratories, pharmacies, and private practices—in Jambi City has increased significantly. However, this growth lacks adequate capacity and understanding regarding medical hazardous waste (B3) governance. This Community Service initiative aimed to enhance healthcare operators' regulatory compliance and understanding toward realizing Green Healthcare practices. Executed collaboratively with the Jambi City Environmental Office (DLH), the program applied a Triple Helix approach involving academia, government, and business sectors across five key stages: (1) participatory preparation and coordination, (2) baseline identification and analysis, (3) development of educational materials and soft technology products (Pocketbook and visual SOPs), (4) dissemination, workshops, and logbook/waste balance sheet simulations, and (5) monitoring and evaluation. The intervention significantly improved operators' knowledge regarding recent regulatory integration of storage permits into Environmental Approval Technical Details (Rintek), source segregation, container color-coding/labeling, and occupational health and safety (OHS) management. The Hazardous Waste Management Pocketbook served as an effective upstream-to-downstream operational guide for healthcare personnel and sanitation staff. Ultimately, this program contributed to waste management cost efficiency, reduced nosocomial infection risks, and strengthened the supervisory role of the Jambi City Environmental Office.

Keywords: Green Healthcare, Medical Hazardous Waste, Healthcare Providers, Waste Management Governance.

Room 7

PHYSICOCHEMICAL CHARACTERIZATION AND EVALUATION OF THE MULTI-BIOACTIVITY OF VIRGIN COCONUT OIL (VCO) FROM LOCAL BENGKULU COCONUT VARIETIES

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Abstract

VCO is a natural oil whose physicochemical properties and biological activities may vary with coconut variety. This study aimed to characterize VCO from three local coconut varieties in Bengkulu, Indonesia (manna, pandan wangi, and kopyor) and evaluate its antioxidant and antibacterial activities. VCO was produced using a heating method and analyzed for yield, organoleptic properties, moisture, free fatty acid (FFA) content, and pH. Antioxidant activity was assessed using the DPPH assay, while antibacterial activity against Salmonella typhi was evaluated by the inhibition zone method. Kopyor yielded the most VCO (44,55%), followed by pandan wangi (24,67%) and manna (17,27%). All samples had characteristic coconut aroma, clear yellowish appearance, and normal coconut-oil taste. Moisture content ranged from 0,02% to 0,18%, whereas FFA content ranged from 0,0855% to 0,2911%; manna VCO exceeded the 0,2% reference limit used in this study. Pandan wangi showed the strongest antioxidant activity, with the lowest IC_{50} value (49,25 mg/L), followed by kopyor (55,82 mg/L) and manna (60,03 mg/L). Kopyor showed the strongest antibacterial activity, with an inhibition zone of 14,60 mm at 100% concentration. These findings indicate that coconut variety influences VCO characteristics and bioactivities, with pandan wangi showing greater antioxidant activity and kopyor stronger antibacterial activity.

Keywords: Antioxidant Activity, Antibacterial Activity, DPPH Assay, Physicochemical Characterization, Virgin Coconut Oil

Room 7

A NATURAL PEDIASTRUM DUPLEX-DOMINANT CONSORTIUM PERSISTS THROUGH PROLONGED NUTRIENT-UNAMENDED STORAGE

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Abstract

The persistence of natural microalgal consortia under prolonged storage without nutrient supplementation remains relevant to the development of low-input cultivation systems. This study investigated the viability and growth recovery of a Pediastrum duplex-dominant freshwater consortium following 37 days of untreated storage. Three natural microalgal consortia were collected from freshwater ponds in Jambi, Indonesia, and maintained in 250 mL bottles without nutrient supplementation, filtration, or chemical preservation. Following storage, the P. duplex-dominant consortium retained visible green pigmentation and biofilm-like structures, whereas the other two consortia exhibited brown discoloration and biomass sedimentation, suggesting reduced viability. The surviving consortium was subsequently cultivated in a 2 L Erlenmeyer flask for approximately seven days, during which the culture developed a progressively greenish appearance. UV–Vis spectrophotometry revealed an increase in optical density at 680 nm from 0.012 to 0.015 within the first 24 h of cultivation. Microscopic examination at 1000× magnification revealed changes in the detectable taxonomic composition, from an initially P. duplex-dominant community containing Melosira and Phacus to a recovered consortium additionally containing Chlorella, Ankistrodesmus, Chlamydomonas, and Euglena. These findings demonstrate the capacity of a natural freshwater microalgal consortium to remain cultivable following prolonged nutrient-unamended storage and suggest its potential for low-input inoculum maintenance.

Keywords: Pediastrum duplex, Microalgal consortium, Storage viability, Growth recovery, Community composition

Room 7

SURFACE FUNCTIONALIZATION OF OIL PALM-DERIVED CELLULOSE NANOCRYSTALS: IMPLICATIONS FOR CU(II) ADSORPTION PERFORMANCE

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Abstract

One effective approach to enhancing adsorption capacity is surface functionalization. Surface functionalization introduces functional groups on the surface. Examining changes in surface charge properties, particularly zeta potential and point of zero charge (pHPZC), helps clarify the relationship between surface functional groups and adsorption capacity. This study investigated how carboxy modification influences the surface charge characteristics and Cu(II) adsorption performance of functionalized nanocellulose. Nanocellulose was derived from oil palm empty fruit bunches (OPEFB) through delignification, bleaching, and acid hydrolysis, followed by 2,2,6,6-Tetramethylpiperidine-1-oxyl (TEMPO)-mediated oxidation using the TEMPO/NaClO/NaBr system. By varying the NaClO dosage, five adsorbent samples with carboxylate contents were obtained. The results showed that increasing carboxylate content led to more negative zeta potential values, although the changes became less pronounced at higher carboxylate contents. In addition, higher carboxylate content shifted the pHPZC toward more acidic values. It provided a wider pH range favorable for Cu(II) adsorption. These changes in surface charge characteristics were associated with improved Cu(II) adsorption, particularly under mildly acidic conditions. At pH 5.8, the sample with the highest carboxylate content (1.60 mmol/g) achieved an adsorption capacity of approximately 100 mg/g. These findings suggest that surface functionalization not only increases the availability of active binding sites but also modifies surface charge characteristics that influence adsorption performance. Overall, this study highlights the role of surface functionalization in developing nanocellulose-based adsorbents for removing heavy metals from aqueous solutions.

Keywords: Surface functionalization, Nanocellulose, Carboxylate content, Cu(II) adsorption

Paralel Schedule

Room 8 (Online) 13:30-15:30 Moderator: Maulina Tanjung, M.Si.		
Time (WIB)	**Presenter**	**Title**
13.30-13.40	Winny Febriyanti (Universitas Lampung)	Transformation Of Cukuh Balak Natural Zeolite Into Zeolite-X For Enhancement Of Deoxygenation Activity As Catalyst For Co-Pyrolysis Of Mixed Waste Plastic And Used Cooking Oil
13.40-13.50	Anak Agung Sagung Dewi Rahadiani (Universitas Udayana)	Spatial Variability Of In-Situ Water Quality In Lake Batur, Bali, Indonesia
13.50-14.00	Q & A	
14.00-14.10	Mutia Fadhila Putri (Universitas Jambi)	Why Micro and Small Enterprises Adopt E-Commerce: Integrating the Technology Acceptance Model and the Theory of Planned Behavior with Trust and Religiosity
14.10-14.20	Zainil Abidin (Universitas Jambi)	Design Of An Interactive Web-Based Thematic Map Of Mineral Resources In Jambi Province
14.20-14.30	Q & A	
14.30-14.40	Anjany Oktarina (Universitas Jambi)	Sulfur-Modified Palm Oil Fuel Ash at 125°C for Mercury Adsorption in Water Systems
14.40-14.50	Lyra Purna Dewi (Universitas Jambi)	Formulation And Physicochemical Characterization Of Liberika Coffee Waste Nanoemulsion And Nanoemulgel As A Body Scrub Base
14.50-15.00	Q & A	
15.00-15.10	Riki Rahman (National Central University, Taiwan)	Evaluation Cell Performance of Membrane Electrode Assembly (MEA) for Hydrogen Production via AEM Water Electrolyzers
15.10-15.20	Ade Adriadi (Universitas Jambi)	Empowering A Women's Plant-Lover Group Through Training On Liquid Organic Fertilizer Production From Household Organic Waste For The Vegetative Growth Phase In Mendalo Indah Village, Muaro Jambi
15.20-15.30	Q & A	

Room 8

TRANSFORMATION OF CUKUH BALAK NATURAL ZEOLITE INTO ZEOLITE-X FOR ENHANCEMENT OF DEOXYGENATION ACTIVITY AS CATALYST FOR CO-PYROLYSIS OF MIXED WASTE PLASTIC AND USED COOKING OIL

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Abstract

This study investigates the conversion of natural zeolite from the Cukuh Balak region (CBZ), Tanggamus Regency, Lampung Province, Indonesia, into zeolite-X. Prior to modification, the CBZ was purified using 1.0 M HCl and 1.0 M HNO_3 solutions. The purified zeolite was then converted into zeolite-X by adjusting the Si/Al ratio to 1.5, followed by calcination at 500, 600, 700, and 800 °C, characterized using XRD and SEM, and finally used as a catalyst for the co-pyrolysis of mixed waste plastic and used cooking oil. XRD characterization revealed that zeolite-X had formed at 500 °C and continued to grow with increasing calcination temperature up to 800 °C. Pyrolysis experiments showed that the original CBZ had very low catalytic activity, whereas the zeolite-X samples exhibited significant improvement. The zeolites also showed promising deoxygenation activity, as indicated by hydrocarbons being the main component of the bio crude oil (BCO) produced, with the highest relative percentage of 71.79% in the BCO obtained using zeolite-X calcined at 800 °C. Overall, the results indicate that natural zeolite from Cukuh Balak has promising potential as an alternative raw material for the production of synthetic zeolites, not limited to the zeolite-X achieved in this study.

Keywords: Bio crude oil, Pyrolysis, Clinoptilolite, Acid activation, Catalytic upgrading

Room 8

SPATIAL VARIABILITY OF IN-SITU WATER QUALITY IN LAKE BATUR, BALI, INDONESIA

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Abstract

Lake Batur is a volcanic endorheic lake in Bali, Indonesia, where water quality may differ between locations. This study assessed spatial variations in physicochemical water quality to support lake monitoring and management. Field measurements were conducted at eight stations on 9 March 2023 using a multiparameter instrument. To ensure comparable observations, the analysis used measurements at a depth of 2 m. The evaluated parameters were water temperature, pH, oxidation–reduction potential (ORP), turbidity, dissolved oxygen (DO), and total dissolved solids (TDS). The data were summarized descriptively and compared across stations. Water temperature ranged from 23.61 to 24.70 °C, pH from 5.06 to 8.56, ORP from 117 to 135 mV, turbidity from 3.4 to 15.7 NTU, and TDS from 1.31 to 1.33 g/L. Station 1 had the lowest pH, while Station 3 had the highest turbidity. Temperature and TDS varied within relatively narrow ranges, whereas pH and turbidity showed clearer differences among stations. These observations suggest that measurements at one station may not represent conditions across the lake. This station-based assessment provides a baseline for selecting monitoring locations and parameters. Because the results come from a single survey, repeated measurements across seasons are required before drawing conclusions about long-term trends or their causes.

Keywords: Lake Batur; Water quality, Spatial variability, In-situ monitoring, Endorheic lake

Room 8

WHY MICRO AND SMALL ENTERPRISES ADOPT E-COMMERCE: INTEGRATING THE TECHNOLOGY ACCEPTANCE MODEL AND THE THEORY OF PLANNED BEHAVIOR WITH TRUST AND RELIGIOSITY

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Abstract

Micro and small enterprises in Jambi Province, Indonesia, remain slow to adopt e-commerce, hindering their digital transformation. This study aims to examine the factors involved in this adoption by articulating the Theory of Planned Behavior and the Technology Acceptance Model with religiosity and trust. The model was built from prior literature and tested by conducting a survey with 155 owners of micro and small enterprises who use e-commerce, validated by using partial least squares structural equation modeling with bootstrapping, including mediation analysis. The measurement model was valid and reliable, and eight of eleven hypotheses were supported. The results show that perceived behavioral control is the strongest determinant of usage intention (β = 0.507), suggesting that owners adopt e-commerce when they feel capable of selling online. Subjective norm strengthens intention (β = 0.195), whereas attitude has only a marginal effect (β = 0.184; p = 0.067). Perceived usefulness and perceived ease of use shape a positive attitude but do not translate directly into intention. Religiosity shows no effect, as uniformly high religiosity among respondents leaves little variation to explain adoption. Trust is the central antecedent, building perceived ease of use, perceived behavioral control and subjective norm, and exerts the largest total effect on intention (0.448). The model explains 58.6% of the variance in usage intention. In conclusion, capability and trust come before perceived benefits and religious values in driving e-commerce adoption among micro and small enterprises. Government agencies and platform providers should therefore combine digital skills training with trust-building initiatives.

Keywords: Technology acceptance, Planned behavior, E-Commerce, Micro and small business

Room 8

DESIGN OF AN INTERACTIVE WEB-BASED THEMATIC MAP OF MINERAL RESOURCES IN JAMBI PROVINCE

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Abstract

Jambi Province, Indonesia, has significant mineral resources, including coal, gold, tin, iron ore, and other industrial minerals. However, information about their spatial distribution remains scattered across separate sources, limiting accessibility and making it difficult to explore mineral information spatially. This study aims to design an interactive web-based thematic mapping application to support the visualization and management of mineral resource distribution across Jambi Province. The research adopts a Research and Development (R&D) approach. At the current stage, the work focuses on system requirements analysis, data flow diagram (DFD) modelling, information architecture, and medium-fidelity user interface and user experience (UI/UX) prototyping. The proposed system architecture incorporates Laravel and Blade for web application development, PostgreSQL/PostGIS for data management, Leaflet.js for interactive map visualization, and Chart.js for statistical presentation. The planned features include interactive mineral location markers, detailed mineral information, filters by mineral type, administrative region, and licensing status, statistical summaries, and data management interfaces for administrators. The design also considers CSV and GeoJSON data export, subject to access policy decisions. The expected contribution is a structured system design that can guide subsequent implementation and testing of an accessible, integrated, and maintainable mineral information application for Jambi Province.

Keywords: Web mapping, Geospatial data, Mineral resources, UI/UX design, Jambi Province

Room 8

SULFUR-MODIFIED PALM OIL FUEL ASH AT 125°C FOR MERCURY ADSORPTION IN WATER SYSTEMS

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Abstract

This study examines the characteristics of sulfur-modified Palm Oil Fuel Ash (POFA) for potential application in mercury adsorption in multicomponent solutions and mining wastewater. POFA was first activated using 1 M NH_4Cl, followed by sulfur impregnation using thiourea in a microwave digester at 125°C and 5 bar. The POFA-to-sulfur mass ratio was 87.5:12.5 for the multicomponent system and 97.5:2.5 for mining wastewater. FTIR characterization was conducted on the modified adsorbent used in the multicomponent system. The results showed absorption bands at 3338.36, 1632.23, 1416.65, 1003.97, 786.34, 601.56, 560.50, and 422.94 cm^{-1}, indicating the presence of hydroxyl groups, adsorbed water, and silicate-based mineral structures. Changes in physicochemical parameters were also observed following treatment. In the multicomponent system, the treated solution showed an electrical conductivity (EC) of 472 µS/cm, total dissolved solids (TDS) of 238 ppm, pH of 9.73, and turbidity of 4.57 NTU. In mining wastewater, treatment resulted in an EC of 1452 µS/cm, TDS of 728 ppm, pH of 9.55, and turbidity of 1.19 NTU, compared with an initial turbidity of 748 NTU. This study provides preliminary information on the characteristics of sulfur-modified POFA and its influence on water quality parameters in two different treatment systems. The findings may serve as a basis for further investigation of POFA-based adsorbents for mercury-contaminated water treatment.

Keywords: Palm Oil Fuel Ash, Sulfur Modification, Mercury Adsorption, FTIR Characterization

Room 8

FORMULATION AND PHYSICOCHEMICAL CHARACTERIZATION OF LIBERIKA COFFEE WASTE NANOEMULSION AND NANOEMULGEL AS A BODY SCRUB BASE

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Abstract

Liberica coffee processing produces coffee husk and spent coffee grounds that contain bioactive compounds but remain underutilized. This study aimed to formulate and characterize nanoemulsion and nanoemulgel containing Liberica coffee waste extracts for body scrub application and to evaluate their antioxidant potential. Coffee husk and spent coffee grounds were extracted separately by maceration using 70% ethanol for 5 × 24 h and concentrated using a rotary evaporator. Five nanoemulsion formulations (F1–F5) with different extract compositions were prepared using isopropyl myristate, Tween 80, ethanol, and distilled water. The formulations were evaluated based on physical stability, transmittance, pH, emulsion type, particle size, polydispersity index (PDI), and FTIR characteristics. The selected nanoemulsions were subsequently incorporated into nanoemulgel formulations and evaluated for organoleptic properties, pH, spreadability, homogeneity, antioxidant activity using the DPPH method, and SPF value. The extraction yields were 0.24% for coffee husk and 3.60% for spent coffee grounds. The nanoemulsions exhibited O/W characteristics with particle sizes ranging from 14.2 to 29.90 nm and PDI values of 0.203–0.520. The nanoemulgels showed semisolid, transparent, soft, and homogeneous characteristics, with pH values of 5.61–6.06. DPPH analysis showed lower absorbance than the control, indicating radical-scavenging potential. The SPF values ranged from 0.048 to 0.077. These findings indicate the potential of Liberica coffee waste as a value-added natural ingredient for nanoemulsion-based body scrub development.

Keywords: Antioxidant; body scrub; liberica coffee waste; nanoemulsion; nanoemulgel.

Room 8

EVALUATION CELL PERFORMANCE OF MEMBRANE ELECTRODE ASSEMBLY (MEA) FOR HYDROGEN PRODUCTION VIA AEM WATER ELECTROLYZERS

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Abstract

Background: Green hydrogen production through anion exchange membrane water electrolysis (AEMWE) is a promising approach to reducing carbon emissions (CO2). However, membrane properties, catalyst selection, and catalyst loading are remains important factors influencing cell performance. Objective: in this study aimed to evaluate the effects of different membrane, catalyst support combinations, and catalyst loading at oxygen evolution reaction (OER) and hydrogen evolution reaction (HER) electrodes on AEMWE single-cell performance. Methods: Membrane electrode assemblies (MEAs) were fabricated using Pt/C and Ni/C catalysts with different loading (2 and 4 mg/cm²). Catalyst inks were prepared through ultrasonic homogenization and deposited onto porous transport layers using spray coating. The electrodes were dried at 50 °C for 2 hours. Single-cell performance was conducted with 1 M KOH at room temperature, with maximum cell voltage operation increased up to 2.2 V. Results: The Sustainion® X37-50 membrane demonstrated a higher current density compared with FAA-3-50 membrane at 2.2V. The Ni/C at OER and Pt/C at HER demonstrated the highest performance among the investigated catalyst combinations. Furthermore, the catalyst loading of 2 mg/cm² Ni/C at the OER and 4 mg/cm² Pt/C at HER achieved ~1050.72 mA/cm², compared with 935.36 mA/cm² for the reversed loading combination. Conclusions: Membrane selection, catalyst combinations, and catalyst loading distribution are important parameters for optimizing AEMWE single-cell performance.

Keywords: Anion exchange membrane water electrolysis, Green hydrogen; membrane electrode assembly (MEA), Catalyst loading; Oxygen evolution reaction (OER); Hydrogen evolution reaction (HER)

Room 8

EMPOWERING A WOMEN'S PLANT-LOVER GROUP THROUGH TRAINING ON LIQUID ORGANIC FERTILIZER PRODUCTION FROM HOUSEHOLD ORGANIC WASTE FOR THE VEGETATIVE GROWTH PHASE IN MENDALO INDAH VILLAGE, MUARO JAMBI

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Abstract

Household organic waste is the largest fraction of Indonesia's municipal waste, while urban housewives who garden at home largely depend on purchased chemical fertilizers and rarely match fertilization to the plant growth phase. This community engagement program aimed to improve the knowledge and skills of a women's plant-lover group in Mendalo Asri Housing, Mendalo Indah Village, Muaro Jambi Regency, in producing liquid organic fertilizer (LOF) from kitchen waste for the vegetative growth phase. A participatory approach was applied through partner coordination, needs assessment, counselling on household organic waste, LOF concepts and nitrogen requirements during vegetative growth, and hands-on training using a 1:3:10 sugar:organic waste:water formulation with 10 mL L^{-1} EM4 bioactivator in jerrycans fitted with a simple water-trap gas outlet. Training was followed by fermentation monitoring. Knowledge was measured with a 10-item multiple-choice pre-test and post-test, and satisfaction with a 10-item, 5-point Likert questionnaire. The mean knowledge score increased from 65.3 ± 29.5 (n = 15) to 93.8 ± 12.0 (n = 16), giving a high normalized gain (N-gain = 0.82). Among 10 matched participants, scores rose from 71.0 to 96.0 (Wilcoxon signed-rank test, p = 0.035). The largest gains were in the ingredient ratio (33.3% to 93.8% correct) and recognising a putrid odour as a sign of failed fermentation (26.7% to 75.0%). Mean satisfaction was 4.46 out of 5, with 93.8% of responses in the satisfied or very satisfied categories; the allocated practice time received the lowest score. Low-cost microbial fermentation, linked to plant nutritional needs and taught through demonstration and practice, effectively strengthened household capacity for organic waste recycling and home gardening.

Keywords: Community empowerment, EM4, Household organic waste, Liquid organic fertilizer, Vegetative phase

Paralel Schedule

Room 9 (Online) 13:30-15:30 Moderator: Yuliana Safitri, M.Si.		
Time (WIB)	**Presenter**	**Title**
13.30-13.40	Fadillah Priatna (Universitas Jambi)	Synthesis And Characterization Of Liberica Coffee Husk Cellulose Hydrogel Modified With Amine-Chitosan Using Polyphenol As A Bio-Crosslinker For Metronidazole Delivery System
13.40-13.50	Rista Mutia Anggraini (Universitas Jambi)	Functional Group Characterization Of Fish Bone Waste-Derived Hydroxyapatite Scaffolds Modified With Honeycomb Biopolymer And Graphene Oxide
13.50-14.00	Q & A	
14.00-14.10	Dawam Suprayogi (Universitas Jambi)	Freshwater Fish Diversity And Spatial Distribution In The Batanghari River Basin Based On GBIF Records
14.10-14.20	Devi Ratnawati (Universitas Bengkulu)	A Fluorescent Turn-Off Chemosensor For Selective And Sensitive Detection Of Hg(Ii) Ions
14.20-14.30	Q & A	
14.30-14.40	Revi Maylina Tanjung (Universitas Jambi)	Synthesis And Characterization Of Pva/Chitosan Nanofibers By The Electrospinning Method
14.40-14.50	Vebria Ardina (Universitas Jambi)	Physicochemical Characterization Of Coal Fly Ash Components Using Ftir And XRD
14.50-15.00	Q & A	
15.00-15.10	Febri Berthalita Pujaningsih (Universitas Jambi)	Application Of Community-Based Wetland Filter Technology For Irrigation Ecosystem Restoration In Muaro Kumpeh
15.10-15.20	Alfrielman Christian Ziliwu (Universitas Persada Indonesia Y.A.I)	Verification And Validation Of An Ai-Based Procurement System For Government Construction Consultancy Cost Estimation
15.20-15.30	Q & A	

Room 9

SYNTHESIS AND CHARACTERIZATION OF LIBERICA COFFEE HUSK CELLULOSE HYDROGEL MODIFIED WITH AMINE-CHITOSAN USING POLYPHENOL AS A BIO-CROSSLINKER FOR METRONIDAZOLE DELIVERY SYSTEM

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Abstract

Liberica coffee husk is an agricultural waste that contains cellulose and phonelic compounds wint potential for utilization as renewable materials in hydrogel development. This study aimed to synthesize and characterize cellulose-chitosan hydrogels from liberica coofe husk with amine modification and different crosslinking agents, and to evaluate their potential for metronidazole delivery. Cellulose was isolated from Liberica coffee husk and modified through an amination process using ethylenediamine. Coffee husk polyphenol extract was subsequently incorporated as a natural crosslinking component, while sodium tripolyphosphate (STPP) was used as a conventional crosslinking agent. Four hydrogel formulations were prepared, namely H1 consisting of amine-modified cellulose with polyphenol extract, H2 consisting of amine-modified cellulose with STPP, H3 consisting of unmodified cellulose with polyphenol extract, and H4 consisting of unmodified cellulose with STPP. The polyphenol extract showed a total phenolic content of 160 mg GAE/g extract. FTIR analysis showed absorption bands at 3332.23 cm^{-1} attributed to O–H/N–H stretching and 1578.85 cm^{-1} associated with N–H vibration, indicating the presence of characteristic functional groups after modification. SEM analysis showed differences in surface morphology among the hydrogel formulations, including irregular, aggregated, and porous structures. At 240 min, the swelling values of H1, H2, H3, and H4 were 707.17%, 748.96%, 711.24%, and 568.49%, respectively. Metronidazole adsorption efficiencies were 9.262%, 8.832%, 47.98%, and 56.38%, respectively. These findings demonstrate that cellulose modification and the type of crosslinking agent influence the swelling behavior, morphology, and metronidazole adsorption performance of the hydrogels.

Keywords: Amine modification, Cellulose, Hydrogel, Liberica coffe husk

Room 9

FUNCTIONAL GROUP CHARACTERIZATION OF FISH BONE WASTE-DERIVED HYDROXYAPATITE SCAFFOLDS MODIFIED WITH HONEYCOMB BIOPOLYMER AND GRAPHENE OXIDE

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Abstract

Hydroxyapatite (HAp) is a bioceramic material with potential applications in bone tissue engineering due to its composition, which resembles that of natural bone minerals. This study aims to develop hydroxyapatite-based scaffolds synthesized from fish bone waste and modified with honeycomb biopolymer and graphene oxide (GO). The addition of the biopolymer and GO is expected to modify the chemical characteristics of the material and support the development of scaffolds for bone biomaterial applications. Characterization was performed using Fourier transform infrared (FTIR) spectroscopy to identify the functional groups present in the material after modification. The FTIR spectra generally indicate the presence of characteristic hydroxyapatite functional groups, particularly phosphate and hydroxyl groups, as well as functional groups associated with the biopolymer and graphene oxide components. This study is expected to provide insights into the chemical characteristics of biogenic hydroxyapatite-based composite scaffolds as a foundation for developing materials for bone tissue engineering.

Keywords: Scaffold, Hydroxyapatite, Fish bone, Grapheme oxide , Honeycomb

Room 9

FRESHWATER FISH DIVERSITY AND SPATIAL DISTRIBUTION IN THE BATANGHARI RIVER BASIN BASED ON GBIF RECORDS

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Abstract

Openly accessible biodiversity databases play an important role in identifying gaps in species documentation. The Batanghari River Basin supports diverse freshwater fish assemblages, yet integrated information on species occurrence remains limited. This study characterized the composition, sources, spatial and temporal distribution, origin, and conservation status of freshwater fish records in the basin, using Global Biodiversity Information Facility (GBIF) data. We intersected Actinopterygii records downloaded from GBIF on 10 September 2026 with the basin boundary and screened them for taxonomic rank, duplicates, coordinates, habitat, and locality type (capture site or market). Species were classified as native or introduced, assigned IUCN categories, and compared with the inventory of Tan and Kottelat (2009). Screening of 1,283 downloaded records retained 538 records of 169 species from 38 families. Of the 297 species in the earlier inventory, 129 (43.4%) had GBIF records. Conversely, 40 of the 169 GBIF species (23.7%), including all seven introduced species, were not listed in that inventory, possibly because of taxonomic changes. Just over half of the records were iNaturalist observations (50.7%), and 21.9% were market records. Records were concentrated in 2020–2026 (53.0%) and within 25 km of Jambi City (75.1%); 7.0% of 10 × 10 km grid cells contained a record. Fourteen species (8.3% of 169) were globally threatened, including 12 of the 162 native species (7.4%). GBIF data provides a useful but uneven source of information, with record sources, locality types, and sampling effort influencing where and when species are documented. Standardized surveys are needed, particularly beyond Jambi City, to address spatial gaps and support freshwater fish monitoring and conservation.

Keywords: Batanghari River Basin, Biodiversity records, Freshwater fish, GBIF, Spatial distribution

Room 9

A FLUORESCENT TURN-OFF CHEMOSENSOR FOR SELECTIVE AND SENSITIVE DETECTION OF HG(II) IONS

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Abstract

Mercury(II) ions are highly toxic environmental pollutants that require sensitive and selective detection methods. This study aimed to develop a novel vanillin-derived N-phenylpyrazoline fluorescent chemosensor for Hg(II) detection through molecular modification and photophysical characterization. The chemosensor was synthesized through three consecutive reactions involving etherification of vanillin, Claisen–Schmidt condensation, and cyclocondensation, yielding the target compound (MPP) at 91.60%. The molecular structure was confirmed using FT-IR, ^{1}H-NMR, ^{13}C-NMR, and mass spectrometry. Its sensing performance was evaluated using UV–Vis and fluorescence spectroscopy in an acetonitrile–water medium. The results demonstrated that MPP exhibited fluorescence emission at 483 nm upon excitation at 352 nm, which was significantly quenched following Hg(II) addition, indicating a turn-off sensing mechanism. The sensor achieved a limit of detection (LOD) of 1.92 μM, a limit of quantification (LOQ) of 6.43 μM, and a binding constant of 62.89 μM^{-1}. Job's plot and spectroscopic analyses revealed a 1:2 binding stoichiometry between MPP and Hg(II). The novelty of this study lies in the integration of a vanillin-derived conjugated structure with an N-phenylpyrazoline fluorophore for Hg(II) recognition. These findings contribute to the development of fluorescence-based chemical sensing through the understanding of molecular interactions, electronic transitions, and fluorescence quenching, offering potential applications in environmental mercury monitoring.

Keywords: vanillin, mercury, fluorescent, chemosensor, N-phenylpyrazoline

Room 9

SYNTHESIS AND CHARACTERIZATION OF PVA/CHITOSAN NANOFIBERS BY THE ELECTROSPINNING METHOD

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Abstract

Nanofibers have attracted considerable attention due to their high surface area and porous structure, making them suitable for various applications, including filtration, sensors, drug delivery, and biomedical materials. Chitosan is a natural polymer with potential for nanofiber fabrication due to its functional groups and biocompatibility. However, its high solution viscosity and limited spinnability make standalone fiber formation challenging. Polyvinyl alcohol (PVA) is commonly used as a supporting polymer because of its good fiber-forming ability and abundant hydroxyl groups. This study aimed to synthesize and characterize PVA/chitosan nanofibers using electrospinning. The process was carried out at a voltage of 20 kV, a spinneret-to-collector distance of 12 cm, and a flow rate of 0.01 mL/min. The resulting nanofibers were characterized using optical microscopy, Scanning Electron Microscopy (SEM), and Fourier Transform Infrared Spectroscopy (FTIR) to examine their morphology, fiber diameter, and functional groups. SEM analysis showed that the PVA/chitosan nanofibers formed continuous and uniform structures with an average fiber diameter of 176.5 nm. FTIR analysis indicated the presence of characteristic functional groups associated with both PVA and chitosan in the nanofibers. These findings demonstrate the successful fabrication of PVA/chitosan nanofibers by electrospinning and provide information on their morphological and chemical characteristics, supporting further research into their potential applications in functional materials.

Keywords: Nanofibers, Electrospinning, Polyvinyl alcohol (PVA), Chitosan, SEM, FTIR

Room 9

PHYSICOCHEMICAL CHARACTERIZATION OF COAL FLY ASH COMPONENTS USING FTIR AND XRD

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Abstract

Coal fly ash is a fine residue from coal power plants. This study used ash from the PT. Petrokimia Gresik power plant. To provide a more comprehensive evaluation, this research integrates both qualitative and quantitative characterizations of the ash before and after HCl activation. The researchers used XRD and FTIR to identify the dominant mineral phases, combined with the AAS method to determine oxide percentages after chemical digestion. The activation process showed clear results: SiO2 increased from 33.33% to 41.67%, and Al2O3 went up from 16% to 17%, while Fe2O3 and CaO levels dropped. FTIR analysis revealed specific chemical bonds with Si-O vibrating at 462 and 1080 cm^{-1}, Al-O at 1088 cm^{-1}, and the silanol group at 1635 cm^{-1}. XRD tests confirmed that quartz and mullite are the main components of the ash. Ultimately, this integrated approach yields essential information on the proportion of primary components, supporting the future utilization of coal fly ash as a value-added material.

Keywords: Alkaline activation, Coal Fly Ash, FTIR, XRD

Room 9

APPLICATION OF COMMUNITY-BASED WETLAND FILTER TECHNOLOGY FOR IRRIGATION ECOSYSTEM RESTORATION IN MUARO KUMPEH

Febri Berthalita Pujaningsih[1], Frastica Deswardani[1], Rista Mutia Anggraini[1], Lucky Zaehir Maulana[1], Alrizal[1], Muhammad Ficky Afrianto[1], Jesi Pebralia[1], Yoza Fendriani[1], Husnul Hamdi[1], Ichy Lucya Resta[1], Samsidar[1], Medison[2], and Refiandy[2]

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Abstract

The densely populated residential area in Lorong Bambu RT.015 RW.003, Muara Kumpeh Village, is currently facing a sanitation crisis due to the accumulation of domestic waste from 100 households, the majority of whom work as culinary vendors. The practice of discharging production waste, including specific organic waste such as jengkol peels directly into irrigation channels has led to extreme environmental degradation. The water has turned black and foul-smelling, serving as a breeding ground for disease vectors that heighten the risk of health issues such as skin irritation, diarrhea, and dengue fever. A low level of technical literacy regarding waste management is the primary factor behind the failure of pollution mitigation in the area. This initiative aims to enhance the community's capacity and ecological awareness through educational and practical interventions. The program introduces appropriate waste treatment solutions based on Constructed Wetland technology and simple physical filtration using natural media. Through hands-on training, residents are empowered to independently build and maintain waste filtration devices to sustainably restore the quality of the irrigation ecosystem. As an output of this program, an average of 86% of participating residents gained a deep understanding of sanitation management and pollutant impacts, along with the installation of a functional wetland filtration prototype at a strategic location in Lorong Bambu.

Keywords: Service, Irrigation, Filter, Wetland, Kumpeh

Room 9

VERIFICATION AND VALIDATION OF AN AI-BASED PROCUREMENT SYSTEM FOR GOVERNMENT CONSTRUCTION CONSULTANCY COST ESTIMATION

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Abstract

The preparation of the Owner's Cost Estimate for government construction consultancy services requires cost estimates aligned with technical requirements, transparent cost structures, and compliance with procurement regulations. Public Procurement Tools (PLUTO) is a public procurement support system that integrates artificial intelligence (AI) to assist procurement activities, particularly the preparation of Owner's Cost Estimates for construction consultancy services. Although PLUTO can generate cost estimates, its functions and the conformity of its outputs require systematic verification and validation. This study aims to verify PLUTO's functions in preparing Owner's Cost Estimates and validate the resulting documents against technical and regulatory criteria. A descriptive quantitative approach is applied through simulations of three distinct government construction consultancy procurement cases, with a separate estimate prepared for each case. Inputs include project identification, location, duration, contract type, personnel qualifications (education, experience, and certification), assignment periods, and non-personnel requirements. Functional verification uses a checklist to assess input processing, generation of detailed cost breakdowns, and consistent application of procedures across the three cases. Validation examines four aspects: technical conformity, regulatory compliance, calculation accuracy, and appropriateness of price references for personnel and non-personnel components. Each indicator is classified as met, not met, or not applicable, and the findings are summarized descriptively. PLUTO can generate Owner's Cost Estimate documents at its initial development stage; however, structured testing results are not yet available. The study proposes a preliminary framework for transparent and accountable AI-assisted cost planning in government construction consultancy procurement.

Keywords: Owner's Cost Estimate; artificial intelligence; functional verification; regulatory validation; construction consultancy services

Paralel Schedule

Room 10 (Online) 13:30-15:50 Moderator: Andrea Hasbullah, S.T., M.Eng.		
Time (WIB)	**Presenter**	**Title**
13.30-13.40	Andicho Haryus Wirasapta (Universitas Jambi)	Environmental Footprint Screening and Hazard-Weighted Prioritization of Electronic Waste in University Engineering Laboratories
13.40-13.50	Fiqri Al Faruqi (Universitas Jambi)	Design and Feasibility Assessment of an Irrigation Canal-Based Micro-Hydropower System for Agricultural Energy Applications in Air Kandis Village
13.50-14.00	Q & A	
14.00-14.10	Andini Vermita Bestari (Universitas Jambi)	Developing a Smart Faculty Data House Model through SINERGI: An Integrated Academic Analytics Approach to Digital Faculty Governance
14.10-14.20	Hisni Rahmi (Universitas Jambi)	Mapping The Water Pollution Index Of The Batanghari River In Muaro Bungo Regency As A Basis For Environmental Control Zoning
14.20-14.30	Q & A	
14.30-14.40	Fitra Wahyuni (Universitas Jambi)	Selecting a Bioactive Fraction of Jernang Resin for Topical Wound Studies: Integrating Antioxidant and Membrane-Stabilizing Activities
14.40-14.50	Andini Vermita Bestari (Universitas Jambi)	From Forest Patches to Living Labs: Integrating Fern Ecological Assessment into Education for Sustainable Development in Oil Palm Landscapes
14.50-15.00	Q & A	
15.00-15.10	Miranty Yudistira (Universitas Jambi)	Enhancing AI Literacy of Kindergarten and Elementary Teachers through Gemini AI Training
15.10-15.20	Fitra Wahyuni (Universitas Jambi)	Assistance for the Batik Diana SME in Jambi through the Production of Natural Dyes from Low-Grade Areca Nut and SEO-Based Digital Marketing
15.20-15.30	Riskal Fadli (Universitas Jambi)	Evaluation Of Mercury (Hg) Removal Efficiency Based On The Comparison Of Inlet And Outlet Concentrations In A Gas Mercury Adsorber
15.30-15.40	Panji Cakra Buana (Universitas Jambi)	Underutilized Pisang Kerap (*Musa salaccensis*) Leaves as a Source of Chlorophyll-Derived Chromophores
15.40-15.50	Q & A	

Room 10

ENVIRONMENTAL FOOTPRINT SCREENING AND HAZARD-WEIGHTED PRIORITIZATION OF ELECTRONIC WASTE IN UNIVERSITY ENGINEERING LABORATORIES

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Abstract

Engineering laboratories keep long-lived electrical and electronic equipment, yet universities rarely know where its end-of-life hazards concentrate. This study screened asset records from 21 laboratories of the Faculty of Science and Technology, Universitas Jambi, in Electrical Engineering, Earth Engineering, and Civil, Chemical and Environmental Engineering. A total of 542 powered units were counted and classified into six categories; instruments and power equipment made up 77.1%, while computers and office devices accounted for 6.1%. Screening by equipment type flagged 88 units (16.2%) carrying refrigerants, mercury lamps, lead-acid batteries, or possible dielectric fluids. Laboratories were then ranked by unit count and by a hazard-weighted score tested under 81 weight combinations. Robotics held the most equipment, but Electrical Conversion held the most flagged units (36.4%) and ranked first or joint first in 61 weight sets, whereas Electronics and Instrumentation, second by size, never ranked above fourth. The Civil, Chemical and Environmental Engineering department held a quarter of all units but 4.5% of flagged ones, and reported failures in 8.1% of its powered equipment. Laboratories should be prioritized by hazard rather than inventory size, with condition and end-of-life status recorded in asset registers.

Keywords: Electronic Waste, Environmental Footprint, University Laboratories, Hazard Prioritization, Inventory Screening

Room 10

DESIGN AND FEASIBILITY ASSESSMENT OF AN IRRIGATION CANAL-BASED MICRO-HYDROPOWER SYSTEM FOR AGRICULTURAL ENERGY APPLICATIONS IN AIR KANDIS VILLAGE

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Abstract

Limited access to reliable electricity in agricultural areas remains a challenge for supporting various farming activities, particularly in rural locations with limited grid infrastructure. Irrigation canals, which provide continuous water flow for agricultural land, have the potential to serve as local renewable energy resources for small-scale hydropower generation. However, the feasibility of utilizing irrigation canal flow for agricultural energy applications needs to be assessed based on the actual hydraulic characteristics of the site before a hydropower system is developed. This study aims to assess the feasibility and develop a preliminary design of an irrigation canal-based micro-hydropower system for agricultural energy applications in Air Kandis Village, Empat Lawang Regency, South Sumatra. The study employs a quantitative and engineering design approach involving field assessment of irrigation canal characteristics, measurement of flow-related parameters, estimation of hydraulic and electrical power potential, preliminary system sizing, and selection of appropriate system components. The hydraulic power potential is determined based on flow rate and effective head, while the estimated electrical output is calculated by considering the expected overall system efficiency. The results of the assessment are used to develop a preliminary micro-hydropower system configuration consisting of a turbine, generator, transmission system, and supporting electrical components. The feasibility assessment focuses on the technical suitability of the available irrigation flow and its potential to support small-scale agricultural energy needs. The study is expected to provide a technical basis for the subsequent development of an irrigation canal-based micro-hydropower prototype and contribute to the utilization of locally available renewable energy resources for sustainable agricultural activities.

Keywords: irrigation canal, micro-hydropower, hydropower potential, feasibility assessment, preliminary design, agricultural energy, air kandis village.

Room 10

DEVELOPING A SMART FACULTY DATA HOUSE MODEL THROUGH SINERGI: AN INTEGRATED ACADEMIC ANALYTICS APPROACH TO DIGITAL FACULTY GOVERNANCE

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Abstract

Digital transformation in higher education requires systematic management of academic information and quality assurance documents to support faculty governance. However, information originating from different units may be stored in various formats and locations, creating challenges in document management and accessibility. This study aims to develop a Smart Faculty Data House model through SINERGI, a platform designed to support digital governance at the Faculty of Science and Technology, Universitas Jambi, Indonesia. The study employed a Design Science Research (DSR) approach, consisting of user requirements analysis, data source identification, conceptual model design, prototype development, and functional testing. The conceptual model and system architecture were developed using Use Case, Activity, and Class Diagrams to represent system functions, workflows, and data structures. Based on these designs, the SINERGI prototype was developed with a landing page, login interface, dashboard, document management modules for the five stages of the PPEPP quality assurance cycle, accreditation status monitoring, and accreditation document repositories. Functional testing was conducted using the black-box testing method to assess whether the main application features operated according to their specifications. The documented test results indicated that the tested functions met the expected requirements. The study produced a conceptual Smart Faculty Data House model and a functional prototype that brings faculty quality assurance and accreditation documentation into a unified platform. These outputs provide a foundation for the continued development of integrated academic information management and analytics-supported faculty governance.

Keywords: academic analytics; digital governance; higher education; Smart Faculty Data House; SINERGI; data management

Room 10

MAPPING THE WATER POLLUTION INDEX OF THE BATANGHARI RIVER IN MUARO BUNGO REGENCY AS A BASIS FOR ENVIRONMENTAL CONTROL ZONING

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Abstract

Illegal Gold Mining (PETI) activities are occurring on a massive scale in Jambi Province, with a major concentration in Bungo Regency. These mining practices contribute significantly to environmental degradation, including landscape alteration and the deterioration of water quality. This study aims to analyze and map the Water Pollution Index (PI) along the Batanghari River in Muaro Bungo Regency. Sampling was conducted at nine monitoring stations, focusing on three primary water quality parameters: acidity (pH), Total Suspended Solids (TSS), and heavy metal Mercury (Hg) concentrations. The evaluation of water quality status was calculated using the Pollution Index method, in accordance with the Decree of the Minister of Environment Number 115 of 2003. The analytical results confirm that the high intensity of PETI in the Batanghari River riparian zone has triggered an escalation in the water's physical parameters. This is evidenced by a surge in TSS concentrations, which exceeded the established environmental quality standard threshold of 50 mg/L. Conversely, the test results for the heavy metal Hg parameter did not indicate concentrations exceeding the permissible threshold. This phenomenon suggests that mercury residues from the amalgamation process may have undergone massive dilution by the river's discharge, precipitated and accumulated in the bottom sediments, or were not yet measurably distributed within the water column during the sampling period. Comprehensively, the Pollution Index calculation across the nine monitoring stations yielded an average value of 4.43, classifying the water quality of the Batanghari River in the study area as lightly polluted.

Keywords: mapping, pollution index, river, zoning, environmental control

Room 10

SELECTING A BIOACTIVE FRACTION OF JERNANG RESIN FOR TOPICAL WOUND STUDIES: INTEGRATING ANTIOXIDANT AND MEMBRANE-STABILIZING ACTIVITIES

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Abstract

Jernang (dragon's blood) resin from Daemonorops draco (Willd.) Blume is used traditionally in Jambi to treat wounds. Its crude ethanolic extract, however, shows only moderate radical scavenging activity (DPPH IC50 ≈ 160 µg/mL), and no fraction has yet been selected by explicit criteria for mechanistic wound research. This study aimed to identify the most suitable bioactive fraction of jernang resin for topical wound studies by integrating antioxidant and membrane stabilizing activities. The 96% ethanolic extract was partitioned into n-hexane (F-Hx), ethyl acetate (F-EA), and aqueous (F-Aq) fractions, which were then screened phytochemically. Antioxidant activity was measured with the DPPH assay using sample-blank correction. Anti-inflammatory potential was assessed with the hypotonicity induced rat red blood cell (RRBC) membrane stabilization assay at 50–1000 µg/mL, with diclofenac sodium as the reference. Erythrocytes were obtained from healthy Wistar rats, the same species used in the subsequent wound model. IC50 values were derived by four-parameter logistic regression (n = 3) and compared by ANOVA (α = 0.05). Fractions are ranked with a weighted decision matrix (RRBC activity 0.5, DPPH activity 0.2, yield 0.2, gel compatibility 0.1). F-EA was flavonoid-rich and gave the highest yield (87.84%) and the strongest DPPH scavenging among the fractions (IC~50~ 115.84 ± 1.70 µg/mL), a 1.4-fold improvement over the crude ethanolic extract (≈160 µg/mL). With these two criteria, F-EA leads the matrix (partial score 1.2 of 1.4); the RRBC membrane stabilization assay, which carries the largest weight, is in progress and will determine the final selection. Integrating both activities is expected to provide a transparent basis for choosing the fraction to be formulated as a 5% topical gel for evaluation of TNF-α modulation in a rat acute excision wound model.

Keywords: Daemonorops draco; dragon's blood; erythrocyte membrane stabilization; DPPH; bioactivity-guided selection

Room 10

FROM FOREST PATCHES TO LIVING LABS: INTEGRATING FERN ECOLOGICAL ASSESSMENT INTO EDUCATION FOR SUSTAINABLE DEVELOPMENT IN OIL PALM LANDSCAPES

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Abstract

The conversion of forests into oil palm plantations has contributed to habitat fragmentation, leaving remnant forest patches that may serve as refugia for biodiversity, including ferns (Pteridophyta). One such forest patch is located in Danau Lamo Village, Muaro Jambi Regency, and offers potential opportunities for biodiversity conservation and field-based science education. However, ecological baseline data and a structured educational framework are needed to support the effective use and management of these areas. This study aimed to assess fern community structure and distribution in a forest patch within an oil palm landscape and to explore its potential as an off-campus living lab supporting Education for Sustainable Development (ESD). The study adopted a transdisciplinary approach that integrated ecological assessment with educational design. Preliminary site surveys were conducted, the study area was spatially documented using SW Maps, and a sampling framework was developed to cover forest interior, forest-edge, and plantation habitats. The ecological assessment included inventories of terrestrial and epiphytic ferns, vegetation and land-cover observations, and measurements of selected environmental variables, including soil microclimatic conditions. Species composition and community structure were analyzed using the Shannon–Wiener diversity index and the Important Value Index (IVI). Principal Component Analysis (PCA) was also used to explore relationships between fern distribution and environmental conditions. The ecological assessment informed the development of a proposed living lab framework incorporating learning zones, field observation stations, interpretive materials, and practical learning modules. This study provided a preliminary foundation for integrating ecological field research with contextual science education in a fragmented tropical landscape. The proposed framework connected biodiversity monitoring, experiential learning, and sustainability-oriented education. It also identified opportunities for future collaboration among universities, landowners, and plantation stakeholders to support biodiversity conservation and environmental learning.

Keywords: Living Lab, Fern Diversity, Education for Sustainable Development (ESD)

Room 10

ENHANCING AI LITERACY OF KINDERGARTEN AND ELEMENTARY TEACHERS THROUGH GEMINI AI TRAINING

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Abstract

Generative artificial intelligence (AI) can support efficient, contextual, and interactive instructional design, yet many early childhood and primary school teachers still lack the literacy and skills to use it. At the Ash Shiddiiqi Foundation in Jambi, Indonesia, teachers mostly prepared teaching materials manually, which was time-consuming and produced limited variety and interactivity. This community service program aimed to improve the AI literacy and practical competence of kindergarten and elementary school teachers in using Gemini AI to develop interactive learning materials. The program comprised six sessions combining conceptual input, demonstrations, hands-on practice, discussion, and mentoring. Teachers learned the Role–Context–Task–Format (RCTF) prompting framework, were introduced to NotebookLM and Canvas, and practiced drafting daily lesson plans, stories, quizzes, and worksheets, followed by a session on AI ethics covering output verification, student data privacy, and copyright. Effectiveness was assessed with a 15-item pre-test and post-test completed by 50 teachers. The mean score rose from 13.64 (90.93%) to 13.96 (93.07%), with the largest gains on items that initially had the lowest rates of correct answers, and every participant produced a lesson plan draft and a learning media prototype. The novelty lies in a context-specific training model that combines RCTF-based prompting, hands-on production of classroom-ready materials, and ethical AI use for early-grade teachers. Although baseline knowledge was already high, the results suggest that structured hands-on training helps teachers turn conceptual understanding into practical skills, and the model can be replicated in other schools.

Keywords: AI literacy, teacher training, Gemini AI, prompt engineering, interactive learning

Room 10

ASSISTANCE FOR THE BATIK DIANA SME IN JAMBI THROUGH THE PRODUCTION OF NATURAL DYES FROM LOW-GRADE ARECA NUT AND SEO-BASED DIGITAL MARKETING

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Abstract

The Batik Diana SME in Seberang Kota Jambi faces two priority problems: dependence on synthetic dyes that risk environmental pollution, and low digital marketing visibility. Meanwhile, Jambi Province has great potential in areca nut, particularly the betara variety (Areca catechu var. Betara), which contains tannin and is potential as a natural dye source. This community service aims to assist the partner in producing environmentally friendly natural dyes from low-grade areca nut and to strengthen digital marketing based on Search Engine Optimization (SEO) and Google Business Profile. The method applied a participatory approach through five stages: preparation, socialization and needs analysis, training and implementation, monitoring–evaluation, and reflection and follow-up. Results in the ongoing stage show that the priority problems were specifically identified, the natural dye production technology from areca nut was validated through maceration and color fixation with mordants, and the partner has adopted this technology in batik production. For marketing, the creation and optimization of a Google Business Profile, a simple website, an online catalog, and a pre-order mechanism were designed to address stock-availability constraints. The activity is expected to improve product quality and differentiation, reduce synthetic dye use, and expand the partner's market reach.

Keywords: areca natural dye; Jambi batik; SME; digital marketing; Google Business Profile

Room 10

EVALUATION OF MERCURY (HG) REMOVAL EFFICIENCY BASED ON THE COMPARISON OF INLET AND OUTLET CONCENTRATIONS IN A GAS MERCURY ADSORBER

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Abstract

Mercury (Hg) is a hazardous contaminant in natural gas processing that can adversely affect process equipment and operational safety. This study aimed to analyze the reduction in Hg concentration and evaluate the Hg removal efficiency of a gas mercury adsorber unit based on a comparison of inlet and outlet concentrations. Gas samples were collected at the inlet and outlet points over a 30-day period using Tedlar bags. Mercury was subsequently trapped using a gold trap (gold amalgamator) and analyzed using a Mercury Analyzer MA-3 Solo at a wavelength of 253.7 nm. The results showed that the Hg concentration at the inlet ranged from 16.11 to 17.95 µg/Nm³, with an average of 17.11 µg/Nm³, whereas the outlet concentration ranged from 0.0127 to 0.0146 µg/Nm³, with an average of 0.0134 µg/Nm³. The reduction in Hg concentration ranged from 16.0960 to 17.9373 µg/Nm³, corresponding to a removal efficiency of 99.91–99.93%, with an average of 99.92%. Although the removal efficiency remained high, the Hg concentration at the outlet was still above the specified standard of 0.001 µg/Nm³ and showed an increasing trend throughout the observation period. This condition was associated with adsorbent saturation, which resulted in a gradual decline in the adsorbent's ability to adsorb and retain mercury. Therefore, periodic monitoring of Hg concentration and evaluation of adsorbent condition are necessary to determine the appropriate timing for adsorbent replacement or regeneration, thereby maintaining optimal gas mercury adsorber performance and ensuring that the Hg concentration in the outlet gas meets the specified specification.

Keywords: mercury; natural gas; gas mercury adsorber; adsorbent; mercury analyzer.

Room 10

UNDERUTILIZED PISANG KERAP (MUSA SALACCENSIS) LEAVES AS A SOURCE OF CHLOROPHYLL-DERIVED CHROMOPHORES

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Abstract

Wild banana species represent an underexplored source of natural pigments with potential applications in the development of chlorophyll-derived chromophores. This study investigated the utilization of pisang kerap (Musa salaccensis) leaves as a source of chlorophyll a for subsequent conversion into pheophorbide a. Fresh leaves (788.207 g) were macerated in 4 L of acetone for 24 h, followed by silica gel impregnation and preparative column chromatography using n-hexane/ethyl acetate mixtures. One-eighth of the impregnated extract, equivalent to 98.526 g of fresh leaves, was processed. The isolated chlorophyll a-enriched fraction was subjected to acid-induced demetallation using 0.5 M HCl for 30 min, followed by alkaline dephytylation using 0.5 M KOH for 1 h. UV–Vis spectroscopic characterization in ethyl acetate revealed characteristic Soret and Qy absorption maxima at 434 and 662 nm for chlorophyll a, 410 and 666 nm for pheophytin a, and 408 and 664 nm for the pheophorbide a-enriched product. These spectral changes supported the sequential transformation of the chlorophyll chromophore. The process yielded 325 mg of pheophorbide a-enriched material, corresponding to an apparent isolated yield of 3.30 mg g^{-1} fresh weight (0.330%, w/w). These findings highlight the potential of underutilized M. salaccensis leaves as an alternative botanical resource for producing value-added chlorophyll-derived chromophores.

Keywords: Musa salaccensis, Chlorophyll a, Pheophorbide a, Pigment valorization, UV–Vis spectroscopy

Paralel Schedule

Room 11 (Online) 13:30-15:40 Moderator: Faisal Asadi, S.T., M.Sc.		
Time (WIB)	**Presenter**	**Title**
13.30-13.40	Jefri Marzal (Universitas Jambi)	A Reproducible Procedural Environment With Moving Obstacles For Multi-Robot Watchman Research
13.40-13.50	Teguh Setyawan (Universitas Jambi)	Synthesis Of Eco-Friendly Composite (Fly Ash/Biochar/Fe3O4/Chitosan)
13.50-14.00	Q & A	
14.00-14.10	Mochammad Arief Hermawan Sutoyo (Universitas Jambi)	Gamified Knowledge Management Systems: A Systematic Literature Review of Effectiveness and Design Risks
14.10-14.20	Akhiyar Waladi (Universitas Jambi)	Small Patches Favor Compact Siamese Encoders: Deforestation Detection in Jambi Province
14.20-14.30	Q & A	
14.30-14.40	Muh Rizal B (Universitas Jambi)	Synthesis And Characterization Of Ce-UiO-66-NH2
14.40-14.50	Fetty Febriasti Bahar (Universitas Jambi)	Development of a Smart Farming-Integrated Productive Housing Model for Rural Households: Design and Prototype Development in Pematang Jering Village, Indonesia
14.50-15.00	Q & A	
15.00-15.10	Husnul Hamdi (Universitas Jambi)	Surface Runoff and Soil Percolation Across Land-Use Groups in the Lower Batanghari Sub-Watershed
15.10-15.20	Uni Baroroh Husnudin (Universitas Jambi)	Conservation of Local Durian in Jambi Tulo Village, Maro Sebo District, Muaro Jambi Regency
15.20-15.30	Rara Ayu Lestary (Universitas Jambi)	Preliminary Characterization of MCC–Chitosan Hydrogels Prepared under Different Freeze–Thaw Conditions
15.30-15.40	Q & A	

Room 11

A REPRODUCIBLE PROCEDURAL ENVIRONMENT WITH MOVING OBSTACLES FOR MULTI-ROBOT WATCHMAN RESEARCH

Jefri Marzal, Akhiyar Waladi, Massudi Mahmuddin, and Niken Rarasati

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Abstract

Comparing watchman algorithms in dynamic environments requires repeatable static layouts and moving-obstacle sequences. This paper describes a procedural generator that records the settings needed to reconstruct both. Seeded Perlin noise defines occupied lattice cells, which are merged into orthogonal polygons after enclosed free cells are filled. A style scheduler varies the spatial arrangement of obstacles, while optional raster edits adjust their boundary complexity. Rectangular moving obstacles follow a persistent random walk, with footprint checks at initialization and each accepted update. We characterize the generated geometry across twelve seeds and examine complexity editing and obstacle motion in worked examples. The requested vertex counts differ from the achieved counts, and maps with similar obstacle densities can have different boundary complexity. The illustrated motion sequence remains collision-free at the sampled states. The resulting benchmark record pairs the requested settings with measured geometry and obstacle positions. The study covers environment construction. Guard placement and route planning remain outside its scope.

Keywords: benchmark environment; procedural generation; Perlin noise; moving obstacles; reproducible simulation

Room 11

SYNTHESIS OF ECO-FRIENDLY COMPOSITE (FLY ASH/BIOCHAR/FE3O4/CHITOSAN)

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Abstract

Natural materials such as coal fly ash, rice husk biochar, Fe3O4 derived from iron sand, and chitosan from shrimp shells can be utilized to create eco-friendly composites. This research aims to develop a high-performance material that repurposes waste while ensuring environmental sustainability. The study involved the synthesis of the individual components coal fly ash, rice husk biochar, Fe3O4 from iron sand, and chitosan from shrimp shells followed by the synthesis of an eco-friendly composite (Fly Ash/Biochar/Fe3O4/Chitosan) incorporating all four materials. The synthesized material was characterized using XRD, FTIR, FE-SEM-EDX, and TEM. The results of the synthesis and characterization indicate that the eco-friendly composite possesses a large pore surface area and magnetic properties, enabling maximum adsorption capacity.

Keywords: Biochar, Chitosan, Eco-friendly composite, Fe3O4, Fly Ash.

Room 11

GAMIFIED KNOWLEDGE MANAGEMENT SYSTEMS: A SYSTEMATIC LITERATURE REVIEW OF EFFECTIVENESS AND DESIGN RISKS

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Abstract

Background: Higher education institutions lose tacit governance knowledge whenever structural officials end their terms. ARUNIKA, a faculty knowledge repository, was built to retain this knowledge, but contributions remain low because sharing is voluntary and former officials have little incentive to document their experience. Gamification is a candidate non-coercive incentive, yet its effectiveness in knowledge management systems (KMS) is uncertain. Objective: This review examined how gamified KMS have developed, whether they improve user engagement, participation motivation and long-term knowledge retention, which game elements are associated with successful systems, and how design risks are mitigated. Methods: A systematic literature review following Kitchenham's guidelines was conducted on IEEE Xplore, ScienceDirect, ACM Digital Library and Google Scholar on 10 October 2026. Of 1,828 records, 31 passed title and abstract screening and eight primary studies (2019–2026) were retained after full-text reading. The studies were appraised with a six-item quality checklist and synthesised narratively. Results: Studies came from Indonesia, Brazil, Saudi Arabia, India, Europe and Finland. Three reported success (Yes), none reported failure, and five gave no effectiveness verdict. Evidence for engagement and motivation was perceptual and short-term (5 weeks to one semester); no study measured long-term knowledge retention. Peer feedback, vote-up ratings, reputation status, expert validation and self-set goals were the elements most consistently linked to positive outcomes. Reported risks were competitive pressure, uncalibrated goals, delayed feedback, imbalanced element placement, declining engagement over time and privacy concerns. Conclusion: Gamified KMS are promising but not yet conclusively effective. For ARUNIKA, recognition-centred, quality-validated, privacy-aware and goal-calibrated gamification is recommended, followed by longitudinal evaluation.

Keywords: gamification; knowledge management system; knowledge sharing; knowledge retention; higher education; systematic literature review

Room 11

SMALL PATCHES FAVOR COMPACT SIAMESE ENCODERS: DEFORESTATION DETECTION IN JAMBI PROVINCE

Akhiyar Waladi[1], Andini Vermita Bestari[1], Nindy Raisa Hanum[1], Yogi Perdana[1], and dan Hasanatul Iftitah[1]

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Abstract

Large pretrained encoders may be poorly matched to the 32-pixel patches used for deforestation monitoring from Sentinel-2. We test whether a large ImageNet-pretrained Siamese encoder outperforms a compact network trained from scratch on small 32×32-pixel bitemporal patches for annual deforestation detection in Jambi Province, Indonesia. We compare all models under one spatial-block protocol on annual Sentinel-2 pairs: training, validation, and test patches come from separate geographic blocks with a guard band at every block boundary, so no input pixel is shared across partitions. A three-layer convolutional network trained from scratch matches the change-detection accuracy of an ImageNet-pretrained ResNet-34 and outperforms ResNet-50, while using a fraction of a percent of their parameters; blockbootstrap confidence intervals on the paired differences support this ordering. The larger models trade precision for recall, multiplying the false alarms that an operational monitoring system must dismiss. The standard ResNet stem downsamples a small patch before its first residual block, reducing the local detail available to the classifier. The larger pretrained encoders provide no accuracy gain at this scale under the training budget used. These results favour compact encoders for annual monitoring on small patches.

Keywords: change detection, Siamese CNN, Sentinel-2, model efficiency, spatial-block validation, deforestation, Jambi Province

Room 11

DEVELOPMENT OF A SMART FARMING-INTEGRATED PRODUCTIVE HOUSING MODEL FOR RURAL HOUSEHOLDS: DESIGN AND PROTOTYPE DEVELOPMENT IN PEMATANG JERING VILLAGE, INDONESIA

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Abstract

Rural households face challenges in optimizing limited residential yard space for productive and sustainable activities. This study aims to develop a productive housing model integrated with smart farming, combining residential functions with household-scale food production in Pematang Jering Village, Indonesia. The study adopts a design and prototype development approach, focusing on architectural design, spatial integration, and the development of a productive housing prototype based on the utilization of residential yards. The proposed model integrates residential spaces, productive yard areas, and smart farming components to promote efficient resource utilization and support household food production. The prototype is developed to assess the technical feasibility of the proposed design and the functional integration of its architectural, spatial, and smart farming components

Keywords: Smart farming; Productive housing; Rural households; Residential yards; Prototype; Sustainable housing.

Room 11

SURFACE RUNOFF AND SOIL PERCOLATION ACROSS LAND-USE GROUPS IN THE LOWER BATANGHARI SUB-WATERSHED

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Abstract

Land use and land cover (LULC) influence rainfall partitioning into surface runoff, soil-water storage, and subsurface flow. This study examined surface runoff and soil percolation across land-use groups in the lower Batanghari watershed, Jambi, Indonesia, using existing Soil and Water Assessment Tool (SWAT) simulation outputs for 2023–2025. The analysis covered 13 subbasins containing 62 terrestrial hydrologic response units, with open-water units excluded. Monthly and annual outputs were summarized using area-weighted means and compared across agricultural land, wetlands, urban land, mixed forest, and shrubland. Spatial distributions were mapped at the subbasin level. Wetlands exhibited the highest mean annual surface runoff at 932.51 mm/year, followed by agricultural and urban land at 738.98 and 732.98 mm/year, respectively. Mixed forest and shrubland exhibited lower runoff at 214.88 and 248.60 mm/year, alongside higher soil percolation at 1,248.06 and 1,178.47 mm/year. Runoff represented 8.33% of precipitation in mixed forest and 34.05% in wetlands, while the corresponding percolation fractions were 48.40% and 28.47%. These contrasts indicate differences in simulated rainfall partitioning among land-use groups. However, they also reflect variations in soil properties, terrain, and rainfall and cannot be attributed exclusively to LULC. The findings identify spatial contrasts relevant to watershed management and field verification. Soil percolation represents water movement below the root zone and should not be interpreted directly as measured infiltration or aquifer recharge.

Keywords: Surface runoff, soil percolation, LULC, SWAT, Batanghari watershed

Room 11

A REPRODUCIBLE PROCEDURAL ENVIRONMENT WITH MOVING OBSTACLES FOR MULTI-ROBOT WATCHMAN RESEARCH

Jefri Marzal[1], Akhiyar Waladi[1], and Niken Rarasati[1]

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Abstract

Comparing watchman algorithms in dynamic environments requires repeatable static layouts and moving-obstacle sequences. This paper describes a procedural generator that records the settings needed to reconstruct both. Seeded Perlin noise defines occupied lattice cells, which are merged into orthogonal polygons after enclosed free cells are filled. A style scheduler varies the spatial arrangement of obstacles, while optional raster edits adjust their boundary complexity. Rectangular moving obstacles follow a persistent random walk, with footprint checks at initialization and each accepted update. We characterize the generated geometry across twelve seeds and examine complexity editing and obstacle motion in worked examples. The requested vertex counts differ from the achieved counts, and maps with similar obstacle densities can have different boundary complexity. The illustrated motion sequence remains collision-free at the sampled states. The resulting benchmark record pairs the requested settings with measured geometry and obstacle positions. The study covers environment construction. Guard placement and route planning remain outside its scope

Keywords: benchmark environment; procedural generation; Perlin noise; moving obstacles; reproducible simulation

Room 11

CONSERVATION OF LOCAL DURIAN IN JAMBI TULO VILLAGE, MARO SEBO DISTRICT, MUARO JAMBI REGENCY

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Abstract

Jambi local durian (Durio zibethinus) possesses rich germplasm resources with high genetic diversity and distinctive taste profiles. However, its population continues to decline due to land conversion and the dominance of modern commercial varieties. This community service program aims to conserve Jambi local durian through community empowerment based on science and technology applications in Jambi Tulo Village, Muaro Jambi Regency. The partner of this activity is the Gerakan Muaro Jambi Bersakat (GMB) community group. The implementation method is structured into five stages: (1) Focus Group Discussion (FGD) for problem identification and mapping of local varieties, (2) hands-on training on grafting techniques using a grafting tool kit, (3) establishment of a germplasm collection demonstration plot in Jambi Kecil Village, (4) varietal labeling (hanging label) and digital mapping based on GPS geotagging using the SW Maps application, and (5) periodic seedling growth monitoring. The results indicate an enhancement in partners' technical skills in vegetative propagation, yielding 32 successfully grafted local superior varieties (such as Belimbing, Tembago, Terung, and Daun), the establishment of a spatially documented demonstration plot, and the fostering of collective awareness in germplasm conservation. This program successfully strengthens community independence, expands collaboration through the signing of an Implementation Arrangement (IA), and generates scientific publication outputs in mass media and national journals.

Keywords: Conservation, Local Durian, Grafting, Geotagging, Jambi Tulo

Room 11

PRELIMINARY CHARACTERIZATION OF MCC–CHITOSAN HYDROGELS PREPARED UNDER DIFFERENT FREEZE–THAW CONDITIONS

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Abstract

The physical properties of hydrogels depend on the interactions between polymer components and the structure of the resulting network. This study investigated the preliminary characteristics of microcrystalline cellulose (MCC)–chitosan hydrogels prepared under different freeze–thaw conditions, focusing on swelling behavior, gel fraction, weight loss, and thickness. The hydrogels were prepared at an MCC-to-chitosan ratio of 50:50 and subjected to four treatments: a non-freeze–thaw control, freeze–thaw pretreatment of MCC, and one or two freeze–thaw cycles applied to the hydrogel formulation. Swelling was evaluated after 1 h of immersion, while gel fraction, weight loss, and thickness were assessed to provide complementary information on hydrogel stability and physical characteristics. The combined evaluation of these parameters is intended to provide insight into the relationship between freeze–thaw processing conditions and the physical properties of MCC–chitosan hydrogels. The findings are expected to establish a preliminary basis for understanding the effects of freeze–thaw treatment on hydrogel characteristics and to inform further development and optimization of these biopolymer-based materials.

Keywords: microcrystalline cellulose; chitosan; hydrogel; freeze–thaw treatment; swelling; gel fraction

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